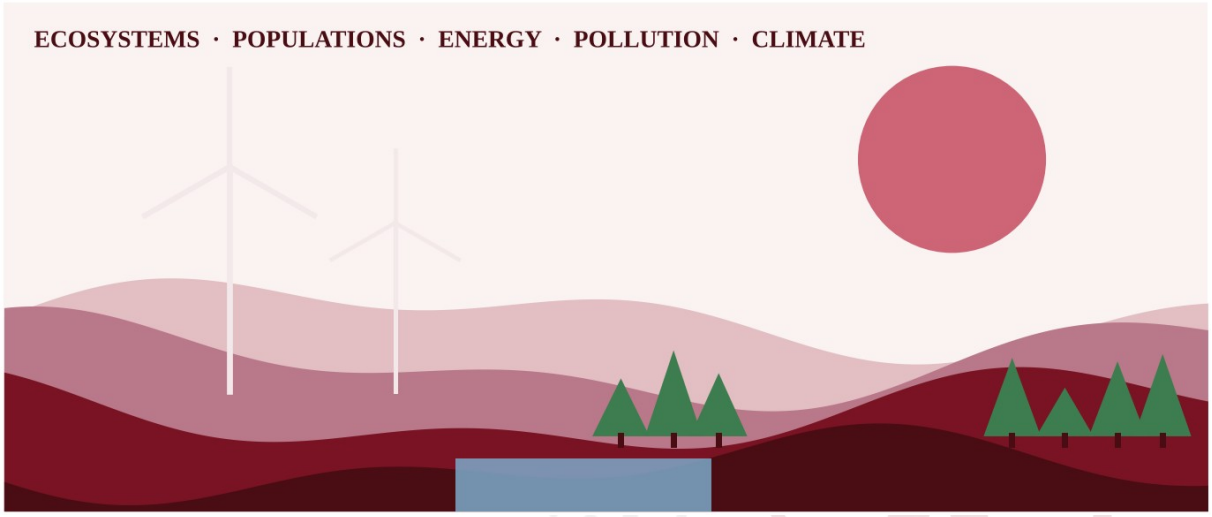
Success4Sure Academy

BOOK FOUR

AP ENVIRONMENTAL SCIENCE

COLLEGE BOARD RELEASED QUESTIONS & COMPLETE SOLUTIONS CLASSIFIED BY UNIT



Released-exam questions sorted by unit, with the verified year on every item and all solutions at the end.

Prepared by Mr. Waseem Allabadi

Published by Success4Sure Academy

2026–2027 Edition | Aligned to the 2026 Course and Exam Description

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Prepared by Mr. Waseem Allabadi. Published by Success4Sure Academy. First edition, 2026–2027.

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About This Book

This book organizes questions from College Board released AP Environmental Science materials by the nine units of the 2026 course. It is a study tool built from three things: a verified year for every question, an original task description, and an original, complete solution.

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How the book is organized

- **Part I — Classified Questions.** Nine units. Inside every unit, multiple-choice (MCQ) entries come first and free-response (FRQ) entries second. Each entry shows its verified examination year.
- **Part II — Complete Solutions.** All solutions are printed at the end of the book, by unit, MCQ first and FRQ second, in the same order as the questions.

Notes on older exams

Exams before 2020 used five answer choices (A–E) in the multiple-choice section; the 2026 exam uses four (A–D). Where a question is based on subject matter that is no longer part of the 2026 course, the item has been left out of this book. Free-response questions are classified by their primary assessed concept, with secondary units recorded internally.

Part I — Classified Questions

> **NOTE**
> Every entry below corresponds to a question from a College Board released AP Environmental Science examination. To protect College Board copyright, the question text and figures are not reproduced: each entry gives the verified examination year and an original description of the task. Use the official released exam or free-response materials from College Board (apcentral.collegeboard.org) alongside this book. Original solutions to every entry are in Part II at the end of this book.

Unit 1 — The Living World: Ecosystems

Section A — Multiple-Choice Questions

Question 1 — 2008
Which ocean region is least likely to contain photosynthetic organisms?

Question 2 — 2008
Organic waste added in different amounts to algae tanks: what is the main purpose of the experiment?

Question 3 — 2008
Which pond/lake zone holds rooted emergent plants such as cattails?

Question 4 — 2008
Why did soil have abundant nitrogen after a nitrogen-poor field was colonized by an invasive plant?

Question 5 — 2010
Matching set (elements): identify which element has marine sediments forming limestone as a major long-term sink.

Question 6 — 2010
Matching set (elements): which element is commercially obtained from seabird guano deposits.

Question 7 — 2010
Ocean cross-section with five depth zones: identify the zone with the greatest biodiversity.

Question 8 — 2010
Same ocean cross-section: which zone has species adapted to the least light and highest pressure.

Question 9 — 2010
Same ocean cross-section: which zone absorbs the greatest total mass of CO2 from the atmosphere.

Question 10 — 2010
What is the main purpose of photosynthesis for a primary producer: oxygen, CO2, glucose, protein or water.

Question 11 — 2010
Which ecosystem has the lowest net primary productivity per square meter per year: taiga, open ocean, temperate grassland, estuary, farmland.

Question 12 — 2010
Which is a characteristic of tropical rain forests (low rainfall, high soil pH, nutrient-poor soils, low diversity, rain shadows).

Question 13 — 2010
Why are decomposers in soil and water important to ecosystems.

Question 14 — 2010
Primary characteristic grasslands share.

Question 15 — 2010
Which ecological concept includes all the others.

Question 16 — 2016
Five climatographs (monthly precipitation and temperature): choose the climate supporting the greatest tree-species diversity.

Question 17 — 2016
Same climatographs: which climate favors dry scrub with frequent fires?

Question 18 — 2016
Why are phytoplankton concentrated in the upper few hundred feet of water?

Question 19 — 2016
Before a solar plant is built, what baseline data would let a student test whether panel shade raises creosote bush mortality?

Question 20 — 2016
Which of five diagrams correctly shows the nitrogen cycle?

Question 21 — 2016
Corn needs 1 ha per 1,000 kcal; beef takes 20 times more land: which equation gives land for 10,000 kcal of beef?

Question 22 — 2016
Explain why the open ocean accounts for such a large share of Earth's total net primary productivity even though its productivity per unit area is low.

Question 23 — 2016
Which is an example of a carbon sink?

Question 24 — 2016
Which test best checks whether hikers' trampling killed alpine plants?

Question 25 — 2016
NPP bar chart for three forest biomes: what explains the variation?

Question 26 — 2024
Matching set: which species type is first to colonize disturbed ground and stabilizes soil.

Question 27 — 2024
World map with lettered locations; which is in a temperate seasonal forest biome.

Question 28 — 2024
Same map: location with highest primary productivity.

Question 29 — 2024
Why local vegetation improved after wolves arrived on the island.

Question 30 — 2024
Why there are far fewer wolves than deer on the island.

Question 31 — 2024
Simplified nitrogen cycle with lettered arrows: which arrow shows nitrogen fixation.

Question 32 — 2024
Caterpillar experiment on air temperature tolerance in a sub-Antarctic region: the dependent variable.

Question 33 — 2024
Elk, wolf and aspen graphs for Yellowstone: likely outcome.

Section B — Free-Response Questions

Question 1 — 2003
Coastal estuaries: two causes of temperature or salinity variation, two ecological and two economic roles of coastal wetlands, three human harms, and a policy plus an economic incentive for one harm. *(Free-response; 10 points)*

Question 2 — 2008
Biogeochemical cycles: describe nitrogen cycling and its main reservoir, one way the phosphorus cycle differs, a pollutant nitrogen or phosphorus compound with its source and effect, and one human disruption of the carbon cycle with two consequences. *(Free-response; 10 points)*

Question 3 — 2014
Carbon and phosphorus cycles: biological removal and return of atmospheric carbon, ocean sinks, a long-term land sink, other human carbon sources, a CO2-related problem with a consequence, and how phosphorus differs and why it is needed. *(Free-response; 10 points)*

Question 4 — 2025
Serengeti ecology: plate boundary type, dominant biome, wildebeest calving location from a migration map, resource partitioning, effect of a proposed highway and a solution, traits of invasive plants, crop rotation, pesticide problems and an alternative pest control. *(Analyze and interpret quantitative data; 10 points)*

Unit 2 — The Living World: Biodiversity

Section A — Multiple-Choice Questions

Question 1 — 2008
A pond sample yields one fish and 17 invertebrates of 10 species; what can be estimated for the invertebrates?

Question 2 — 2008
After a year of nearby development, which resampling result best signals harm to the pond?

Question 3 — 2008
Which is the best example of a keystone species?

Question 4 — 2008
Why does a wet winter in Mediterranean regions lead to worse fires the next summer?

Question 5 — 2010
Why native pollinators serve native plants better than honeybees.

Question 6 — 2016
Same world map: where is tropical forest fragmentation by people driving major biodiversity loss?

Question 7 — 2016
Succession graph of tree biomass after clear-cutting: at which time is species richness highest?

Question 8 — 2016
Same graph: which tree does best in full sun?

Question 9 — 2016
Why do small isolated islands lose species at higher rates than large near islands?

Question 10 — 2016
Beak sizes of finches on two islands differ although the species is the same: choose the best hypothesis.

Question 11 — 2016
Which question is best for investigating why alpine meadow natives decline while invasive grasses spread?

Question 12 — 2016
Which fact best supports a citizens' group claim that oak-hickory forest will not be harvestable in 50 years?

Question 13 — 2016
Which hypothesis shows economic benefit of plant diversity in pastures?

Question 14 — 2024
Matching set: species type whose abundance or scarcity signals changing abiotic conditions.

Question 15 — 2024
Matching set: bumblebee whose decline alters community structure and cuts biodiversity.

Question 16 — 2024
Which strategy best maintains regulating ecosystem services of coastal wetlands.

Question 17 — 2024
Why domesticated honeybee colonies declined while wild colonies did not.

Question 18 — 2024
Same article: how the rivers provide cultural ecosystem services.

Question 19 — 2024
Which ranking orders biomes from greatest to least biodiversity.

Question 20 — 2024
Monarch article with bar graph: approximate percent decline in western monarchs from 1997 to 2019.

Question 21 — 2024
Same article: which reason the author uses to explain the decline.

Question 22 — 2024
Same article: strategy the author says would help monarchs.

Question 23 — 2024
Which ecosystem service category covers recreation, tourism and aesthetic enjoyment.

Question 24 — 2024
Which example best shows an indicator species.

Question 25 — 2024
Manatee populations with different genetic diversity: best conclusion.

Section B — Free-Response Questions

Question 1 — 2003
Whooping crane and California condor recovery: causes of original decline, protection measures, characteristics that slow recovery, and an argument for and against protecting an endangered species. *(Free-response; 10 points)*

Question 2 — 2011
Article on mountain pine beetle outbreaks and declining managed honeybees: why fire suppression favors beetles, two ways beetle damage worsens climate change, physical forest changes after tree death, economic effects of bee collapse, another ecosystem service. *(Free-response; 10 points)*

Question 3 — 2013
Biodiversity: two traits of high-biodiversity ecosystems, two human activities that reduce it with mechanisms, practical solutions for each, one natural cause of loss, and two ecological benefits. *(Free-response; 10 points)*

Question 4 — 2016

Newspaper article on white-nose syndrome devastating little brown bats: why disease rarely drives hosts extinct, threats to tiny populations, slow recovery, effects on other organisms, forest ecosystem services, and why human emerging diseases are increasing. *(Free-response; 10 points)*

Question 5 — 2023

Satellite track of a manatee crossing a shipping channel with water temperatures: temperature range used, why power-plant discharge attracts manatees, harm to other species, seagrass loss effects, an estuary pollution fix, photochemical smog in summer, and fuel-cell car drawbacks. *(Analyze and interpret quantitative data; 10 points)*

Question 6 — 2024

Succession diagram from bare rock to forest: process name, stage where pioneers appear, soil formation, a cultural service, then a clear-cut versus intact watershed stream study (hypothesis, independent variable, comparison forest, temperature, golf-course effects). *(Design an investigation; 10 points)*

Unit 3 — Populations

Section A — Multiple-Choice Questions

Question 1 — 2008
From a table of population and land area, which country has the highest population density?

Question 2 — 2008
Which country has a population about three times that of the U.S.?

Question 3 — 2008
Which practice would have the biggest impact on global sustainability?

Question 4 — 2008
In the demographic transition, during which phase is population growth fastest?

Question 5 — 2008
Which statement about total fertility of a society is true?

Question 6 — 2008
Best description of a population with a stable age distribution?

Question 7 — 2008
Fremont Center's power demand in 40 years at 3.5% growth with constant per capita demand?

Question 8 — 2008
True of K-strategists compared to r-strategists?

Question 9 — 2008
Why remain concerned about environmental impact despite slowing population growth?

Question 10 — 2008
Doubling time of a population growing 2% per year?

Question 11 — 2010
Choose, from five population-versus-time graphs, the one for growth at biotic potential.

Question 12 — 2010
Choose, from the same five graphs, the one for a population that grew and then leveled off at carrying capacity.

Question 13 — 2010
Which best characterizes exponential population growth.

Question 14 — 2010
Population momentum is most likely when a large share of people are in which age group.

Question 15 — 2010
Which pair of statistics most likely indicates the highest quality of life.

Question 16 — 2010
Country with negligible migration: birth rate fell from 44 to 16 and death rate from 17 to 5; what stage is it in.

Question 17 — 2010
Doubling time for a population growing 1.75 percent a year.

Question 18 — 2010
How is the crude growth rate r calculated.

Question 19 — 2010
Global population graph to 2150: what will it most likely do.

Question 20 — 2010
Characteristic of r-selected species.

Question 21 — 2016
Table of five countries with fertility, infant mortality and life expectancy: which is most likely shrinking in population?

Question 22 — 2016
Same table: which country most likely offers women limited education?

Question 23 — 2016
On a growth graph, what does the horizontal line labeled K show?

Question 24 — 2016
Five age-structure diagrams: which population most likely has a negative growth rate?

Question 25 — 2016
Besides birth control access, what else is a likely cause of the projected decline in world growth rate?

Question 26 — 2016
Which describes demographic changes at the start of transition from preindustrial to industrial?

Question 27 — 2016
Which policy would promote above-replacement fertility?

Question 28 — 2016
Map of population doubling times: which statement is best supported?

Question 29 — 2016
What does a K-selected strategy produce?

Question 30 — 2016
Which statement is consistent with graphs of meat consumption and population?

Question 31 — 2016
Which growth rate is paired with the correct doubling time?

Question 32 — 2024
Population graph over 100 years with labeled points; which point shows carrying capacity K.

Question 33 — 2024
Same graph: which point shows rapid increase when resources are abundant.

Question 34 — 2024
Same graph: which point shows the culmination of a population dieback.

Question 35 — 2024
Deer and wolf census table: what happened to the wolf population after 1985.

Question 36 — 2024
Same table: year of greatest deer population.

Question 37 — 2024
Demographic transition graph: what helps a country move from stage 2 to stage 3.

Question 38 — 2024
Squirrels with Type II survivorship: what that implies about mortality.

Question 39 — 2024
Why people older than 64 now outnumber children under 5 globally.

Question 40 — 2024
Method for percent increase of 1998 San Francisco rainfall (119.94 cm vs 57.20 cm average).

Question 41 — 2024
Four age structure diagrams: which has replacement-level fertility.

Question 42 — 2024
Wolves and elk feedback: why elk herd size changes through time.

Question 43 — 2024
Birth and death rate table: most stable population (assuming constant rates).

Question 44 — 2024
Same table: natural rate of increase for Jamaica.

Question 45 — 2024
Which combination of birth/death and migration gives greatest population increase.

Question 46 — 2024
Matching set (carrying capacity, total fertility rate, demographic transition model, density-independent factor): shows birth and death rates shifting as a country develops.

Question 47 — 2024
Same set: the total number of individuals a habitat can sustain.

Question 48 — 2024
Same set: the value that, at replacement level, leaves population size relatively stable.

Question 49 — 2024
Same set: what happens to the ecosystem if a population exceeds this value and resources become depleted.

Question 50 — 2024
Age structure diagram of Country Z: which age range is the largest share.

Question 51 — 2024
Same diagram: claim about total fertility rate best supported.

Question 52 — 2024
Same diagram: likely socioeconomic concern.

Question 53 — 2024
Best explanation of why a population overshooting carrying capacity can crash.

Question 54 — 2024
Age structure with pyramid shape and large prereproductive cohort: environmental problem in next 10 years.

Question 55 — 2024
Doubling times from annual percent change in population.

Section B — Free-Response Questions

Question 1 — 2003
Fictional country Industria: plot birth and death rates over time, compute 1950 growth rate and 1977 birth rate, reasons for falling death rate and sticky birth rate, and project a population using doubling time. *(Analyze and interpret quantitative data; 10 points)*

Question 2 — 2010
Termite wood-consumption table (mg/day/termite) across five temperatures and three humidities; find optimum, scale methane emissions per hectare and for a changed climate, explain population limits, post-clearing termite boom and bust, another way deforestation warms climate. *(Analyze and interpret quantitative data; 10 points)*

Question 3 — 2017
Haiti: reason for deforestation and a strategy to reduce it, how deforestation alters stream water quality, other forest benefits, then a fertility and life-expectancy data table and factors that could explain falling fertility and rising life expectancy. *(Free-response; 10 points)*

Question 4 — 2022

Charlotte, North Carolina growth and coal dependence: health effect of coal pollutants, coal-ash disposal problem and clay liners, percent population change 2013-2019, doubling year from a 1.88% rate, annual water use, and cutting outdoor water use. *(Analyze an environmental problem doing calculations; 10 points)*

Question 5 — 2025

Chickadees (K-selected) and nonnative yard plants: reproductive strategy, spider decline effect, reading spider counts and insect trend (Figure 1), growth-rate evidence for a 25% threshold (Figure 2), ant-diversity study in mowed park versus grassland, resilience, paved playground effect, road fragmentation. *(Design an investigation; 10 points)*

Unit 4 — Earth Systems and Resources

Section A — Multiple-Choice Questions

Question 1 — 2008
Map matching: world location with least naturally available freshwater per person.

Question 2 — 2008
Map matching: region that relies heavily on desalination for its freshwater supply.

Question 3 — 2008
Which atmospheric layer in the temperature profile holds the protective ozone layer?

Question 4 — 2008
Which atmospheric layer contains most of the atmosphere's molecules?

Question 5 — 2008
Which layer is mainly responsible for Earth's surface weather?

Question 6 — 2008
Which atmospheric layer has the lowest air pressure?

Question 7 — 2008
Order of soil particles by increasing size.

Question 8 — 2008
In a soil profile diagram, which lettered layer has the most organic material?

Question 9 — 2010
True statement about the stratosphere.

Question 10 — 2010
Which does NOT exert a major influence on Earth's overall climate.

Question 11 — 2010
Soil sample passes 98 of 100 ml of water; which composition is likely.

Question 12 — 2010
Which soil property most influences water flow through the sample.

Question 13 — 2010
Anchovy catch table: low 1998 catch most likely from what.

Question 14 — 2010
Cause of high 1998 sea surface temperature.

Question 15 — 2010
Which best describes leaching.

Question 16 — 2016
World plate map with five lettered locations: which shows chains of volcanic islands formed above subducting plates?

Question 17 — 2016
Same plate map: which location has a rising non-volcanic mountain range from continental collision?

Question 18 — 2016
Same plate map: where is new crust forming at a divergent boundary?

Question 19 — 2016
Which process is NOT powered by the Sun: plant growth, ocean currents, hurricanes, plate tectonics, or cloud formation?

Question 20 — 2016
Which graph shows soil organic matter falling over time under long-term cropping without manure or fertilizer?

Question 21 — 2016
Soil texture triangle: read the sand, silt and clay percentages at marked point T.

Question 22 — 2016
Same triangle: which point's texture best suits lining a landfill?

Question 23 — 2016
Why are most deserts near 30° north and south latitude?

Question 24 — 2016
How does El Niño cause a drop in southern California brown pelican numbers?

Question 25 — 2016
A family of five switches from 5 to 2 gallons-per-minute showerheads (8 min each daily): water saved in 30 days?

Question 26 — 2016
What is the most direct result of surface ocean circulation?

Question 27 — 2016
Why are two soil samples dark and high in C and N while four are low?

Question 28 — 2016
Which latitude has 24 hours of daylight on December 21?

Question 29 — 2024
Matching set: periodic Pacific surface-temperature change altering global rainfall, wind and ocean circulation.

Question 30 — 2024
Matching set: dry conditions on the leeward side of a mountain range.

Question 31 — 2024
Four block diagrams: which shows a divergent boundary with seafloor spreading.

Question 32 — 2024
Graph of percent of soil eroded versus soil life span: years to erode 60% under standard farming.

Question 33 — 2024
Monthly precipitation graph east and west of mountains: best-supported statement.

Question 34 — 2024
If Earth's axial tilt were zero, how would mid-latitude climate be characterized.

Question 35 — 2024
Soil of 35% clay, 40% sand, 25% silt: which description of properties applies.

Question 36 — 2024
Component best added to poorly drained soil.

Question 37 — 2024
Best soil for citrus needing mixed composition and high drainage.

Question 38 — 2024
Seismic shockwave article: author's primary claim.

Question 39 — 2024
Same article: evidence supporting the claim that the pattern is produced by waves of a specific frequency.

Question 40 — 2024
Same article: how the pattern could influence urban development.

Question 41 — 2024
Atmosphere altitude-temperature graph: line dividing stratosphere from mesosphere.

Question 42 — 2024
Map of tectonic plates near Japan: why Tokyo is subject to earthquakes.

Question 43 — 2024
Map with ages of seamounts: pattern from Detroit to Hawaii.

Section B — Free-Response Questions

Question 1 — 2003
Article about invasive earthworms consuming forest leaf litter: roles of leaf litter, three abiotic changes, link to takeover by exotic grass, and a controlled experiment design. *(Design an investigation; 10 points)*

Question 2 — 2004

Summer extension-service intern: describe one chemical and one physical soil test and how results guide sustainable farming, one pro and con of inorganic fertilizer, two erosion-control practices, and a humus-rich biome. *(Free-response; 10 points)*

Question 3 — 2014

Pacific Ring of Fire: what happens at subduction zones and why volcanoes form, primary succession and soil formation on new volcanic ground, tsunami generation and ecological harm, and transform-boundary behavior during and between earthquakes. *(Free-response; 10 points)*

Question 4 — 2016

Soil: how two climate factors affect formation rate, biotic and abiotic components of the A horizon, nitrate in groundwater from farming, acid deposition effects and remediation, and two ways climate change degrades soil. *(Free-response; 10 points)*

Question 5 — 2021

Soil erosion rate versus slope for four farming practices (graph), then a road-construction test of straw bales versus grass for sediment control (hypothesis, control, variables, turbidity data), and fish that lay eggs in streambed gravel. *(Design an investigation; 10 points)*

Question 6 — 2024

Colorado River drought: desert landscaping plant and extra benefit, prescribed burn purpose and drawback, then Lake Powell calculations: maf at 36% full, snowmelt share in a low-flow year, and households supported by 9.60 maf. *(Analyze an environmental problem doing calculations; 10 points)*

Question 7 — 2025

El Niño-Southern Oscillation maps and North American jet stream: sea surface condition, which phase, regional climate contrast, urban flooding cause, solution and extra benefit, temperate forest versus savanna climate, post-fire process, fire pollution, sustainable forestry. *(Analyze and interpret quantitative data; 10 points)*

Unit 5 — Land and Water Use

Section A — Multiple-Choice Questions

Question 1 — 2008
Which environmental problem is associated with abandoned coal mines?

Question 2 — 2008
Which commercial tree-harvest method most fragments landscapes and harms biodiversity?

Question 3 — 2008
Greatest potential risk of using biological pest controls?

Question 4 — 2008
Gold extraction in remote regions often contaminates the environment with what?

Question 5 — 2008
Which statement about farm-raised salmon is true?

Question 6 — 2008
Best illustration of the pesticide treadmill?

Question 7 — 2008
Which agricultural practice leads to dangerous disease strains?

Question 8 — 2008
Best illustration of the tragedy of the commons?

Question 9 — 2008
Effects on water cycle when forest is converted to urban land?

Question 10 — 2008
Which practice is allowed under USDA organic production?

Question 11 — 2008
Which best describes mountaintop removal?

Question 12 — 2008
Why is vegetarianism cited as helping with deforestation?

Question 13 — 2010
Matching set (elements): which element in the mineral pyrite drives acid mine drainage.

Question 14 — 2010
Which best describes acid mine drainage.

Question 15 — 2010
Which switch from large-scale conventional farming most reduces farmworker and neighbor exposure to toxic chemicals.

Question 16 — 2010
Farmer needs increasing pesticide concentrations to control a pest; best explanation.

Question 17 — 2010
Why does salmon farming threaten wild salmon.

Question 18 — 2010
Why biological control of insects is effective.

Question 19 — 2010
Best representation of worldwide water use by sector.

Question 20 — 2010
Problem best addressed by contour plowing.

Question 21 — 2010
Correct pairing of agricultural practice and consequence.

Question 22 — 2016
Which statement about the Ogallala Aquifer is correct?

Question 23 — 2016
Graph of glyphosate use and herbicide-resistant weed species both rising: which statement explains it?

Question 24 — 2016
Which is a concern about growing more genetically modified crops?

Question 25 — 2016
Table of five reclaimed mine sites: choose the one with the best forest reclamation after ten years.

Question 26 — 2016
New suburban housing replaces farms: what is its biggest effect on the local water supply?

Question 27 — 2016
Which homeowner step would help protect creek water quality?

Question 28 — 2016
Bar graph of land use in four watersheds: which statement is likely true?

Question 29 — 2016
Which equation computes income from a farm that is 40 % trees and 60 % vegetables?

Question 30 — 2016
Which problem is fracking most likely to cause?

Question 31 — 2016
Best example of the tragedy of the commons?

Question 32 — 2016
Three bar graphs of irrigation water, money and return per liter: which statement is best?

Question 33 — 2024
Which statement best describes a benefit of clear cutting.

Question 34 — 2024
Compost study graphs of sediment loss and runoff: short-term effects of compost.

Question 35 — 2024
Same study: average sediment loss with compost applied previously, years 4–6.

Question 36 — 2024
Same study: identify the control group.

Question 37 — 2024
Graphs of Mexico's freshwater per capita and population, 1950–2025: supported conclusion.

Question 38 — 2024
Predicted water temperature and dissolved oxygen change after clear cutting along a stream.

Question 39 — 2024
How salinization occurs.

Question 40 — 2024
Farming technique used specifically to prevent soil erosion.

Question 41 — 2024
Recycled wastewater versus surface water irrigation soil table: interpret results.

Question 42 — 2024
Which situation most favors pesticide resistance in a greenhouse.

Question 43 — 2024
Why aquaculture has increased in recent years.

Question 44 — 2024
Which factor most likely explains flash flooding in urban streams.

Question 45 — 2024
Graph of GE crop adoption: approximate percent change in HT corn acres from 2005 to 2007.

Question 46 — 2024
Same graph: time period with greatest rate of increase in Bt corn.

Question 47 — 2024
Advantage of genetically modified crops such as those in the graph.

Question 48 — 2024
Disposing of anthracite coal mine tailings in a nearby lake: likely consequence.

Question 49 — 2024
Coastal wetland loss table: which cause accounts for most freshwater acres lost.

Question 50 — 2024
Same table: conclusion about wetland loss.

Question 51 — 2024
Which method gives percent of 40 million wetland acres lost to urban development.

Question 52 — 2024
Which diagram shows traditional furrow irrigation.

Question 53 — 2024
Spearfishing removes herbivorous fish from a reef: effect.

Question 54 — 2024
Difference between CAFOs and free-range farms.

Question 55 — 2024
Reason for annual catch limits in US commercial fisheries.

Section B — Free-Response Questions

Question 1 — 2002
Colorado River water diversion: two environmental problems, argue for urban use and for natural-area flow, and another large diversion project with two problems. *(Free-response; 10 points)*

Question 2 — 2005
Global meat production grew from 52 to 240 billion kg while population rose 2.6 to 6.0 billion: per capita figures, percent change, efficiency of grain vs meat, environmental consequences, and health effects of low-meat diets. *(Analyze an environmental problem doing calculations; 10 points)*

Question 3 — 2005
Surface coal mining reclamation: restoration steps, why arid regions are harder, sulfur impact and remedy, two other impacts of coal use, and why per capita coal use may rise. *(Free-response; 10 points)*

Question 4 — 2006
Georges Bank groundfish harvest graph: five-year period of steepest decline and rate, two commercial fishing practices and how they deplete stocks, a law or treaty managing marine harvest, and another commons with impacts and management. *(Analyze and interpret quantitative data; 10 points)*

Question 5 — 2008

Gold mining calculation on a block diagram of 20 blocks with gold percentages: total gold volume, percent gold, break-even price per gram from a given cost and density, mine-waste uses, and post-abandonment problems. *(Analyze an environmental problem doing calculations; 10 points)*

Question 6 — 2011

Table of farmland needed to feed humanity 1900 to 2060: plot and curve it, estimate when a 4.00 billion ha arable limit is reached, soil properties and quality, ways to cut land demand, how soil salinization occurs and is prevented. *(Analyze and interpret quantitative data; 10 points)*

Question 7 — 2012

Pesticide debate framing: design a lab test of whether a new pesticide kills minnows (hypothesis, method, control, dependent variable), results that would reject the hypothesis, describe integrated pest management with two tactics and a benefit, two non-pest yield practices. *(Design an investigation; 10 points)*

Question 8 — 2014

Combined sewer overflow: two runoff pollutants, runoff volume from a 200 by 100 m parking lot and from 20 percent impervious city area after 5 cm rain, bypass volume at the plant, ways to cut runoff, a paving problem. *(Analyze an environmental problem doing calculations; 10 points)*

Question 9 — 2015

Urban sprawl: causes, health effects, describe the trend between population density and petroleum use on a graph, smart-growth measures, ways to reduce harm to wildlife from roads and sprawl, and urban food production. *(Analyze and interpret quantitative data; 10 points)*

Question 10 — 2016

Iron ore, pig iron and steel: four mass calculations from global production data and steel recycling savings, plus environmental problems of abandoned coal mines, a mitigation method, and why surface mining is cheaper. *(Analyze an environmental problem doing calculations; 10 points)*

Question 11 — 2016

Iron and steel supply chain: calculate rock waste, pig iron from all reserves, ore reserve used for steel, coal saved by recycling, then abandoned mine problems, a mitigation method, and why surface mining costs less. *(Analyze an environmental problem doing calculations; 10 points)*

Question 12 — 2018

School renovation article: define carbon footprint, link a heating system to it and reduce it, two green-roof benefits, cafeteria practices, native-plant landscaping, and an environmental benefit of plant-based flooring over synthetic carpet. *(Free-response; 10 points)*

Question 13 — 2021

Pesticide use in cotton: human-health benefit, exposure route, a 1960-2015 table of application rate versus pests trapped, evidence of resistance, effects on non-target wildlife, crop rotation, a lower-pesticide alternative and its human benefit. *(Analyze and interpret quantitative data; 10 points)*

Question 14 — 2022

Urban heat islands and rooftop solar: cause, a solution and extra benefit, percent change in cell efficiency, change in yearly electricity cost, annual kWh from 30 panels, and seasonal sunlight difference between hemispheres. *(Analyze an environmental problem doing calculations; 10 points)*

Question 15 — 2023

Four-plot trial of three genetically modified bean strains versus unmodified beans on a floodplain: control, research question, less irrigation, runoff and soil-temperature table, fertilizer-use hypothesis, post-burn succession, diversity and recovery, invasive beetle and its control. *(Design an investigation; 10 points)*

Question 16 — 2023

Coal-mine overburden piles restored naturally versus technically (plant-species graph): readings and trends, claim about animal richness, early colonizers, tundra versus forest recovery, then a wheatgrass, wood-chip and bare-soil runoff test, sand effects, and mine-drainage impact and control. *(Design an investigation; 10 points)*

Question 17 — 2024

Environmental costs of animal protein (global warming potential and land use graphs): land for chicken protein, mealworm survivorship type and advantage, chicken versus pork impact, why beef differs, cattle grazing near streams, sustainable erosion control, and climate effects on crops. *(Analyze and interpret quantitative data; 10 points)*

Unit 6 — Energy Resources and Consumption

Section A — Multiple-Choice Questions

Question 1 — 2008
Map matching: region where the world's largest hydroelectric dam sits on one of the longest rivers.

Question 2 — 2008
Which listed energy source is renewable?

Question 3 — 2008
Which listed fuel has proven reserves expected to be largely used up within about 50 years?

Question 4 — 2008
What solar system type is required for high-temperature water or steam for industry and turbines?

Question 5 — 2008
Which is the best example of an energy storage element in a solar energy system?

Question 6 — 2008
Ethiopia's per capita electricity use is about what percent of the U.S. value, from the table?

Question 7 — 2008
Primary environmental advantage of natural gas over gasoline as a vehicle fuel?

Question 8 — 2008
From a graph of fossil fuel use, what is true of 1965–1980 growth?

Question 9 — 2008
If 1/16 of a radioactive waste remains after 200 million years, what is its half-life?

Question 10 — 2008
Crude oil components are separated in refineries by differences in what?

Question 11 — 2008
Table of three towns' populations, growth rates and per capita demand: years until the 2,000 MW plant cannot meet demand?

Question 12 — 2008
What percent of total available power is West Fremont using now?

Question 13 — 2008
Congress's current recommendation for permanent storage of high-level radioactive waste in the U.S.?

Question 14 — 2008
Which has the largest proven coal reserves?

Question 15 — 2008
Which explanation is consistent with oil consumption doubling from 1960 to 1970?

Question 16 — 2010
Matching set (energy sources): which source has noise and release of corrosive chemicals as drawbacks.

Question 17 — 2010
Matching set (energy sources): which source is used more in the least developed countries than in highly developed ones.

Question 18 — 2010
Matching set (energy sources): which source converts sunlight directly into electricity.

Question 19 — 2010
Calculation: a 2.0 W clock runs 8,600 hours a year; find annual energy use in kWh.

Question 20 — 2010
Best evidence that the Aswan High Dam was an ecological disaster (archaeological flooding, sardines, power supply, upstream diversion, trapped silt).

Question 21 — 2010
Fuel-source graph (1850-2000): year coal's share was greatest.

Question 22 — 2010
Same graph: approximate year oil and gas became dominant.

Question 23 — 2010
Why nuclear power cannot fully replace oil and gas.

Question 24 — 2010
Primary cause of the decline in wood fuel in the 1800s.

Question 25 — 2010
Most successful solution to fish mortality caused by dams.

Question 26 — 2010
Best example of cogeneration.

Question 27 — 2016
Same world map: which region has been the world's largest petroleum exporter for roughly forty years?

Question 28 — 2016
Which statement correctly describes how a fuel cell vehicle works?

Question 29 — 2016
Why does a hybrid electric vehicle use less fuel per mile than an ordinary car?

Question 30 — 2016
Graph of US electricity sources 1950–2010: pick the most valid inference.

Question 31 — 2016
Which is an advantage of nuclear power for electricity?

Question 32 — 2016
What is hydraulic fracturing used to do?

Question 33 — 2016
Using a decay curve of moles against time, estimate the half-life.

Question 34 — 2016
A major limitation of photovoltaic cells?

Question 35 — 2016
Compare CO_2 released by a 30 mpg and a 20 mpg car over 60,000 miles at 20 lb CO_2 per gallon.

Question 36 — 2024
Island formed by a hot spot on a divergent boundary: which energy source its geology supports.

Question 37 — 2024
Why biomass such as wood is often used for fuel in developing countries.

Question 38 — 2024
Disadvantage of hydraulic fracturing.

Question 39 — 2024
Renewable source that still releases CO_2 and other greenhouse gases.

Question 40 — 2024
Which expression gives annual electricity cost from monthly use and price per kWh.

Question 41 — 2024
Diagram of heated groundwater driving a turbine: type of energy production.

Question 42 — 2024
Same diagram: why water at location X is hot.

Question 43 — 2024
Same diagram: how the hot water is used to generate electricity.

Question 44 — 2024
Which is an advantage of generating electricity from solar energy.

Question 45 — 2024
Diagram of a home photovoltaic system with lettered parts: which part converts sunlight to electricity.

Question 46 — 2024
Same diagram: the part that converts DC from a battery into AC for appliances.

Question 47 — 2024
Same diagram: the part allowing electricity use on cloudy days or at night.

Question 48 — 2024
Greatest disadvantage of generating energy with wind turbines.

Question 49 — 2024
Correct sequence of events to produce electricity in a fossil-fuel power plant.

Question 50 — 2024
Most likely negative effect of a large solar panel farm in a desert.

Question 51 — 2024
Which list contains only renewable energy sources.

Question 52 — 2024
Fukushima nuclear incident: likely long-term health effect.

Question 53 — 2024
Percent increase in solar generation from 25 to 90 million MWh.

Section B — Free-Response Questions

Question 1 — 2002
Electric vehicles versus gasoline cars: two environmental benefits, estimate petroleum saved if 10% of 150 million cars are electric, point vs. dispersed pollution sources, and two policies to encourage adoption. *(Analyze an environmental problem doing calculations; 10 points)*

Question 2 — 2004
Small town replacing a 12 MW coal plant with 10 wind turbines: compute plant annual output, household total, compare, cost per kWh of wind over 25 years, and two benefits plus two costs. *(Analyze an environmental problem doing calculations; 10 points)*

Question 3 — 2004
Radioactive waste: low versus high-level properties and storage, characteristics of an ideal deep burial site, two other management options with feasibility, and a sublethal health effect of radiation. *(Free-response; 10 points)*

Question 4 — 2005
Arctic refuge oil development: days of US supply from 10 billion barrels at 20 million barrels per day, fragility of tundra, activities with impacts, and two end uses with conservation measures. *(Analyze an environmental problem doing calculations; 10 points)*

Question 5 — 2006

Arizona family weighing a photovoltaic system: one benefit and one cost, persuasive choice between stand-alone and grid-connected, two ways to promote PV use, and two passive solar approaches. *(Free-response; 10 points)*

Question 6 — 2008

Projected population and energy-use table for developing and developed countries: social reasons for LDC growth, percent-increase calculations, reasons for the different energy trends, how fossil fuels become electricity, and sustainable alternatives. *(Analyze an environmental problem doing calculations; 10 points)*

Question 7 — 2011

Scatter graph of GDP per capita against electricity use per capita for nine countries: geothermal electricity generation, China's large total CO2 despite low per-capita use, health-harming coal-power pollution, sugarcane ethanol versus tar sands production, drawbacks and sustainability. *(Analyze and interpret quantitative data; 10 points)*

Question 8 — 2012

Article on hydraulic fracturing of shale gas: two water-related problems, two benefits of natural gas over coal, two non-water drawbacks, one economic benefit, and two environmental impacts of nuclear power. *(Free-response; 10 points)*

Question 9 — 2012

School district heated by fuel oil plans to offset carbon by conserving forest: one alternative energy source with environmental pros and cons, two ecological benefits of intact forest, then four calculations of carbon stored, emitted, hectares needed and cost. *(Analyze an environmental problem doing calculations; 10 points)*

Question 10 — 2013

Compares battery electric and gasoline vehicles: government incentives, fuel cost per mile for each, charging cost from a 4.0 kW charger over 7.0 h, difference in CO2 per 100 miles, and economic effects of more BEVs. *(Analyze an environmental problem doing calculations; 10 points)*

Question 11 — 2014

Nuclear plant proposal for a river town: judge two claims (solid waste, greenhouse gases), construction-phase environmental problems, normal-operation pollutant, its ecological effect and control system, electricity-saving steps, and an accident example. *(Free-response; 10 points)*

Question 12 — 2017

Dams: how a hydroelectric dam generates electricity, two economic benefits besides power and flood control, ecological value of floodplain flooding, benefits and drawbacks of dam removal, and beaver dams as a keystone species. *(Free-response; 10 points)*

Question 13 — 2018

Offshore wind farm: environmental benefit, an economic effect, another ocean energy source, then calculations for 200 turbines of 4 MW and 2.0 million MWh annual demand: 80 percent of demand, revenue at $0.20 per kWh, and operating hours needed. *(Analyze an environmental problem doing calculations; 10 points)*

Question 14 — 2019

Oil-sands (bitumen) mining under boreal forest in Alberta: forest benefits, environmental costs of extraction and transport, and three unit-conversion calculations (days and years to extract 73 billion barrels; water needed for synthetic crude). *(Analyze an environmental problem doing calculations; 10 points)*

Question 15 — 2021

Homeowner designs a test of rooftop solar panel compass direction (variables, hypothesis, extra variable), then hydroelectric dam diagram: turbine energy step, stored potential energy, pairing with solar or wind, climate risks, reservoir benefit and silt effect. *(Design an investigation; 10 points)*

Question 16 — 2021

Replacing aging coal plants with natural gas: why coal is nonrenewable, environmental and economic gains of gas, preventing CO poisoning, then percent change in PM2.5, kilograms of PM2.5 no longer removed by logging, and added hospital admissions. *(Analyze an environmental problem doing calculations; 10 points)*

Question 17 — 2022

Hydraulic fracturing: well-water methane versus distance to a fracking well (graph), how fracking fluid works, a geologic effect, tragedy of the commons, groundwater overuse and saltwater intrusion, and a policy to cut oil use. *(Analyze and interpret quantitative data; 10 points)*

Question 18 — 2023

Crude oil and gold-ore mining: marine impact, a pollutant from burning oil, a personal fuel-saving action and its health benefit, then grams of gold in a deposit, value from 1,000 t, and ore needed for 100,000 phones. *(Analyze an environmental problem doing calculations; 10 points)*

Question 19 — 2024

Coal-fired plant diagram: smokestack pollutant, reduction method, process at X, coal delivery emissions, solar basics and latitude, then pesticide, IPM and fertilizer runoff solution. *(Free-response; 10 points)*

Question 20 — 2024

Climate mitigation with nuclear and natural gas: nuclear fuel, cooling-water impact, nuclear kWh from an 18.9% share, air-quality gain from gas, tree planting benefit, homes powered by a gas deposit, and CO_2 saved by replacing coal. *(Analyze an environmental problem doing calculations; 10 points)*

Question 21 — 2025

New coal-powered town: surface mining type, mining and downwind problems, a policy to cut coal impacts, pounds of coal for 11,000 homes, percent change in bird richness 7.5 to 6.0, and doubling year at 5.38% growth. *(Analyze an environmental problem doing calculations; 10 points)*

Unit 7 — Atmospheric Pollution

Section A — Multiple-Choice Questions

Question 1 — 2008
Which listed greenhouse gas forms in the lower troposphere through photochemical reactions?

Question 2 — 2008
Which listed energy source is chiefly responsible for acid rain in the northeastern U.S.?

Question 3 — 2008
Which listed material occurs naturally in bedrock and leads to radon formation?

Question 4 — 2008
An arrow on a pH scale (near pH 5.6) represents which of the choices?

Question 5 — 2008
How many times greater is the H^+ concentration at pH 4 than at pH 7?

Question 6 — 2008
Which device reduces SO_2 emissions from coal-burning power plants?

Question 7 — 2008
Which statement about the Clean Air Act is true?

Question 8 — 2008
Which strategy appropriately removes radon from a home?

Question 9 — 2008
Which factors contribute to smog formation in the Los Angeles basin?

Question 10 — 2008
Which method removes particulates from smokestack emissions?

Question 11 — 2008
Which are direct products of hydrocarbon fuel use in engines?

Question 12 — 2010
Which health outcome is NOT a likely consequence of indoor air pollution (asthma, lung cancer, skin cancer, flu-like symptoms, respiratory disease).

Question 13 — 2010
Which component of photochemical smog is a reddish-brown, pungent gas.

Question 14 — 2010
Pond pH fell from 8.2 to 5.2 over fifty years; which other changes are expected (oxygen, heavy metals, fish species).

Question 15 — 2010
Effective short-term remediation of the acidified pond.

Question 16 — 2010
Most likely effect of acid deposition on a northern forest.

Question 17 — 2010
Where is danger from radon greatest.

Question 18 — 2010
Which air pollutant is correctly paired with a major effect.

Question 19 — 2010
Which is a primary air pollutant produced by burning coal.

Question 20 — 2010
Most important factor in tree decline in northeastern US.

Question 21 — 2010
Bar graph of precipitation pH: how many times more acidic NE US than South Pacific.

Question 22 — 2010
Primary cause of acid precipitation in northeastern US.

Question 23 — 2016
Same pollutant list: which is the leading cause of lung cancer in the US among these (after smoking)?

Question 24 — 2016
Which pair of exhaust pollutants do catalytic converters reduce?

Question 25 — 2016
Diagram of warm air over a layer of cold air trapping pollution over a town: name the phenomenon.

Question 26 — 2016
Which soil component most likely neutralizes acid rain?

Question 27 — 2016
Diagram of gases over a city on a smoggy day: which explains the pattern in NO concentration?

Question 28 — 2016
If hydrocarbon emissions were reduced, what would most likely happen to smog constituents?

Question 29 — 2016
What would most likely reduce ground-level ozone in a city?

Question 30 — 2016
Health impact most closely associated with atmospheric particulates?

Question 31 — 2016
Which links a health effect to fine particulates correctly?

Question 32 — 2016
Health effect associated with thermal inversion?

Question 33 — 2024
Matching set: local reversal of normal temperature decrease with altitude.

Question 34 — 2024
Indoor air pollutant that is the second leading cause of US lung cancer.

Question 35 — 2024
Strategy to reduce noise pollution affecting western bluebirds in forests and farmland.

Question 36 — 2024
Same diagram: pollutant most likely released to the atmosphere.

Question 37 — 2024
Unit conversion: gallons of gasoline needed to release 200,000 tons of lead at 2 g lead per gallon.

Question 38 — 2024
Calculating motorcycle particulate matter per year (250 miles/month, 0.021 g/mile).

Question 39 — 2024
Graph of six pollutants 1980–2018: statement best supporting Clean Air Act effectiveness.

Question 40 — 2024
Method to reduce indoor carbon monoxide from a gas stove.

Question 41 — 2024
Same set: law that led to extensive federal and state limits on emissions from stationary and mobile sources.

Question 42 — 2024
Which environmental problem noise pollution most likely causes for wildlife.

Question 43 — 2024
Rain pH 4.2–4.4 downwind of which energy source.

Question 44 — 2024
Graph of NO, NO_2 and O_3 by location: highest NO.

Question 45 — 2024
Same graph: how O_3 differs between locations.

Question 46 — 2024
Same graph: strategy to reduce NO_2.

Section B — Free-Response Questions

Question 1 — 2018

Household biomass burning: two indoor pollutants and a respiratory illness, a realistic approach to reduce harm, why young children are more at risk, and sources and exposure-reduction methods for two of asbestos, radon and mold. *(Free-response; 10 points)*

Question 2 — 2018

Burning biomass indoors for cooking and heat: pollutants and a resulting illness, a practical mitigation, why under-five children are more at risk, then sources and exposure controls for two of asbestos, radon, mold. *(Free-response; 10 points)*

Question 3 — 2019

Visibility loss in four national parks (graph): worst park, primary versus secondary pollutants and how each enters or forms, percent increase in Great Smoky Mountains visibility, two government actions, and two non-air impacts of heavy visitation. *(Analyze and interpret quantitative data; 10 points)*

Question 4 — 2025

Air pollution and energy use: a particulate source, vapor-recovery nozzle, acid rain and commuting, percent change in mileage (22 to 36 mpg), extra miles on 14 gallons, school heating fix, and LED lighting energy for 28,000 bulbs. *(Analyze an environmental problem doing calculations; 10 points)*

Unit 8 — Aquatic and Terrestrial Pollution

Section A — Multiple-Choice Questions

Question 1 — 2008
Graph of BOD and DO downstream of a sewage spill: at which point are fish populations probably smallest?

Question 2 — 2008
Which choice is NOT used to deal with the effects of eutrophication in lakes?

Question 3 — 2008
Why did dissolved oxygen fall as more organic waste was added to the tanks?

Question 4 — 2008
Which change would best improve the validity (reliability) of the waste/DO tank experiment?

Question 5 — 2008
Which element most often limits primary production in freshwater lakes?

Question 6 — 2008
Nitrogen fertilizer in the Midwest contributes to which problem?

Question 7 — 2008
Which federal law deals with cleanup of abandoned hazardous waste sites?

Question 8 — 2008
What produces the majority of atmospheric mercury?

Question 9 — 2008
Put steps of water screening, disinfection, flocculation and settling in correct order.

Question 10 — 2008
What process makes toxins retained in tissue especially risky for top-of-food-chain organisms?

Question 11 — 2008
Which statement about U.S. sewage treatment plants is true?

Question 12 — 2008
Which factors (hydrology, soil type, species composition) define a wetland?

Question 13 — 2008
A dose-response graph shows breast cell cancer induction falling as drug concentration rises: does it support 'the dose makes the poison'?

Question 14 — 2008
Best description of the relationship in the anastrozole graph?

Question 15 — 2008
Why are more contaminants leached from crushed solid waste in water than from large pieces?

Question 16 — 2008
Greatest component by weight of U.S. municipal solid waste?

Question 17 — 2008
Primary waste disposal method in the U.S.?

Question 18 — 2008
Which is a nonpoint source of pollution?

Question 19 — 2008
Which law gives EPA cradle-to-grave control of hazardous waste?

Question 20 — 2008
Air-travel growth graph: which correlated phenomenon is the most direct consequence?

Question 21 — 2008
Why is DDT still found in U.S. residents' tissue after the 1972 ban?

Question 22 — 2008
Daphnia survival table/graph: pH at which 50 percent survive?

Question 23 — 2008
How would a control group help in the Daphnia experiment?

Question 24 — 2008
From the Daphnia data, predict normal pH of wild habitat.

Question 25 — 2010
A large sewage spill into a river most likely causes a sharp what.

Question 26 — 2010
Which is the best example of gray water.

Question 27 — 2010
Rats showed steadily increasing effects as toxin concentration rose; what does this suggest.

Question 28 — 2010
Persistent estrogen-mimicking chemicals such as PCBs would most affect which group.

Question 29 — 2010
Why do most plastics pose a special problem in the waste stream.

Question 30 — 2010
Which characteristically produces non-point-source pollution.

Question 31 — 2010
Factors increasing spread of infectious disease, EXCEPT which one.

Question 32 — 2010
How West Nile virus is acquired.

Question 33 — 2010
Significant limitation of rat tests for cancer risk.

Question 34 — 2016
Matching set on pollutants: identify the heavy metal formerly blended into US gasoline to raise octane.

Question 35 — 2016
Same pollutant list: which is the radioactive gas that seeps from uranium-bearing rock?

Question 36 — 2016
Same pollutant list: which metal makes large predatory ocean fish a health risk for people who eat them often?

Question 37 — 2016
Which item is NOT household hazardous waste: batteries, bug spray, motor oil, oil-based paint, or food scraps?

Question 38 — 2016
How might agricultural herbicide runoff contribute to amphibian declines?

Question 39 — 2016
Table of stream measurements: which site is likely just downstream of construction?

Question 40 — 2016
Storm runoff from the new development into a creek likely causes what?

Question 41 — 2016
Cause of recent cyanobacteria blooms in Lake Erie?

Question 42 — 2016
Experiment on ammonium sulfate and mung bean germination: which statements about the design are true?

Question 43 — 2016
Besides fertilizer, vehicle exhaust and industrial air pollution, a major source of nitrogen pollution?

Question 44 — 2016
Postawko Creek table: which site is immediately downstream from a new dairy farm?

Question 45 — 2016
Most likely route for mercury into the food web?

Question 46 — 2016
How does eutrophication kill fish?

Question 47 — 2016
60 % of 2 kg per day is compostable: amount per person in a week?

Question 48 — 2016
Best description of bioaccumulation?

Question 49 — 2024
Same article: assumption the author makes about plastic.

Question 50 — 2024
Same article: best evidence that the study was the most thoroughly researched.

Question 51 — 2024
Same article: most effective strategy to stop the patch from growing.

Question 52 — 2024
Same article: average weight of one piece of plastic (88,000 tons, 1.8 trillion pieces).

Question 53 — 2024
Table of toxicity ratings of four insecticides to four organisms: which is most toxic overall.

Question 54 — 2024
Environmental problem from soil erosion affecting aquatic environments.

Question 55 — 2024
Why increased nutrients can create dead zones in Chesapeake Bay.

Question 56 — 2024
Farm change most likely to reduce nutrient input to a downstream wetland.

Question 57 — 2024
Advantage to humans of eating fish from lower trophic levels.

Section B — Free-Response Questions

Question 1 — 2002
Brine-shrimp toxicity test with copper sulfate: plot dose data on a semi-log graph, find LD50, find threshold level, and argue for and against extending results to humans. *(Analyze and interpret quantitative data; 10 points)*

Question 2 — 2004
Editorial about mercury from coal: source and transport, two ways to cut emissions, why predatory fish carry more risk, and another toxic metal with sources and a sublethal effect. *(Free-response; 10 points)*

Question 3 — 2005

Article on emerging and re-emerging infectious diseases: transmission and control for a new and an old disease, an environmental driver, a rationale for a quoted statement, and a disease behind rising US mortality after 1980. *(Analyze and interpret quantitative data; 10 points)*

Question 4 — 2006

City brownfield cleanup: problems of excavating soil, using plants to decontaminate with a pro and con, environmental and societal benefits of reclamation, and one hazardous-waste reduction method and one legal disposal method. *(Free-response; 10 points)*

Question 5 — 2008

Municipal-landfill siting scenario with a newspaper article and a three-tract map: benefits of a local landfill, design features, choosing a tract with justifications, and a relevant law. *(Free-response; 10 points)*

Question 6 — 2010

News-article scenario on local toxic pollutants (PCBs, mercury, lead): source, exposure route and health effect, two control steps, child susceptibility, CO2 capture cost calculation, coal-plant emission control technology, reasons for offshoring manufacturing. *(Free-response; 10 points)*

Question 7 — 2010

Newspaper article on children harmed by PCBs, mercury and lead near industrial sites in India and China: one chosen pollutant (source, exposure, health effect, controls), child vulnerability, CO2 capture cost for one gadget, coal-plant emission controls, reasons for offshoring. *(Free-response; 10 points)*

Question 8 — 2012

Wetlands: two defining characteristics, completing a wetland food-web arrow diagram and explaining why eagles need many hectares, two economic benefits, one harmful human activity, and how wetlands mimic primary and secondary wastewater treatment. *(Free-response; 10 points)*

Question 9 — 2013

Gulf Coast wetlands: human activities that change sediment delivery, harm from sediment loss, effects of Mississippi fertilizer runoff on the Gulf, economic costs, and a non-fertilizer-reduction way to cut nutrient flow. *(Free-response; 10 points)*

Question 10 — 2015

Everglades restoration: two human activities reducing freshwater inflow, phosphorus sources, effects and reduction, one climate-change impact each on water quantity, quality and habitat, and ecological and economic restoration benefits. *(Free-response; 10 points)*

Question 11 — 2015

Mobile-phone e-waste: percent increase in sales from 30 to 180 million, gold used at 0.03 g each, mass not recycled at 0.1 kg each with 10 percent recycled, mercury health effect, reasons for exporting e-waste, a retailer change. *(Analyze an environmental problem doing calculations; 10 points)*

Question 12 — 2015

Oil spills: read the high-estimate Deepwater Horizon volume from a bar graph, coastal environmental and economic impacts, dispersant pros and cons, a cleanup method, another oil source, and a petroleum substitute in a product. *(Analyze and interpret quantitative data; 10 points)*

Question 13 — 2016

Graph of US municipal solid waste 1960–2012: reason per-capita waste changed after 2000, percent increase in total MSW from 1980, drawbacks of incineration, landfills and composting, landfill-gas electricity steps, and tire-pile health problems. *(Analyze and interpret quantitative data; 10 points)*

Question 14 — 2016

US municipal solid waste graph: cause of a change in per-person waste 2000-2012, percent rise in total MSW 1980 to 2012, incineration and landfill drawbacks, landfill-gas power steps, tire pile health risk, composting benefit and drawback. *(Analyze and interpret quantitative data; 10 points)*

Question 15 — 2017

Microbeads and nutrients in wastewater: effects of ingested microbeads, how nitrates degrade water quality, primary-treatment plastic removal, a disinfection method, sludge on farmland pros and cons, and mangrove loss and services. *(Free-response; 10 points)*

Question 16 — 2022

Snapping-turtle nests upstream and downstream of a mercury-emitting factory (nest map): best site, dependent variable, research question, role of upstream sites, effect of removing bank vegetation, mercury effects, methylmercury origin, claim testing, farming and soil. *(Design an investigation; 10 points)*

Question 17 — 2022

Gulf of Mexico dead zone: least-extensive watershed land use, how upstream farming and cities feed the zone, effects on marine life and coastal economy, summer growth, and designing a study of riparian buffers on stream nutrients. *(Design an investigation; 10 points)*

Question 18 — 2023

Landfill methane and beef: methane problem, decomposition factors, a fix and extra benefit, then yearly methane from 31.2 million cattle, pasture hectares for one day, and extra people fed if corn replaces beef land. *(Analyze an environmental problem doing calculations; 10 points)*

Question 19 — 2024

Five zones of a stream with dissolved oxygen and biological oxygen demand (diagram): lowest-oxygen zone, BOD-DO relationship, likely point-source zone, dependent variable and hypothesis, control choice, winter sampling, raw sewage and bacteria, abiotic factor, persistent pollutants. *(Design an investigation; 10 points)*

Unit 9 — Global Change

Section A — Multiple-Choice Questions

Question 1 — 2008
Which listed gas is the most abundant naturally occurring greenhouse gas?

Question 2 — 2008
Which listed greenhouse gas comes only from human activity?

Question 3 — 2008
How does burning fossil fuels raise net atmospheric carbon?

Question 4 — 2008
Global climate change occurs because rising greenhouse gases in which layer absorb which radiation?

Question 5 — 2008
Why are CFCs potent greenhouse gases despite low concentrations?

Question 6 — 2008
CO_2 rose from 278 ppm to 383 ppm: approximate percent increase?

Question 7 — 2008
What has most directly increased Australian skin cancer rates?

Question 8 — 2008
Why are introduced species like zebra mussels a concern?

Question 9 — 2008
Which is part of a positive feedback in climate change?

Question 10 — 2010
Matching set (laws and treaties): which one includes a provision allowing private landowners to alter land under a habitat conservation plan.

Question 11 — 2010
Matching set (laws and treaties): which agreement addressed stratospheric ozone depletion.

Question 12 — 2010
Bar graph of bird and mammal extinctions by century: how many more mammal extinctions occurred in 1901-2000 than 1801-1900.

Question 13 — 2010
Most likely reason the most bird extinctions occurred in 1901-2000.

Question 14 — 2010
Which action best illustrates a climate-change mitigation approach.

Question 15 — 2010
Which statement about invasive exotic species is FALSE.

Question 16 — 2010
Global warming is likely to cause all of these EXCEPT which one (ice sheet, plant range, insects, UV radiation, sea level).

Question 17 — 2010
Energy-flow diagram of the greenhouse effect: units contributed by greenhouse gases.

Question 18 — 2010
Same diagram: units of energy absorbed by Earth's surface.

Question 19 — 2010
Same diagram: units of energy returned to space from Earth and atmosphere.

Question 20 — 2010
Which action most likely reduces US contribution to rising atmospheric CO2 (organic farming, stop logging, oil to coal heating, cars to rail, ozone protection).

Question 21 — 2010
Which compound is NOT involved in stratospheric ozone depletion.

Question 22 — 2010
Which greenhouse gas comes only from human sources.

Question 23 — 2010
Four-tank UV experiment on polar phytoplankton: which inference is supported by results.

Question 24 — 2010
Most valid conclusion that can be inferred from the UV experiment.

Question 25 — 2010
Which two-tank comparison best shows natural-level UV inhibits photosynthesis.

Question 26 — 2010
What would most add to the reliability of the experiment series.

Question 27 — 2010
Which statements about ozone are true (acid-rain precursor, harmful, absorbs UV).

Question 28 — 2010
Forest fires may emit up to 40 percent of annual global emissions of which gas.

Question 29 — 2010
Why small island populations are prone to extinction.

Question 30 — 2010
Which stratospheric substance increases albedo by reflecting sunlight.

Question 31 — 2016
Same world map: which region is warming fastest on average because of human-caused climate change?

Question 32 — 2016
How will warming over 200 years probably alter the range of red pine, now limited to cool northern areas?

Question 33 — 2016
Which sequence is a positive feedback loop in the Arctic under climate change?

Question 34 — 2016
Which is NOT a likely result of Arctic warming?

Question 35 — 2016
Impact of CO_2 and volcanic stratospheric aerosols?

Question 36 — 2016
Why does anthropogenic water vapor not matter much for warming?

Question 37 — 2016
Why will infectious diseases spread with climate change?

Question 38 — 2016
Which global change is expected from more atmospheric CO_2?

Question 39 — 2024
Article on the Great Pacific Garbage Patch: the author's claim about it.

Question 40 — 2024
Experiment with algae in three tanks at 5, 15 and 25 °C: identify the independent variable.

Question 41 — 2024
Best example of biological control of rabbits in Australia.

Question 42 — 2024
Why ocean acidification harms corals.

Question 43 — 2024
Diagram of UV splitting O_2 and ozone, ozone recombining: what process is shown.

Question 44 — 2024
Which molecule interferes with the process in the ozone diagram.

Question 45 — 2024
Cheetah vulnerable from lack of genetic diversity: which process caused its loss.

Question 46 — 2024
Montana trout article: author's primary claim.

Question 47 — 2024
Same article: evidence used to conclude warming could greatly affect trout fishing.

Question 48 — 2024
Which response most likely increases survival in northern hemisphere as temperatures rise.

Question 49 — 2024
Which species would most likely become invasive after introduction.

Question 50 — 2024
Best strategy to prevent spread of zebra mussels to uninfected water.

Question 51 — 2024
Impact of Arctic tundra changes on global climate.

Question 52 — 2024
Matching set of laws and agreements: the 1987 global agreement phasing out ozone-depleting substances such as CFCs.

Question 53 — 2024
Same set: law protecting essential habitat and banning import and export of listed plants and animals.

Question 54 — 2024
Which substance is most responsible for reducing UV-B radiation reaching Earth's surface.

Question 55 — 2024
Invasive cactus moths: which adaptation most makes control almost impossible once infested.

Question 56 — 2024
Table of threats to endangered amphibians: which strategy reduces threat to the greatest number of species.

Question 57 — 2024
Which action most likely leads to stratospheric ozone depletion.

Question 58 — 2024
Same table: best explanation for saltwater wetland changes.

Question 59 — 2024
Graph of northward winter range shifts of 305 bird species: best hypothesis.

Question 60 — 2024
Kudzu: economic advantage of this invasive vine.

Question 61 — 2024
Stacked graph of endangered vertebrate classes in 2021: which class has the most.

Question 62 — 2024
Which approach gives percent change in endangered species from 2007 to 2021.

Question 63 — 2024
West Nile virus concern at higher elevations in Colorado.

Question 64 — 2024
Robotic bass to limit invasive mosquito fish: advantage.

Section B — Free-Response Questions

Question 1 — 2006
Ice-core graphs of CO_2 and temperature over 200,000 years: change between two dates, ratio, prediction with the ratio, assumption, causes of a 200 ppm rise, and two other warming gases. *(Analyze and interpret quantitative data; 10 points)*

Question 2 — 2010
Termite wood-consumption table by temperature and humidity: find optimum, scale methane emissions over a forest area and under a warmer, drier scenario, explain population limits, post-clearing density boom and bust, and another climate effect of deforestation. *(Analyze and interpret quantitative data; 10 points)*

Question 3 — 2010
Zebra mussel invasion in the Great Lakes with a range map: why mainly eastern, how lakes get colonized and one prevention method, an ecosystem impact, another invasive species, a biological-control example with its downside, and two traits that favor invaders. *(Free-response; 10 points)*

Question 4 — 2010
Sea-level rise tied to warming at 3.0 mm per year: calculate rise over 50 years, two warming processes that raise sea level, two estuary impacts, one economic loss for coastal residents, two policies to discourage coastal settlement. *(Analyze an environmental problem doing calculations; 10 points)*

Question 5 — 2010
Zebra mussels spreading in the Great Lakes and a shaded range map: why mostly eastern, how they reach isolated lakes and prevention, ecosystem impact, another invasive species, biological-control example with downside, two traits that make invaders successful. *(Free-response; 10 points)*

Question 6 — 2010
Sea level rising 3.0 mm/yr with warming: compute 50-year rise in meters, two warming-driven causes of rise explained, impacts on estuaries, a coastal economic cost, and two government policies to discourage coastal settlement. *(Free-response; 10 points)*

Question 7 — 2011

Graph of dissolved ocean CO2 and pH since 1850; explain acidification and rising uptake, then calculate yearly volume, mass and 20% projected loss of coral-reef CaCO3 from given area, growth rate and density; coral-loss impact; another global marine problem. *(Analyze an environmental problem doing calculations; 10 points)*

Question 8 — 2013

Ozone in two atmospheric layers: UV absorbed by the stratospheric layer and a health benefit, the formation/breakdown balance, CFC destruction chemistry and why recovery is slow, tropospheric ozone formation and its impacts. *(Free-response; 10 points)*

Question 9 — 2016

White-nose syndrome in little brown bats: why disease rarely drives extinction, threats to tiny populations, slow recovery biology, effects of bat decline on other organisms, forest ecosystem services, and a reason emerging human diseases are rising. *(Free-response; 10 points)*

Question 10 — 2017

Declining large mammals: plot elephant counts over time, percent loss 1970 to 2000, projected 2022 herd after a 20 percent decline from 2015, a K-strategist trait that raises extinction risk, and two conservation strategies. *(Analyze and interpret quantitative data; 10 points)*

Question 11 — 2018

Arctic food web diagram: name a producer, primary consumer and secondary consumer, what arrows mean, effects of less sea ice on polar bear hunting, the albedo feedback, and migration reasons plus a commercial threat and fix for right whales. *(Free-response; 10 points)*

Question 12 — 2019

Piping plover nesting on Assateague Island (two nest maps): preferred habitat, nest change, cause, protecting nests, barrier-island functions, a human threat to coastal habitat, and economic effects of and local defenses against sea-level rise. *(Analyze and interpret quantitative data; 10 points)*

Question 13 — 2019

Graph of Mauna Loa atmospheric CO_2 and nearby ocean pH since 1992: read values for 2005, relate CO_2 to ocean CO_2 and pH, write the reaction with water, name the problem, explain harm to shell-builders, and name another reef threat. *(Analyze and interpret quantitative data; 10 points)*

Question 14 — 2021

Subdivision planned on forest used by wood thrush: why specialists suffer from fragmentation, cowbird relationship, benefits of keeping forest, a mitigation proposal, and area and percentage calculations for a 275-territory target. *(Analyze an environmental problem doing calculations; 10 points)*

Question 15 — 2021

Global amphibian species-richness map: richest latitudes, biome and why, a cause of deforestation, richness and resilience, other causes of decline, a species-protection law, and a development plan that protects frogs in a wetland. *(Analyze and interpret quantitative data; 10 points)*

Question 16 — 2022

Lizard counts on Pacific islands with and without mongooses (graph by habitat): richest habitat, mongoose effect, island biogeography, invasive-species harm and control, forest services, effect of sugarcane, monoculture problems and legume rotation. *(Analyze and interpret quantitative data; 10 points)*

Question 17 — 2023

Spread of Asian carp from the Mississippi basin toward the Great Lakes (map): a water body invaded, distribution change, ecosystem-service impact, control measure and extra benefit, primary consumer, arrow meaning, food-chain effects, blue-pike tragedy of the commons, and aquaculture drawbacks. *(Analyze and interpret quantitative data; 10 points)*

Question 18 — 2024

Forest cultural service and water filtration, urbanization effects, oil-degrading bacteria in tanks at three temperatures (dependent variable, hypothesis, replicates, aeration), then pH versus CO_2 graph 1850 to 2100 and coral reefs. *(Design an investigation; 10 points)*

Question 19 — 2024

Land-use change since 1700 (graph): year of most forest, grassland versus grazing, overgrazing problem, forest change and CO_2, water problems of cities and responses, why bobcats vanished from farmland, and risks and improvements for small reintroduced populations. *(Analyze and interpret quantitative data; 10 points)*

Question 20 — 2025

Common Tern pairs versus sea-level rise (graph) and wading-bird species at four sites (table): reading and trend, refuting a hypothesis, sea-level cause, hypothesis, variable, PCB runoff, wetland service, sediment and urbanization effects. *(Design an investigation; 10 points)*

Part II — Complete Solutions

> **NOTE**
> Solutions are written by Success4Sure Academy. For multiple-choice items, the official answer key letter refers to the choices printed in the original exam (older exams used five choices, A–E; the 2026 exam uses four). Free-response solutions show complete reasoning and the scoring ideas.

Unit 1 — The Living World: Ecosystems

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: E. Photosynthesis needs sunlight, which penetrates only the upper tens of meters. The abyssal zone of the deep ocean floor is in permanent darkness, whereas intertidal, upwelling, pelagic surface and euphotic zones all receive light.

Question 2 — 2008

Official key: E. The independent variable is the amount of organic waste and the dependent variable is dissolved oxygen, so the experiment's aim is to see how waste affects DO. It is not a measure of algal oxygen production or CO_2 use.

Question 3 — 2008

Official key: C. The littoral zone is the shallow near-shore area with enough light for rooted, emergent plants. The limnetic zone is open water, profundal is deep, benthic is the bottom, and riparian is the bank.

Question 4 — 2008

Official key: C. Nitrogen-fixing bacteria in soil and in root nodules convert atmospheric N_2 into ammonia and related compounds that plants can use, which is why plants colonizing poor soils can leave it enriched.

Question 5 — 2010

Official key: A. Limestone is calcium carbonate built from the shells and skeletons of marine organisms, so seafloor sediment locks away carbon for geologic time. The keyed element is carbon; the other listed elements are not stored mainly as carbonate rock.

Question 6 — 2010

Official key: D. Seabird droppings from fish-eating birds are rich in phosphate because phosphorus moves up the food chain from fish. Guano is mined as fertilizer, so the keyed element is phosphorus.

Question 7 — 2010

Official key: A. The keyed zone is the shallow nearshore area at the water's edge, where light, nutrients and varied habitat support the richest variety of life. Deeper zones have fewer niches because of darkness and pressure.

Question 8 — 2010

Official key: E. Light vanishes with depth and pressure rises about one atmosphere per 10 m, so the deepest zone holds organisms adapted to darkness and extreme pressure.

Question 9 — 2010

Official key: B. CO2 uptake in the ocean is dominated by phytoplankton photosynthesis in the sunlit surface layer over the continental shelf and open water. The large area of this lit zone gives the biggest total uptake.

Question 10 — 2010

Official key: C. Producers use light to convert CO2 and water into glucose, the energy-rich sugar that fuels their metabolism and builds biomass. Oxygen is a by-product, not the goal.

Question 11 — 2010

Official key: B. Open ocean has very few nutrients, so productivity per unit area is low even though its total area is huge. Estuaries and farmland are nutrient-rich and far more productive per square meter.

Question 12 — 2010
Official key: C. Most nutrients in a rain forest are held in living biomass because heavy rain and rapid decomposition leave little in the soil. The soils are thin and nutrient-poor, even though diversity and rainfall are very high.

Question 13 — 2010
Official key: C. Decomposers break dead matter into simple inorganic nutrients that producers can take up again. This recycling keeps nutrients cycling and is the essential service they provide.

Question 14 — 2010
Official key: D. Seasonal temperature and rainfall cycles support abundant grass growth. Neither deserts nor forests share this pattern, and the 30° latitude limit and overgrazing statements are incorrect.

Question 15 — 2010
Official key: B. An ecosystem includes the community and its abiotic environment, which in turn contains populations and species. Genus, species and population are all parts of a community, which is part of an ecosystem.

Question 16 — 2016
Official key: B. Tropical rainforest has constant warmth near 25–30 °C and heavy rain every month, giving the highest productivity and the most tree species. The keyed graph shows flat high temperature with large precipitation in all months.

Question 17 — 2016
Official key: C. Chaparral, the shrubland of Mediterranean-type climates, has mild wet winters and hot dry summers; dry summer vegetation burns often. The keyed graph shows rain concentrated in winter months and almost none in summer.

Question 18 — 2016
Official key: B. Phytoplankton photosynthesize, so they must stay in the sunlit (photic) zone; light fades quickly with depth. Hiding from predators, being benthic, or eating zooplankton do not explain their position since they are producers floating in the water column.

Question 19 — 2016
Official key: D. Shade affects plants by reducing the light they receive, so measuring the direct sunlight reaching creosote bushes before construction gives the baseline to compare with after shading. Noise, global CO_2, erosion and rain are not the variable in the hypothesis.

Question 20 — 2016
Official key: A. The correct cycle has atmospheric N_2 fixed into ammonium, nitrified to nitrite and then nitrate, taken up by plants, and returned to the air as N_2 by denitrification. The keyed diagram follows that order; the others have impossible conversions.

Question 21 — 2016
Official key: B. Land per kcal of corn is 1 ha/1,000 kcal; each kcal of beef requires 20 kcal of corn. Multiplying 1 ha/1,000 kcal by 20 kcal corn per kcal beef by 10,000 kcal beef gives 200 ha, and the units cancel correctly.

Question 22 — 2016
Official key: C. Per unit area, open-ocean productivity is low because nutrients are scarce. The ocean covers about 70% of Earth's surface, though, so the total productivity summed over that huge area (mostly by microscopic phytoplankton) is very large. The area covered, not the size of the individual producers, drives the total.

Question 23 — 2016
Official key: A. A sink absorbs or stores more carbon than it releases. Buried marine sediment stores carbon for long periods, whereas deforestation, burning and decomposition release CO_2, and lawn nutrient uptake is not long-term storage.

Question 24 — 2016
Official key: D. A valid test compares plants where the suspected cause (trampling) is present with similar areas where it is absent. Counting hikers or comparing hikers' traits does not measure effects on plants.

Question 25 — 2016

Official key: D. Net primary productivity rises with warmth and abundant rainfall, as in rain forest compared with northern conifer forest. Longitude is a position on Earth that does not itself control plant growth.

Question 26 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Pioneer species (lichens, mosses, hardy weeds) are the first colonizers of bare or disturbed ground. Their roots and decaying tissue build and hold soil so later species can establish. Keystone, indicator and invasive species do not play this early-succession role.

Question 27 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Temperate seasonal (deciduous) forest occurs at mid-latitudes with four seasons, such as the eastern United States where location A lies. The Sahara, the equatorial area and the high northern location fit desert, rainforest and boreal/tundra instead.

Question 28 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Warm temperatures, abundant rain and year-round sunlight make tropical rainforest near the equator (location D) the most productive biome. Desert and cold northern areas have far lower net primary productivity.

Question 29 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Wolves ate deer, so grazing pressure on plants fell and vegetation recovered, a trophic cascade. Wolves are carnivores, agriculture did not cause it, and wolves do not start primary succession.

Question 30 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Only about 10 percent of energy passes to the next trophic level, so the predator level can support far fewer individuals. Size, specialist/generalist status and invasiveness do not explain the ratio.

Question 31 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Nitrogen fixation converts atmospheric N_2 into ammonium that organisms can use, which is the arrow running from N_2 to NH_4^+ (A). The arrows between ammonium and organic matter are assimilation and ammonification.

Question 32 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The dependent variable is the response measured, here caterpillar survival rate. Air temperature is manipulated, and region and coloration are not the outcome of interest.

Question 33 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** After wolves were reintroduced, elk declined and aspen height increased, a trophic cascade.

Section B — FRQ Solutions

Question 1 — 2003

(a) Causes with the direction of change: tides mix ocean and river water, raising salinity at high tide and lowering it at low tide; river and runoff pulses (storms, snowmelt) add fresh water and lower salinity; seasonal sunlight and air temperature warm shallow water in summer and cool it in winter, and evaporation then raises salinity. *(1 point each, 2 max)*

(b) Ecological: nursery habitat for fish and shellfish; filtering of sediment, nutrients and toxins; buffering floods and storm waves; high productivity and biodiversity; bird nesting and migration stops. Economic: fisheries and aquaculture; tourism and recreation; storm and flood protection of property; reduced water-treatment costs. *(2 + 2 points)*

(c) Three harms with explanation: filling and development destroys habitat; fertilizer, pesticide and sewage runoff cause eutrophication and oxygen loss; dredging, dams and diversions alter flow and bury benthic life; industrial effluent adds toxins or heat; invasive species or overharvesting disrupt food webs. *(3 points)*

(d) For filling and development: a zoning or law forbidding building in wetlands (Clean Water Act permits, coastal-zone rules) plus an economic incentive such as a property-tax reduction for undeveloped wetland, conservation easements, or fines/fees on development. *(1 point policy, 1 point incentive, linked to the chosen harm)*

Question 2 — 2008

(a) The atmosphere (N_2 gas) is the main nitrogen reservoir. Cycling: nitrogen-fixing bacteria (soil and legume root nodules) and lightning convert N_2 to ammonia/ammonium; nitrifying bacteria oxidize this to nitrite and then nitrate; plants absorb nitrate/ammonium and build proteins and DNA; consumers obtain nitrogen by eating; decomposers return ammonium; denitrifying bacteria convert nitrate back to N_2. *(1 point for the atmosphere; up to 3 for the processes.)*

(b) The phosphorus cycle has no significant gas phase, its main reservoir is rock (sediments), and it depends far less on bacteria. *(1 point for any one.)*

(c) Nitrate (or ammonium) from fertilizer, or phosphate from detergents/fertilizer/manure. Human activity: fertilizer applied to fields washes off into streams. Problem: excess nutrients cause algal blooms and cultural eutrophication, with oxygen depletion and fish kills. *(1 point each for compound, activity, problem.)*

(d) Burning fossil fuels (or deforestation) adds CO_2 faster than sinks remove it. Consequences: enhanced greenhouse warming and rising sea level; ocean acidification; more extreme weather such as drought and stronger storms. *(1 point for the disruption; 1 point each for two consequences.)*

Question 3 — 2014

(a)(i) Photosynthesis: autotrophs take in CO_2 and build it into sugars and other organic molecules. *(1.)* (ii) Cellular respiration or decomposition converts organic carbon back to CO_2 (or CH_4 under anaerobic conditions) released to the air. *(1.)*

(b)(i) Two ocean sinks: CO_2 dissolves in seawater, where it forms carbonic acid and bicarbonate; marine organisms use carbon for shells and skeletons (calcium carbonate) that sink and become limestone; phytoplankton fix carbon and it sinks. *(2.)* (ii) A long-term land sink: peat or wetland sediments, old-growth wood, soil organic matter, or carbonate rock. *(1.)*

(c) Other human activities: deforestation and burning biomass (removes sink and releases carbon); cement production, landfills and livestock produce CO_2 or methane. *(2.)*

(d) Problem: enhanced greenhouse effect causing global warming; consequence: sea-level rise and habitat loss, shifting climate zones, or ocean acidification dissolving shells. *(1 + 1.)*

(e)(i) Phosphorus has no significant atmospheric phase; it cycles slowly through rock, soil and water and is released by weathering. *(1.)* (ii) Needed for DNA, RNA, ATP and cell-membrane phospholipids (also bones and teeth). *(1.)*

Question 4 — 2025

(A) A divergent boundary (East African Rift, plates moving apart). *(1)* **(B)** Savanna (tropical grassland with scattered trees). *(1)* **(C)** The southern part of the park (the January-March positions on the migration loop, near the southern plains), where new grass is growing. *(1)* **(D)** Wildebeest eat long grass and gazelles eat short grass, so they avoid direct competition for the same niche. *(1)* **(E)** The highway cuts across the migration route in the north, causing collisions, barriers and habitat fragmentation. *(1)* **(F)** Underpasses or overpasses, wildlife-crossing structures, fencing, or rerouting around the migration path. *(1)* **(G)** Invasive species grow and reproduce quickly, spread widely, and lack natural predators. *(1)* **(H)** Rotation restores soil nitrogen, reduces fertilizer, and improves soil health. *(1)* **(I)** Heavy pesticide use can contaminate water, kill non-target species, bioaccumulate, or breed resistance. *(1)* **(J)** Integrated pest management, biological control, hand weeding, or resistant crops. *(1)*

Unit 2 — The Living World: Biodiversity

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: A. Species richness is simply the number of different species, and the sample directly shows 10 invertebrate species. Productivity, evenness, disturbance and stability would need biomass measurements, abundance by species, or repeated sampling over time.

Question 2 — 2008

Official key: E. Development typically causes runoff, siltation and nutrient pollution that eliminate sensitive species, so a drop in overall biodiversity is the best sign of an adverse effect. The other choices describe changes that would not specifically indicate harm, or would be opposite to what pollution tends to cause.

Question 3 — 2008

Official key: A. Sea otters eat sea urchins that graze kelp, so they maintain kelp forests and the community they support despite being few in number. Removal of otters leads to urchin barrens.

Question 4 — 2008

Official key: A. Heavy rain boosts growth of chaparral and grasses, creating more dry fuel by summer, so fires are larger or more intense.

Question 5 — 2010

Official key: D. Native pollinators coevolved with native plants.

Question 6 — 2016

Official key: B. Tropical forests of central Africa (the Congo basin) are being cut into isolated patches by logging, farming and roads. Smaller fragments hold fewer species and expose edges to disturbance, so biodiversity falls. The keyed region is the equatorial African one.

Question 7 — 2016

Official key: B. Species richness is highest when the greatest number of species are present at once. Around 100 years the pioneer sweet gum is still declining while oak, hickory, beech and dogwood have all established, so most species overlap then.

Question 8 — 2016

Official key: D. Sweet gum rises first and peaks early, then disappears as the canopy closes, showing it is a pioneer needing full sunlight. Later species such as beech and dogwood tolerate shade.

Question 9 — 2016

Official key: D. Island biogeography predicts that small islands support fewer resources and fewer niches, so populations are small and vulnerable. Isolation also reduces recolonization, so extinction outpaces immigration.

Question 10 — 2016

Official key: E. Because the birds are one species and beak size is genetic, a consistent difference points to natural selection by different food on each island, such as seeds of different hardness. Distance of ten kilometers does not explain the pattern with movement, since the birds would mix.

Question 11 — 2016

Official key: A. A good research question is specific and testable about a plausible mechanism. Air-pollution nitrogen fertilizing soils plausibly favors fast-growing invasive grasses, whereas wolves, shrub abundance or ozone-raised soil pH are weakly connected to alpine meadows.

Question 12 — 2016

Official key: B. Oak and hickory are late-successional trees; after clear-cutting, fast-growing pioneer species come first and it takes many decades or even centuries before large oaks are again present. This undermines the harvest plan.

Question 13 — 2016

Official key: A. For ranchers, an economic benefit means more forage or income. Higher net primary productivity in diverse pastures supports more livestock, whereas lower infiltration or fewer birds are not benefits.

Question 14 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** An indicator species is sensitive to specific conditions (for example oxygen, pH or pollution), so its presence or decline reveals ecosystem change. The other categories describe disproportionate influence, early colonization or harmful spread, not signaling.

Question 15 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** A keystone species affects its community far more than its abundance would predict; losing pollinators such as bumblebees changes plant reproduction and ripples through the community. Indicator, pioneer and invasive categories do not capture this outsized role.

Question 16 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Wetlands buffer storms and filter water; requiring new development to stay farther from shore keeps wetlands intact. Filling wetlands, adding fishing pressure or building upstream dams would degrade or starve them.

Question 17 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Selective breeding narrows genetic diversity in managed colonies, leaving them more vulnerable to sudden change, while wild colonies retain diversity. The other choices misstate poaching, farmland or invasive competition.

Question 18 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Recreation for anglers is a cultural service. Resilience, water supply and flood regulation are other categories.

Question 19 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Biodiversity is highest in tropical rainforest, then temperate forest, then grassland, and lowest in tundra, so that order is correct. Other orders place a low-diversity biome first.

Question 20 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The graph shows roughly 1.2 million in 1997 and about 29,000 in 2019. The decline is (1,200,000 − 29,000) ÷ 1,200,000 ≈ 98%.

Question 21 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The article blames urban sprawl and the spraying and plowing associated with agriculture. Invasive wasps and overharvesting of milkweed are not mentioned.

Question 22 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The article recommends protecting overwintering groves and growing milkweed and nectar plants along the migration route, i.e. preserving habitat and food sources. Invasive control, fossil-fuel reduction and captive breeding are not proposed.

Question 23 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Cultural services are the nonmaterial benefits such as recreation and beauty. Provisioning supplies goods, regulating controls processes, and supporting sustains others.

Question 24 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** A freshwater invertebrate that declines when dissolved oxygen falls signals environmental change. Beaver, moss and a mouse illustrate keystone, pioneer and competition.

Question 25 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Higher genetic diversity gives the Antillean population more variation to adapt to threats. The other claims go beyond the data.

Section B — FRQ Solutions

Question 1 — 2003

(a) Two declines, each with its cause: habitat loss (wetland drainage or farmland conversion for cranes; shrinking range and loss of food for condors), hunting and egg collecting, and lead poisoning from ammunition in carrion eaten by condors. *(1 point each, must be described)*

(b) Examples: Endangered Species Act protection and critical-habitat refuges (both); captive breeding and release (both, condors especially); bans on DDT or lead shot; CITES trade limits; wetland protection and migratory-bird treaties (cranes). Name which species benefited. *(2 points; a third possible for elaboration)*

(c) Slow-recovery traits: low reproductive rate (one egg or chick, few per lifetime), late maturity, long parental care, large range or specialized habitat needs, tiny population with low genetic diversity. Link each to slow rebound. *(1 point each)*

(d) For: protecting a species preserves biodiversity and a role in its ecosystem (scavenger, keystone predator) and supports ecotourism income. Against: protection can cost public money and restrict land use and development, and funds could save more species elsewhere. Each side needs a supported argument for the same species. *(4 points: 2 for, 2 against)*

Question 2 — 2011

(a)(i) *(1 pt)* Without fires, dense stands of older, even-aged or stressed trees survive, giving beetles abundant food and weak hosts.

(a)(ii) *(2 pts)* Dead trees take up no CO_2, so a carbon sink shrinks; their decay releases stored carbon as CO_2; and dead timber fuels wildfires that release more. (Beetles' own respiration does not count.)

(b) *(2 changes + 2 linked impacts)* For example: more sunlight reaches the ground, so seedlings and shade-intolerant species spread while shade-lovers decline and snow melts faster; loss of canopy and roots increases erosion and runoff, so streams silt up and nutrients wash away; loss of tree structure removes nesting and shelter, lowering animal diversity.

(c) *(2 pts)* Fewer pollinators raise crop prices and the cost of renting hives or hand-pollinating; lower crop yields and honey output cut farm revenue and jobs.

(d) *(service 1 pt, benefit 1 pt)* Any different ecosystem service with a human benefit, for example wetlands and soils filtering water (cheaper drinking-water treatment), natural predators controlling pests (less pesticide), or forests regulating floods (less property damage).

Question 3 — 2013

(a) Two shared traits: many different species with large numbers of individuals each, complex food webs and many niches, high genetic variation. *(2 points.)*

(b) Two activities with mechanisms: clearing land for development or logging removes and fragments habitat; introducing invasive species lets them outcompete natives; monoculture agriculture replaces diverse natives; fertilizer or sewage runoff lowers dissolved oxygen so only tolerant species persist; overharvesting removes keystone species or top predators. *(1 per activity, 1 per mechanism; 4 points.)*

(c) A practical solution for each, linked: cluster development or smart growth and wildlife corridors for development; selective cutting and replanting for logging; border inspections and public education for invasives; buffer strips or advanced sewage treatment for runoff; quotas and enforcement for harvest. *(1 each; 2 points.)*

(d) One natural factor: volcanic eruption or asteroid impact darkening the sky, wildfire, drought, hurricane damage to coastal nurseries, or a new disease. *(1 point.)*

(e) Two benefits: greater resilience and stability of the ecosystem; pollination and pest control; natural water and air purification; more genetic material for adaptation; faster nutrient cycling by soil organisms. *(2 points.)*

Question 4 — 2016

(a) Why disease seldom causes extinction. Any one valid reason: genetic variation means some individuals resist and survive to reproduce; hosts and pathogens co-evolve toward lower lethality; a pathogen that kills all hosts loses its own habitat; and deaths lower population density, so transmission slows. *(1 point: one correct explanation)*

(b)(i) Two threats to a very small population. Examples: hard to find mates when individuals are scarce or widely dispersed; low genetic diversity (inbreeding, a bottleneck, one disease able to hit nearly all individuals); competition for roosts or food from similar species; loss of group benefits such as shared warmth or protection; vulnerability to a disturbance such as a drought or fire. *(2 points: 1 each, first two only)*

(b)(ii) Slow recovery. Little brown bats have one or two pups a year, mature late and live long (long generation time), so population growth is slow even with high survival. Heavy parental care also limits number of offspring. *(1 point: one biological trait with explanation)*

(c) Effects on other organisms. Examples: more insects (bat prey) and more insect-borne disease or crop pests; fewer predators of bats because less food; fewer plants that bats pollinate or whose seeds they disperse; less guano feeding cave communities; less fungus (WNS) if bats are fewer; competitors with similar niches gain; trophic cascade. *(2 points: 1 each, first two only)*

(d) Two forest ecosystem services. Any two, each with a human benefit: timber and fuel for construction and paper; oxygen production for breathing; soil protection and water filtration giving clean drinking water and flood control; carbon sequestration slowing climate change; habitat and biodiversity supplying food, medicine; recreation and culture. *(2 points: 1 each service linked to a benefit)*

(e) Increase in emerging human diseases. A reason plus explanation, e.g. climate warming lets disease vectors such as mosquitoes survive in new areas; more global travel moves pathogens quickly between regions; greater contact with wildlife through habitat clearing exposes people to zoonotic pathogens; antibiotic resistance produces new strains; denser populations pass disease more easily. *(2 points: 1 reason, 1 correct explanation of how it raises disease)*

Question 5 — 2023

(a) Most movement occurred in water at about 13-19 °C (read from the diagram). *(1)*

(b) The discharge is warm (about 24-26 °C), above the 20 °C survival threshold. *(1)*

(c) Warm effluent lowers dissolved oxygen, can exceed other species' tolerance range, and may shift reproductive timing. *(1)*

(d)(i) Manatees are herbivores that need much food, have a slow reproductive rate, and are tied to warm water, so seagrass loss means starvation and slow recovery. *(1)*

(d)(ii) Less producer biomass, so less energy reaches every higher trophic level. *(1)*

(e)(i) Permeable pavement, vegetated buffers, less fertilizer, runoff barriers, or better wastewater treatment. *(1)*

(e)(ii) Less nutrient load reduces eutrophication, improves water clarity and biodiversity, supports fisheries or ecotourism. *(1)*

(f) Summer has stronger sunlight and heat that speed the reactions of NO_x and VOCs that form ozone, and trees emit more VOCs. *(1)*

(g) Ozone harms plant tissue and photosynthesis, stunts growth and irritates animals' lungs and eyes. *(1)*

(h) Hydrogen fuel needs substantial (often fossil) energy to make, is flammable, requires new fueling infrastructure and costly vehicles. *(1)*

Question 6 — 2024

(a) Primary succession. *(1)*

(b) Stage II. *(1)*

(c) Weathering by wind and rain breaks rock, and pioneers such as lichens break rock and add organic matter when they die. *(1)*

(d) Recreation, aesthetic, spiritual or educational value. *(1)*

(e) "Clear-cutting forest B will raise stream temperature (or turbidity) relative to forest A." *(1)*

(f) Independent variable: forest condition (intact versus clear-cut). *(1)*

(g) Forest A is the control against which forest B is compared. *(1)*

(h) Without shade, more sunlight and eroded sediment warm the water. *(1)*

(i) Fertilizer runoff raises nitrate and phosphate and can lower dissolved oxygen via algal blooms; alternatively, new grass reduces turbidity. *(1)*

(j) Fertilizer or pesticide runoff can cause eutrophication and oxygen loss or toxicity that kills lake organisms. *(1)*

Unit 3 — Populations

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: B. Density is population divided by area. Bangladesh has 144 million people on 0.14 million km², about 1,030 per km², far above India (~300), China (~135) and the others.

Question 2 — 2008

Official key: E. The United States has 300 million people, so three times is about 900 million to 1 billion. India, at 1,000 million, matches; China is about 4.3 times.

Question 3 — 2008

Official key: D. Of the options, reducing human population size reduces total demand for all resources and total waste at the root, while recycling, efficient vehicles, replanting and ecotourism each address only one resource or sector.

Question 4 — 2008

Official key: C. In the transitional stage, death rates have fallen because of better food, sanitation and medicine, but birth rates remain high, so the gap and growth rate are greatest.

Question 5 — 2008

Official key: E. Total fertility rate is the average number of children per woman, and it falls as societies move through the demographic transition because of education, urbanization, access to contraception and lower infant mortality. The other statements confuse it with growth rate or replacement level, or reverse correlations.

Question 6 — 2008

Official key: D. A stable age distribution means the proportions of each age class stay the same over time, even if the population is growing or shrinking in size. A growing population with constant age-class proportions fits this.

Question 7 — 2008

Official key: D. Current demand is 200,000 × 5 kW = 1,000 MW. Growth: 1,000 MW × $(1.035)^{40} \approx 1{,}000 \times 3.96 \approx 3{,}960$ MW, about 4,000 MW.

Question 8 — 2008

Official key: D. K-strategists have long lifespans, few offspring with parental care, late maturity and stable populations near carrying capacity. Early maturity, many young, invasiveness and rapid cycles describe r-strategists.

Question 9 — 2008

Official key: A. Rising living standards in developing countries increase per capita consumption of resources and energy, so total impact grows even as population growth slows.

Question 10 — 2008

Official key: B. Doubling time ≈ 70 ÷ growth rate (%) = 70 ÷ 2 = 35 years.

Question 11 — 2010

Official key: B. Biotic potential is the maximum growth rate with unlimited resources, which gives exponential growth: a curve that bends steadily upward. The keyed graph is the J-shaped curve. A straight rising line would be linear growth, not exponential.

Question 12 — 2010
Official key: E. A population limited by resources rises quickly and then flattens at the carrying capacity, producing the S-shaped (logistic) curve. The keyed graph levels off at a plateau.

Question 13 — 2010
Official key: C. Exponential growth means a constant percentage increase per time period, so the population doubles after equal time intervals. Equal numeric increases would describe linear growth.

Question 14 — 2010
Official key: A. Momentum occurs when many people are younger than reproductive age, because they will have children soon and the population keeps growing even if fertility drops. Older groups do not produce this effect.

Question 15 — 2010
Official key: B. High life expectancy and low infant mortality reflect good health care, nutrition and sanitation. The other pairs each pair a favorable with an unfavorable indicator.

Question 16 — 2010
Official key: D. Both rates are falling and births are far above deaths, with net growth about 11 per thousand, which is still positive. That pattern is the industrial stage, since birth rates have fallen but remain above death rates.

Question 17 — 2010
Official key: B. Rule of 70: 70 ÷ 1.75 = 40 years. Dividing 70 by the percent growth rate is the standard shortcut for doubling time, so the other values correspond to different growth rates.

Question 18 — 2010
Official key: E. Growth rate equals birth rate minus death rate, ignoring migration. Multiplying by life expectancy or adding the rates would not give a rate of natural change.

Question 19 — 2010
Official key: C. Reading the graph, population is about 5 billion in 1950 and near 10 billion by 2150, which is a doubling. It levels off rather than declining or growing exponentially.

Question 20 — 2010
Official key: C. R-selected species mature quickly and produce many offspring with little parental care. These traits suit unstable habitats and match rapid-colonizer strategies, the opposite of slow growth and long lives.

Question 21 — 2016
Official key: C. Replacement-level fertility is about 2.1 births per woman. The keyed country has a fertility rate below that (1.8) while having long life expectancy, so deaths will eventually exceed births without immigration. Countries at or above 2.1 will not shrink.

Question 22 — 2016
Official key: A. High fertility (6.5), very high infant mortality and low life expectancy together are typical where women have little schooling and few options, since education lowers fertility and child deaths. The keyed country shows the most extreme values on all three.

Question 23 — 2016
Official key: C. K is the carrying capacity: the maximum population size the environment can sustain indefinitely given limiting resources. A logistic growth curve levels off at K.

Question 24 — 2016
Official key: D. A pyramid-shaped diagram signals future growth, a straight-sided column signals stability, and a top-heavy shape with a narrow base signals decline. Diagram D alone has more people in older age groups than in the youngest, so deaths and aging will outweigh births.

Question 25 — 2016
Official key: D. Rising female employment and education typically delay marriage and reduce the number of children per woman, so the growth rate falls. Fewer pandemics, more births in developed nations or longer lifespans would push growth up, not down.

Question 26 — 2016

Official key: E. In the early industrial stage of the demographic transition, improvements in sanitation, food supply and medicine reduce the death rate first, while birth rates stay high from tradition. The gap between births and deaths produces rapid growth.

Question 27 — 2016

Official key: E. Fertility above replacement means more than about 2.1 children per woman. Incentives such as priority housing for larger families encourage births, while promoting small families, abstinence or removing tax benefits does the opposite.

Question 28 — 2016

Official key: B. Doubling time is inversely related to growth rate (70 ÷ doubling time ≈ rate). India's 41–80 years is a faster rate than the 81–200 years shown for China and Australia. The map gives no data on population size, density, development or farmland.

Question 29 — 2016

Official key: A. K-selected species live near carrying capacity, with long lives, slow maturity and few young. By investing heavily in each offspring they increase survival, unlike r-selected species that produce many unprotected young.

Question 30 — 2016

Official key: D. The top graph shows North American per-capita consumption (about 320 g/day) is higher than in the other regions. Population growth does not by itself raise per-capita consumption, so the other claims aren't supported.

Question 31 — 2016

Official key: B. Use the rule of 70: doubling time = 70 ÷ growth rate (%). At 2 %, 35 years. Check others: 1 % gives 70 years, 3.5 % gives 20 years, 5 % gives 14 years.

Question 32 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The dashed line at 1,000 individuals marks K, the population the environment can sustain; point A sits on that line. Points B (rapid growth), C (overshoot peak) and D (dieback trough) lie off it.

Question 33 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Early on the curve rises steeply as resources are plentiful and density-dependent limits are weak, which is the exponential phase at point B. The population is still well below K there.

Question 34 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** After overshooting K at the peak (C), resources are depleted and numbers crash to the low trough, point D. A dieback ends at that minimum before recovery toward K.

Question 35 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** After 1985 the deer fell and then leveled near 700, so food for wolves became scarce and the wolf population fell from 25 and stayed low. Carrying capacity was not simply 'reached with no change', and wolves did not go extinct.

Question 36 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Reading the deer column, the peak is 2,200 in 1982. Later years are lower.

Question 37 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Stage 3 begins when birth rates fall, and birth control access (with education and women's opportunities) lowers fertility. Vaccines, sanitation and food mostly lowered death rates, which drove stage 2.

Question 38 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** A Type II curve declines linearly, meaning the chance of dying is about the same at every age. Type III would show highest early mortality and Type I highest late mortality.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Countries moving through the demographic transition have lower fertility and longer life, so populations age. The global population is not yet shrinking.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Percent increase = (new − original) ÷ original × 100% = (119.94 − 57.20)/57.20 × 100% ≈ 110%. Dividing by the new value or adding is wrong.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** A roughly rectangular (columnar) structure with similar-sized young and middle cohorts indicates a stable population at replacement. The wide-base is growth and the narrow base is decline.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Predation reducing prey, then fewer wolves, then prey recovering is a negative feedback loop. It is not mutualism, the 10% rule or overshoot.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Natural increase is birth minus death: Bermuda 0.5 per 1,000, far smaller than Trinidad 4.1, Jamaica 8.3 and Haiti 15.8. Bermuda is most stable.

Question 44 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** 15.91 − 7.59 = 8.32 per 1,000 people.

Question 45 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Births exceeding deaths plus immigration exceeding emigration both add people, giving the largest growth.

Question 46 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The demographic transition model tracks how birth and death rates change in stages as a country industrializes and develops. Carrying capacity and fertility rate are single values, and a density-independent factor is a limit on populations, so none of them illustrate a multi-stage change in both rates.

Question 47 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Carrying capacity is the largest population an environment can support over time given food, water, space and other resources. Fertility rate and the transition model describe births or stages, not a limit set by resources.

Question 48 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Total fertility rate is the average number of children per woman; at about 2.1 (replacement level) each generation replaces itself, so numbers stay roughly constant apart from migration. Carrying capacity is a resource limit, not a birth measure.

Question 49 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Exceeding carrying capacity means more individuals than the resources can support, so food, water and space run short and the habitat becomes unsuitable. That is the defining consequence of overshoot.

Question 50 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Ages 40–74 cover seven wide age bands, more people than ages 0–19, 20–39 or 75 and higher. The diagram bulges at middle age.

Question 51 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** A narrow base relative to the older cohorts indicates births below replacement level. Option B states that conclusion, though its reasoning is loosely worded.

Question 52 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** A large older population raises demand for retirement homes and healthcare. Childcare, new-workforce jobs and family planning relate to young populations.

Question 53 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** After overshoot, resources are depleted and density-dependent factors such as starvation and disease raise mortality, causing a dieback. Growth does not simply continue.

Question 54 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** A large young cohort means rapid growth and rising consumption, producing more waste. Elderly-population and shrinking-population effects belong to other shapes.

Question 55 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Doubling time ≈ 70 ÷ growth rate: Afghanistan 29 yr, Brazil 87 yr, Indonesia 70 yr, Uganda 70 ÷ 3.3 ≈ 21 yr.

Section B — FRQ Solutions

Question 1 — 2003
(a) Time on the x-axis (1855–1990), rate per 1,000 on the y-axis. Birth rate points: 1855 = 43, 1875 = 43, 1895 = 37, 1950 = 22, 1990 = 10. Death rate: 1855 = 41, 1875 = 20, 1895 = 12, 1950 = 12, 1977 = 10, 1990 = 10. Label both curves. *(3 points: axes, birth curve, death curve)*

(b) Growth rate 1950: r = (22 − 12) ÷ 1,000 = 0.010 = **1% per year**. In 1977 the growth rate is 0.4% and death rate is 10/1,000 = 1.0%: birth rate = 0.4% + 1.0% = 1.4% = **14 per 1,000**. *(1 point each)*

(c) Death-rate drop: better sanitation and clean water, improved nutrition and food supply, medical care and hygiene (antibiotics not yet available). High birth rate: children provided farm labor and old-age support and offset infant deaths; slow decline because cultural norms, women's education and employment, and access to contraception changed gradually. *(2 + 1 + 1 points)*

(d) 1895 growth rate = (37 − 12) ÷ 10 = 2.5%. Doubling time = 70 ÷ 2.5 = 28 yr. Time from 1895 to 1951 = 56 yr = 2 doublings. P = $2.5 \times 10^6 \times 2 \times 2$ = **1.0×10^7** people. (Exponential check: $2.5 \times 10^6 \times e^{(0.025 \times 56)} \approx 1.01 \times 10^7$.) *(1 point)*

Question 2 — 2010
(a)(i) Scan the table for the largest consumption rate: 0.27 mg/day at **30 °C and 90% relative humidity**. *(1 pt, both values needed.)*

(a)(ii) Termites in the forest = 2,000 ha × 4.5×10^7 termites/ha = 9.0×10^{10} termites. Methane = 9.0×10^{10} termites × (70 kg CH_4/yr ÷ 1,000 termites) = **6.3×10^9 kg CH_4/yr**. *(1 pt setup with units, 1 pt answer.)*

(a)(iii) Treat emission as proportional to wood consumption. At 35 °C and 50% the rate is 0.09 versus 0.27 at the optimum, a ratio of 1/3. Emission = (0.09 ÷ 0.27) × 6.3×10^9 kg = **2.1×10^9 kg CH_4/yr**. *(1 pt setup, 1 pt answer; mental math accepted.)*

(a)(iv) *(reason + linked explanation)* Temperature and humidity are limiting factors with a tolerance range. Outside it (for example 40 °C, where feeding stops and gut microbes may die) termites feed and reproduce less, or plants that supply food decline, so the population shrinks.

(b)(i) *(1 pt)* Clearing leaves a large pulse of dead wood and plant debris, a sudden food surplus, so density rises.

(b)(ii) *(1 pt)* The surplus is used up and the population overshoots carrying capacity and crashes from starvation and competition; the open, hotter ground may also exceed their temperature tolerance.

(c) *(change + effect)* Removing trees lowers photosynthetic uptake and destroys a carbon sink; burning or decay of the wood releases CO_2; machinery burns fossil fuel; or lost shade and evapotranspiration raise local temperatures. Link the change to more greenhouse gas or more warming.

Question 3 — 2017

(a)(i) Reason: forests are cleared for farmland, fuelwood or charcoal income, and enforcement is weaker. *(1.)*
(ii) Strategy: enforce logging limits, promote efficient stoves or alternative fuels, sustainable forestry, or ecotourism. *(1.)*

(b) Change: warmer, more turbid stream water. Without canopy shade, sunlight warms the water; without roots and litter, soil erodes and washes in. (Alternatives: lower dissolved oxygen, more nutrient runoff.) *(1 + 1.)*

(c) Two other benefits: habitat and biodiversity; carbon storage and oxygen release; local climate regulation; soil conservation; air filtration. *(2.)*

(d) Table shows fertility 5.2 → 2.7 children, life expectancy 55 → 64 years, infant mortality 85 → 48 per 1,000. Factor: better access to health care or vaccines cuts deaths from treatable disease; clean water and sanitation reduce water-borne disease. *(1 + 1.)*

(e) A factor lowering fertility: more education and job opportunities for women, delaying marriage and births and increasing awareness of family planning; with lower infant mortality parents expect children to survive. *(1 + 1.)*

Question 4 — 2022

(a) Coal pollutants cause respiratory disease, heart disease, neurological damage or birth defects. *(1)*

(b)(i) Ash leaches toxic metals into groundwater or spills into waterways. *(1)*

(b)(ii) Clay is much less permeable than bare soil, so it slows leaching. *(1)*

(c)(i) Percent change = (857,425 − 757,278) ÷ 757,278 × 100 ≈ +13.2%. *(2)*

(c)(ii) Rule of 70: doubling time = 70 ÷ 1.88 ≈ 37 years; 2019 + 37 = **2056**. *(2)*

(c)(iii) 90 gal/person/day × 841,611 people × 365 days ≈ 2.76×10^{10} gallons. *(2)*

(d) Xeriscape or drought-tolerant plants, skip nonessential washing, smart or drip irrigation, rain barrels or gray water, water at night. *(1)*

Question 5 — 2025

(A) K-selected species have few, well-cared-for offspring, long life, late maturity and stable populations near carrying capacity. *(1)* **(B)** Fewer spiders means less predation on insects, so insect (herbivore) populations can rise, then eat more plants. *(1)* **(C)** About 2.0 spiders per sample at 25% nonnative plants (read from Figure 1). *(1)* **(D)** Insect numbers decline as the percentage of nonnative plants rises. *(1)* **(E)** In Figure 2 the growth rate is at or above 0 until about 25% nonnative plants and negative beyond, so below 25% populations are stable or growing. *(1)* **(F)(i)** Does mowing (site type) affect ant species richness? *(1)* **(F)(ii)** Dependent variable: number of ant species per trap (richness). *(1)* **(G)(i)** The grassland has more species (5 vs 3), so more functional redundancy and genetic variety, making recovery from fire or flood more likely. *(1)* **(G)(ii)** Pavement offers little habitat or food, so fewer ant species would be found than in grassland. *(1)* **(H)** Roads divide habitat, causing vehicle deaths, restricted movement, smaller isolated populations or edge effects. *(1)*

Unit 4 — Earth Systems and Resources

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: D. Freshwater availability per person is lowest in arid regions with large populations and little renewable runoff. The keyed location is the Middle East/Arabian Peninsula, which gets very little rainfall and has few permanent rivers. The other marked locations are humid or river-rich regions with far more water per person.

Question 2 — 2008
Official key: D. Desalination removes salt from seawater and needs large energy inputs, so it is used mainly where freshwater is extremely scarce and energy is cheap or plentiful. The Persian Gulf/Middle East region (point D) fits this, with oil-funded desalination plants supplying most municipal water. Other regions have enough rivers or groundwater.

Question 3 — 2008
Official key: B. The ozone layer sits in the stratosphere, where ozone absorbs ultraviolet radiation and warms the air, so temperature rises with altitude there. On the profile, this is the layer above the troposphere where temperature increases.

Question 4 — 2008
Official key: A. Air density and pressure fall with height, so the lowest layer, the troposphere, holds roughly 75–80 percent of the atmosphere's mass and molecules.

Question 5 — 2008
Official key: A. Weather arises from rising and sinking air, evaporation, condensation and clouds, all of which occur in the troposphere, the layer touching the surface.

Question 6 — 2008
Official key: E. Air pressure decreases with altitude, so the outermost layer (exosphere) has the least pressure and is where the atmosphere thins into space.

Question 7 — 2008
Official key: B. Clay particles are the smallest (<0.002 mm), silt is intermediate, and sand is the largest of the three. So increasing size is clay, silt, sand.

Question 8 — 2008
Official key: A. The topmost layer (the O/A horizon) accumulates decaying leaves and humus, so organic matter is greatest at the surface and falls off with depth toward bedrock.

Question 9 — 2010
Official key: D. The stratosphere contains the ozone layer, the highest natural ozone concentration. Weather and most water vapor lie in the troposphere.

Question 10 — 2010
Official key: E. Solar input, atmospheric gases, land-ocean distribution and Earth's rotation and orbit shape global climate. Seasonal temperature inversions are local weather phenomena.

Question 11 — 2010
Official key: E. Mostly sand drains fastest because it has large pores.

Question 12 — 2010
Official key: C. Permeability describes how easily water moves. Color, pH, salinity and fertility do not control how fast water moves.

Question 13 — 2010
Official key: C. In 1998 the surface was very warm, indicating weakened upwelling that carries nutrients up, so productivity and catch fell.

Question 14 — 2010
Official key: B. El Niño brings warm water to the Peruvian coast. It is a recurring warm-water event in the equatorial Pacific, unlike eutrophication or salinization.

Question 15 — 2010
Official key: A. Leaching is water moving down through soil, dissolving and carrying material away. It is different from erosion, which moves soil particles, and from precipitation.

Question 16 — 2016
Official key: E. Island arcs form where oceanic plate descends beneath another oceanic plate and the melting produces curved chains of volcanoes. The keyed location lies in the western Pacific, a classic subduction setting.

Question 17 — 2016
Official key: C. When two continental plates collide neither subducts easily, so crust crumples and thickens, building mountains such as the Himalayas without volcanoes. The keyed location is on the India–Eurasia boundary.

Question 18 — 2016
Official key: B. At divergent boundaries plates pull apart and magma rises to make new crust. The Mid-Atlantic Ridge is the clearest example on the map, and the keyed location sits along it.

Question 19 — 2016
Official key: D. Solar energy drives photosynthesis, winds and ocean circulation, storms and evaporation that forms clouds. Plate tectonics is driven by heat from Earth's interior and mantle convection, so it is the exception.

Question 20 — 2016
Official key: C. Continuous cultivation removes plant material and speeds decomposition, so with no organic inputs the soil organic matter declines. The keyed graph shows a downward trend, not a rise or a flat line.

Question 21 — 2016
Official key: D. Reading along the three triangle axes, point T sits at about 10 % clay, 30 % silt, and the remaining 60 % is sand. Percentages must total 100, which also rules out other rows.

Question 22 — 2016
Official key: D. A landfill liner must block leachate, so a clay-rich soil with tiny, tightly packed particles and very low permeability is best. Point W is the one with the highest clay percentage.

Question 23 — 2016
Official key: C. In the Hadley cell warm air rises at the equator and drops moisture as rain, then flows poleward, cools and sinks near 30°. Sinking air warms and dries, giving low humidity and desert conditions.

Question 24 — 2016
Official key: A. Fish-eating birds depend on a steady fish supply near their nests. When warm El Niño water reduces fish near the coast, adults cannot find enough food for chicks, and breeding success and numbers fall.

Question 25 — 2016
Official key: D. Savings per minute: 5 − 2 = 3 gal. Total shower time: 5 people × 8 min × 30 days = 1,200 min. Water saved: 3 gal/min × 1,200 min = 3,600 gallons.

Question 26 — 2016
Official key: A. Surface currents driven by winds and Earth's rotation move warm water poleward and cold water toward the tropics, moderating climates. They do not cause lake turnover, restrict the jet stream or reduce ocean acidification.

Question 27 — 2016
Official key: A. Topsoil (the A horizon) contains dark humus with high organic carbon and nitrogen. The B horizon accumulates minerals but holds little organic matter, so it is lighter and low in C and N.

Question 28 — 2016
Official key: E. At the December solstice, the Sun is overhead near the Tropic of Capricorn, so the polar region of the south is tilted toward the Sun and receives continuous daylight south of the Antarctic Circle. Point E lies on that circle.

Question 29 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** El Niño is the warm phase of ENSO, a recurring shift in Pacific surface temperatures that changes weather patterns worldwide. Hadley cells, thermal inversions and rain shadows are fixed or local features and are not periodic Pacific phenomena.

Question 30 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Air rises over the windward side, cools and drops its moisture, then descends dry on the leeward side. This creates a rain shadow and desert-like land downwind.

Question 31 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Divergent boundaries have plates moving apart with a rift where new crust forms; diagram D shows plates moving in opposite directions with a gap. The others show a volcano arc, transform sliding or subduction.

Question 32 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Reading the standard-farming line at 60% gives roughly 1,000 years.

Question 33 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** West of the mountains gets much more rain than the east year-round, consistent with a rain shadow on the east side.

Question 34 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Seasons occur because of the 23.5° tilt. With no tilt, sunlight at each latitude would be the same all year, so there would be no seasonal temperature variation.

Question 35 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** On the soil triangle this mix is clay loam, which has moderately low water intake, high water retention and medium-to-high erosion susceptibility. Option B matches intake and retention.

Question 36 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Sand has high drainage and very high intake, so adding it improves drainage. Silt and clay hold water.

Question 37 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Sandy loam combines sand, silt and clay with moderately high drainage. Sand is not mixed and clay or sandy clay loam drain slowly.

Question 38 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The article's takeaway is that mapping the four-leaf pattern tells where low-frequency shaking, and thus building damage, is likely greatest.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Data showing low-frequency waves spread in four directions but barely register elsewhere is the direct evidence.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Because buildings are sensitive to low frequencies, planners would avoid placing large buildings near fault zones in the clover pattern.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The stratopause is where temperature peaks around 50 km, marked by line C.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Tokyo lies near the intersection of several converging plates, with subduction causing earthquakes.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** A stationary hot spot with plate motion produces the age pattern.

Section B — FRQ Solutions

Question 1 — 2003

(a) Leaf litter is a protective, nutrient-recycling layer. It decomposes into humus and nutrients for plants, holds moisture, lowers erosion, shelters seedlings and decomposer organisms, and insulates the soil. One statement must relate directly to plants. *(2 points)*

(b) Three described changes (not just listed) if worms remove the litter in a year: soil moisture falls as evaporation rises; erosion and runoff increase (with sediment in streams); soil fertility and humus drop, and soil temperature swings widen. Also pH change or compaction. *(3 points)*

(c) Pick one change and tie it to invaders: with drier, less fertile, more exposed soil, generalist exotic plants with broad tolerance (such as stilt grass) out-compete natives adapted to moist, humus-rich conditions. *(1 point; must be based on a part (b) change)*

(d) Hypothesis: forest plots with worms will have thinner litter depth and lower soil moisture than worm-free plots. Procedure: set matched plots of equal area in the same forest; control plots without added worms, experimental plots with a stated number of worms; same period (for example one year); several replicate plots; control other variables (slope, canopy, rainfall). Data: measure the dependent variable quantitatively, such as litter mass or depth and soil moisture percentage, at set intervals; compare means statistically. *(1 hypothesis naming a measurable factor, 1 design with control, experimental and one specific (area/time/sample size), 1 elaboration such as replication, 1 quantitative data)*

Question 2 — 2004

(a) Chemical test: soil pH (or N-P-K, salinity); physical test: texture (sand/silt/clay by settling or ribbon test) or water-holding capacity. Use: acidic soil needs lime; low nitrogen suggests legume rotation; sandy soil needs organic matter or more frequent irrigation. *(3 points)*

(b) Advantage: nutrients are concentrated, immediately available and easy to apply, raising yield. Disadvantage: runoff causes eutrophication; no organic matter is added; energy-intensive to make; N_2O emission. *(1 each)*

(c) Contour plowing or terracing slows runoff; no-till keeps residue and roots; windbreaks, cover crops, strip cropping hold soil. Link each to reduced erosion. *(2 points)*

(d) Temperate deciduous forest or grassland. Humus forms from partly decomposed leaves, roots and organisms. It holds water and nutrients, improves aeration and structure, buffers pH and supports soil life. *(1 biome, 1 origin, 2 benefits; 3 maximum for the part)*

Question 3 — 2014

(a)(i) At a subduction zone, denser oceanic plate is forced beneath the other plate, often forming a deep trench. *(1.)* (ii) The sinking slab heats and releases water, melting overlying mantle; the buoyant magma rises through the crust and erupts, building volcanic arcs and island chains. *(1 for melting, 1 for rising magma.)*

(b)(i) Primary succession is colonization of bare rock, ash or sand where no soil previously existed. *(1.)* (ii) Pioneers such as lichens and mosses secrete acids and wedge into rock, weathering it into mineral particles; as they die and decompose, organic matter mixes with these particles, forming a thin soil that later plants can use. *(1 + 1.)*

(c)(i) A sudden vertical slip of the seafloor during a subduction earthquake displaces a huge volume of water, which radiates outward as waves growing taller in shallow water. *(1 + 1.)* (ii) Tsunamis destroy coastal habitats (mangroves, reefs), drown land organisms, or contaminate soils with salt water. *(1.)*

(d)(i) At the moment of an earthquake, locked plates suddenly slip past each other horizontally, releasing stored energy. *(1.)* (ii) Between quakes the plates stay stuck by friction while stress and strain accumulate. *(1.)*

Question 4 — 2016

(a) Two climate factors and soil formation rate. Temperature: warmth speeds weathering and microbial decomposition so soil forms faster, while cold slows biological activity (though freeze-thaw can crack rock). Precipitation: moderate to high rainfall increases weathering and biological activity; very heavy rain can erode soil and reduce formation, and very dry conditions slow it. Wind can add or remove particles. *(2 points: 1 each description, not just naming a factor)*

(b) A horizon. (i) Biotic: humus, roots, earthworms, fungi or bacteria. (ii) Abiotic: sand, silt, clay, water, air, minerals or nutrients. *(2 points)*

(c)(i) Practice raising nitrates. Heavy fertilizer application (or improperly sealed feedlots or manure lagoons). *(1 point)*

(c)(ii) How. Nitrate is soluble and not held by soil particles, so water percolating through the soil carries it down into groundwater (or runoff carries it to waters that recharge aquifers). *(1 point)*

(d)(i) Effect of acid deposition on plants. Acidic soil leaches away nutrient cations such as calcium and magnesium, releases toxic aluminum, damages roots and lowers the soil's buffering, so growth declines. *(1 point)*

(d)(ii) Remediation. Add crushed limestone (lime) to neutralize the acid. *(1 point)*

(e) Two ways climate change degrades soil. Hotter, drier conditions cause desertification; higher evaporation of irrigation water causes salinization; heavier rain increases erosion and leaching; warmer soil breaks down organic matter faster; sea level rise brings salt water to coastal soils. *(2 points: 1 each)*

Question 5 — 2021

(a)(i) Question: how do slope and land-use or tillage practice (crop rotation, no-till, cover crops) affect annual soil erosion? *(1)*

(a)(ii) Crop rotation is the practice that stays within tolerable loss on a 15% slope. *(1)*

(a)(iii) A cover crop gives lower erosion than no-till alone, allowing steeper land to stay within the tolerable limit. *(1)*

(a)(iv) Natural erosion agents: running water, rainfall or wind. *(1)*

(b)(i) Dependent variable: amount of sediment run-off or discharge. *(1)*

(b)(ii) Add an untreated bare plot, or measure sediment before and after treatment. *(1)*

(b)(iii) An uncontrolled factor: slope, rainfall amount, soil type, plot size, distance to stream, road material or time since rain. *(1)*

(b)(iv) Turbidity was measured as sight depth: straw plot 12 cm, grass plot 28 cm. Deeper sight depth means clearer water, so grass trapped more sediment and the hypothesis (straw is better) is rejected. *(1)*

(c)(i) Type III: very high early mortality, few survive to adulthood. *(1)*

(c)(ii) Sediment smothers eggs and cuts oxygen, removes spawning gravel, carries toxins, warms water (less dissolved oxygen) or clouds water, impairing reproduction. *(1)*

Question 6 — 2024

(a) Cacti, succulents or other drought-tolerant plants. *(1)*

(b) Native plantings support wildlife, need less fertilizer, pesticide and mowing, or reduce fuel use. *(1)*

(c) Burning removes fuel (undergrowth), lowering intensity and creating a firebreak. *(1)*

(d) Drawbacks: displaces wildlife, emits CO_2 and particulates, post-burn runoff, helps invasive species. *(1)*

(e) 25.16 maf × 0.36 = 9.06 maf. *(2)*

(f) 9.60 maf × 0.36 × 0.50 = 1.73 maf. *(2)*

(g) 9.60 maf × 3.26×10^{11} gal/maf = 3.13×10^{12} gal; ÷ 5.0×10^{4} gal/household ≈ 6.3×10^{7} households. *(2)*

Question 7 — 2025

(A) The eastern equatorial Pacific is cooler than average. *(1)* **(B)** La Niña (stronger trade winds, cool east, warm west). *(1)* **(C)** Region A (northwest) is cooler with greater chance of precipitation; Region B (south) is warmer with lower chance of precipitation. *(1)* **(D)** Impervious surfaces and storm drains cause rapid runoff with little infiltration. *(1)* **(E)** E.g. permeable pavement, green roofs, rain gardens, retention basins or wetlands. *(1)* **(F)** Extra benefit: groundwater recharge, urban cooling, habitat, cleaner runoff or aesthetic value. *(1)* **(G)** Temperate seasonal forest has moderate rain with cold winters and warm summers; savanna is warm year-round with a distinct dry season and less rain. *(1)* **(H)** Secondary succession. *(1)* **(I)** Smoke releases particulates, CO, NO_x, VOCs and CO_2. *(1)* **(J)** Prescribed burns or thinning to cut fuel, or selective harvesting and firebreaks. *(1)*

Unit 5 — Land and Water Use

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: B. Rainwater percolating through exposed spoil heaps dissolves sulfide minerals such as pyrite to form sulfuric acid, producing acid mine drainage that acidifies streams and mobilizes metals. Thermal pollution, ozone smog and alkaline or nutrient discharges are not the typical consequence.

Question 2 — 2008

Official key: B. Clear-cutting removes every tree in an area, leaving patches of habitat isolated, exposing soil to erosion and eliminating forest-interior species. Selective cutting and shelterwood cutting leave much of the canopy. Slash-and-burn is not a commercial logging method, and gleaning is gathering leftover crops.

Question 3 — 2008

Official key: A. An introduced control organism may not be specific and may attack non-target, beneficial or native species, as happened with some introduced predators. The other choices describe cost or effectiveness issues, not ecological risk.

Question 4 — 2008

Official key: B. Gold is commonly leached from ore with cyanide solutions, so spills and tailings ponds release toxic cyanide to waters and soil. Ozone, CFCs, asbestos and selenium are not characteristic of gold recovery.

Question 5 — 2008

Official key: D. Crowded net-pen salmon often suffer outbreaks of parasites like sea lice, which can spread to wild fish. They are less genetically diverse, do escape, pollute surrounding water with waste and are kept at higher densities.

Question 6 — 2008

Official key: B. The treadmill occurs when pests evolve resistance, farmers use more or stronger pesticides, and selection produces still more resistant pests. The keyed choice is increased pesticide use to eradicate genetically resistant pests.

Question 7 — 2008

Official key: E. Routine low-dose antibiotics in crowded livestock select for antibiotic-resistant bacteria, which can spread to people. The other practices don't directly create resistant pathogens.

Question 8 — 2008

Official key: A. Open-access fisheries are shared resources where each user gains from taking more while the cost of depletion is spread across all, so stocks like those in the North Atlantic collapse.

Question 9 — 2008

Official key: A. Impervious surfaces mean less infiltration and less groundwater recharge, more runoff, and fewer plants mean less transpiration and evaporation.

Question 10 — 2008

Official key: C. Pheromone-based mating disruption is a nonchemical-toxin approach permitted for organic crops, while synthetic nitrate fertilizers, strychnine, sewage sludge and GMOs are prohibited.

Question 11 — 2008

Official key: B. Mountaintop removal strips overburden from a summit with heavy equipment to reach coal seams, pushing waste into adjacent valleys.

Question 12 — 2008

Official key: B. Eating at a lower trophic level wastes less energy, so a given area of farmland can feed more people.

Question 13 — 2010

Official key: C. Pyrite is iron sulfide. When mining exposes it to air and water, the sulfur oxidizes and forms sulfuric acid that leaches into streams. The keyed element is sulfur.

Question 14 — 2010
Official key: B. Water that flows through exposed sulfide minerals in coal or metal mines becomes acidic and carries toxic metals into streams. It is leakage of that contaminated water, not an acid-based mining method or a result of acid rain.

Question 15 — 2010
Official key: B. Integrated pest management relies on monitoring, biological control and targeted use, so far less pesticide is applied. No-till, contour plowing and cover crops protect soil but do not primarily reduce chemical exposure, and crop dusting increases it.

Question 16 — 2010
Official key: C. Insects with inherited resistance survive and reproduce, so resistance spreads through the population by natural selection. Acquired traits are not inherited.

Question 17 — 2010
Official key: C. Dense pens concentrate parasites and disease that spread to passing wild fish. The other options misstate the main threat.

Question 18 — 2010
Official key: D. Biological control usually targets one pest and the control organism can persist. It does not promote resistance and a pest explosion.

Question 19 — 2010
Official key: E. Agriculture uses about 70 percent, industry about 20 percent and domestic uses about 10 percent.

Question 20 — 2010
Official key: D. Plowing along contours slows runoff and reduces soil erosion. It does not address nutrient or salt problems directly.

Question 21 — 2010
Official key: A. Flooded rice paddies are anaerobic and release methane.

Question 22 — 2016
Official key: D. The Ogallala lies beneath the US Great Plains (High Plains) and supplies irrigation water there; withdrawals far exceed its slow recharge, so it is being depleted. The other options place it on the wrong continents or in the wrong region.

Question 23 — 2016
Official key: C. Heavy repeated spraying kills susceptible weeds but leaves resistant individuals that reproduce, so resistance spreads. This is natural selection, and explains why the count of resistant weeds climbs with usage.

Question 24 — 2016
Official key: B. Planting a few uniform GM varieties widely can narrow the genetic diversity of crops, making them more vulnerable to pests or disease. The other options are false claims about yield or soil limits.

Question 25 — 2016
Official key: E. Compare indicators of the goal (a forest). Site 5 has 65 % tree cover, the fastest growth (20 cm/yr) and the least grass, meaning trees are outcompeting the seeded grass. Site 1 has mostly grass and little tree cover.

Question 26 — 2016
Official key: A. Pavement and roofs keep water from soaking into the ground, so aquifers receive less recharge. Runoff to streams actually increases, and impervious surfaces increase, rather than decrease, erosion in streams.

Question 27 — 2016
Official key: E. Rain barrels hold roof runoff so less flows quickly to the creek, reducing flooding, erosion and pollutant loads. Fertilizer before a storm and pet waste in drains add nutrients and pathogens, and channelizing speeds flow and worsens erosion.

Question 28 — 2016
Official key: B. Watersheds with high residential and commercial land cover have more impervious surfaces, so runoff reaches streams quickly and flow fluctuates strongly with rainfall. Watershed IV is mainly agricultural, so more algae from nutrient runoff, not excellent quality, is likely there.

Question 29 — 2016
Official key: D. Each crop's income per hectare must be multiplied by the number of hectares it occupies. With total hectares times 0.4 for trees and 0.6 for vegetables, the two terms are added to give total income, so units of dollars result.

Question 30 — 2016
Official key: A. Hydraulic fracturing injects fluids underground, and leaks, spills or poor well casings can contaminate drinking-water aquifers. Ozone depletion, eutrophication, denitrification and desertification are not typical results.

Question 31 — 2016
Official key: A. The tragedy of the commons occurs when a shared, unowned resource is overused or degraded because each user gains while costs are shared. Polluting the open Gulf waters fits; privately owned lakes are not commons.

Question 32 — 2016
Official key: B. From the third graph, vegetables return $1,800 per million liters compared with $300 for livestock and pasture. Livestock uses the most water and earns the most money in total but yields the lowest return per liter.

Question 33 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Clear cutting removes all trees at once, which frees land for crops or other uses and is the benefit listed. The other statements are wrong: trees emit VOCs that do not set stratospheric ozone, clear cutting releases carbon and often lowers fertility, and it is not a sustainable annual harvest.

Question 34 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Years 1–3 show about 200 lb/acre sediment loss with compost versus about 620 without, and about 1.4 versus 3.4 acre-inches of runoff. Compost therefore decreased both.

Question 35 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The bar for compost applied previously reads about 300 lb/acre.

Question 36 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** A control receives no treatment, so the fields with no compost are the control. The measured variables are not groups.

Question 37 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Per-capita water falls from about 13,000 to about 2,500 m^3 as population rises from about 30 to about 150 million, so availability per person decreased as population grew. The graphs do not address deforestation, quality or recreation.

Question 38 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Losing shade raises water temperature, and warmer water holds less dissolved oxygen, so temperature increases and DO decreases.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** High evaporation leaves dissolved salts in soil that build up, especially with irrigation. Fertilizer nitrates, drip systems and snowmelt are not the usual explanation.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** No-till agriculture leaves residue and roots in place, holding soil. Pesticides, slash-and-burn and flood irrigation do not target erosion.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** RWW soils have higher salinity (2,752 versus 960 ppm) and sodium, which harms many plants, so the results support the hypothesis. Equal pH and organic matter do not refute it.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** If a few pests survive an application and reproduce, resistant individuals pass on genes, selecting for resistance. Rotating pesticides and predators slow it.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Demand for fish rose while wild catches fell, so farmed fish fill the gap. Aquaculture does not clearly improve water quality or reduce disease.

Question 44 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Pavement, roofs and other impervious surfaces stop infiltration and send stormwater rapidly into streams. Increased infiltration would reduce flooding.

Question 45 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** HT corn rose from about 25% to about 50%: (50 − 25) ÷ 25 × 100 = 100%.

Question 46 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The Bt corn line rises most steeply from 2006 to 2007, the steepest slope among the options.

Question 47 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Bt crops are engineered to produce a protein toxic to certain insect pests, reducing pest damage. The other claims about herbicide use, diversity and fossil fuels are not supported.

Question 48 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Tailings can contain heavy metals that leach into the water and contaminate the lake. Radioactive products, hypoxia and pH above 7 are not the typical outcome.

Question 49 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The largest freshwater entry is tree farming at 180,000 acres, larger than urban development (100,000) or the other entries.

Question 50 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Freshwater losses total roughly 340,000 acres versus about 104,000 for saltwater, so freshwater wetlands lost more total acres.

Question 51 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Percent lost = (urban loss) ÷ (total) × 100, where urban loss = 100,000 + 3,000 acres, giving 103,000 ÷ 40,000,000 × 100 ≈ 0.26%.

Question 52 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Furrow irrigation sends water along channels between ridges planted with crops, shown in diagram B.

Question 53 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Fewer herbivores allow algae to overgrow and smother coral, reducing diversity.

Question 54 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** CAFO meat is generally cheaper because of high density and low land use. The other statements reverse the facts.

Question 55 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Limits avoid exceeding maximum sustainable yield so the stock persists.

Section B — FRQ Solutions

Question 1 — 2002
(a) Two problems: (i) less flow carries less sediment, nutrients and organisms downstream, lowering productivity in the delta/estuary, while dams trap silt upstream; (ii) reduced flow warms the water and lowers dissolved oxygen, concentrates pollutants and salts, blocks fish migration and exposes banks to erosion. *(1 point each described; commentary shows 3–4 points available)*

(b) For cities: drinking water, sanitation and public health, industry and jobs, hydroelectric power; people's needs come first. For nature: wetlands and riparian habitat, fisheries and biodiversity depend on flow, ecosystems provide services (flood control, water filtration), and species have intrinsic value. *(arguments for both sides, each supported)*

(c) Examples: Aswan High Dam (Nile): the lake behind it floods land and loses water to evaporation, silt no longer fertilizes the floodplain, and downstream erosion and salinity rise. Mono Lake (water exported to Los Angeles): the lake level fell, salinity rose and bird/brine-shrimp habitat shrank. Aral Sea, California Water Project, Owens Valley also work. *(1 point for naming the project, 1 point per impact)*

Question 2 — 2005
(a) 1950: 5.2×10^{10} kg ÷ 2.6×10^{9} people = 20 kg per person. 2000: 2.4×10^{11} kg ÷ 6.0×10^{9} = 40 kg per person. *(1 point each)*

(b) (40 − 20) ÷ 20 × 100 = **100% increase**. *(setup 1, answer 1)*

(c) Land: grain eaten directly needs only cropland; meat needs pasture plus land to grow feed, so a hectare of grain feeds more people. Energy: only about 10% of energy passes up each trophic level, so feeding crops to livestock wastes most of it, and raising, processing and cooling meat takes extra fossil fuel. *(1 land, 1 energy; must compare)*

(d) Overgrazing causing erosion and desertification; manure runoff causing eutrophication and low oxygen; deforestation for pasture lowering habitat and releasing CO_2; livestock methane warming the climate; heavy water use. *(2 points)*

(e) Advantage: less saturated fat and cholesterol, lower heart-disease risk. Disadvantage: risk of protein, iron or vitamin B_{12} deficiency without careful planning. *(1 each)*

Question 3 — 2005
(a) After replacing overburden: regrade to the original contours, spread topsoil and add nutrients or organic matter, replant native or fast-growing cover, then monitor for years. *(3 points)*

(b) Low rainfall and poor soils make plants slow to establish; bare ground then erodes by wind and water and reclamation takes longer. *(1 + 1)*

(c) Sulfur minerals exposed to water and air form sulfuric acid (acid mine drainage), acidifying soil and streams and mobilizing heavy metals. Remedy: add crushed limestone, cover waste to keep it dry, or use retention ponds. *(1 + 1)*

(d) Acid deposition from SO_2/NO_x; CO_2 adding to global warming; mercury and particulates harming health; toxic ash disposal; thermal pollution. *(1 each)*

(e) Coal is abundant and cheap in the US, so it can replace costlier fuels as oil supplies tighten; and rising affluence raises energy use per person. *(1 + 1)*

Question 4 — 2006

(a) Steepest drop: 1965–1970, from about 700 thousand to about 200 thousand metric tons: (700,000 − 200,000) ÷ 5 yr = **100,000 metric tons per year**. *(1 + 1)*

(b) Bottom trawling: drags nets or lines along the seafloor, destroying habitat and catching nontarget species (bycatch). Drift/gill nets or purse seines: huge nets catch whole schools and bycatch, and lost nets keep killing. Long-lining or sonar also acceptable. *(1 + 1 each)*

(c) Magnuson-Stevens Act: regional councils set quotas, size limits and seasons and protect habitat to rebuild stocks. (Also ESA, Marine Mammal Protection Act, CITES, UN Law of the Sea.) *(1)*

(d) Atmosphere: fossil-fuel burning raises greenhouse gases; manage by emissions limits or agreements. (Or aquifers, Antarctica, public forests.) *(3 points)*

Question 5 — 2008

(a) Block counts from the diagram: 7 blocks with 2×10^{-4} % gold, 7 blocks with none, 6 blocks with 1×10^{-4} %. Total volume = 20 blocks × 200 m^3 = 4,000 m^3 (also 25 m × 16 m × 10 m). Volume of gold = 1,400 m^3 × (2×10^{-6}) + 1,200 m^3 × (1×10^{-6}) = $2.8 \times 10^{-3} + 1.2 \times 10^{-3}$ = **4.0×10^{-3} m^3**. *(1 point: setup; 1 point: answer.)*

(b) Percent gold = 4.0×10^{-3} m^3 ÷ 4,000 m^3 × 100 = **1×10^{-4} %**. *(1 point: setup; 1 point: answer.)*

(c) Mass = 4.0×10^{-3} m^3 × 10^6 cm^3/m^3 × 19 g/cm^3 = 7.6×10^4 g. Break-even price = \$1,520,000 ÷ 7.6×10^4 g = **\$20 per gram**. *(1 point: setup; 1 point: answer.)*

(d) Two uses, e.g., crushed waste rock as road-base or fill for mine access roads; aggregate for concrete; re-processing to recover other minerals. *(1 point each; 1 extra for added detail.)*

(e) Two problems, e.g., acid mine drainage and toxic leachate (such as cyanide) reaching groundwater and streams; loss of topsoil and vegetation; habitat destruction and altered drainage; unsightly tailings. *(1 point each; 1 extra for added detail.)*

Question 6 — 2011

(a) *(2 pts)* Plot the five points (1900, 0.40), (1940, 0.60), (1980, 1.25), (2020, 2.50), (2060, 4.75) with years on the x-axis and billion hectares on the y-axis, then draw one smooth upward-curving line through them.

(b) *(1 pt)* Read across from 4.00 billion ha on the curve. Between 2020 (2.50) and 2060 (4.75) the straight-line estimate is about 2047; because the curve bends upward, a reading near **2045 to 2050** is consistent.

(c) *(2 properties, each with its role, 4 pts)* Examples: soil texture or particle size, which sets drainage, aeration and water-holding; pH, which controls nutrient availability and plant tolerance; organic matter or humus, which stores water and nutrients; nutrient content (N, P, K); porosity or compaction, which govern root oxygen and infiltration; salinity, which draws water out of roots.

(d) *(2 pts)* Raise yields per hectare (improved or genetically modified varieties, fertilizer, crop rotation, intercropping) or lower demand for farmland (eat lower on the food chain, cut food waste, slow population growth, protect soil from erosion so land stays productive).

(e)(i) *(1 pt)* In irrigated dry regions, water evaporates or is taken up by plants and leaves dissolved salts behind in the topsoil; salty groundwater can also wick up by capillary action.

(e)(ii) *(1 pt)* Flush the soil with enough low-salt water, install drainage tiles, switch to drip irrigation, or plant salt-tolerant or salt-removing crops.

Question 7 — 2012

(a)(i) *Hypothesis (1 pt):* If minnows are exposed to higher concentrations of product X, then more of them will die (or: product X concentration has no effect on minnow mortality, as a null hypothesis).

(a)(ii) *Method (1 pt):* Set up at least three identical tanks (same water, temperature, fish size and number, feeding), for example 0 (no product X), low and high concentration of product X, with replicates, and observe for a fixed time such as 96 hours.

(a)(iii) *Control (1 pt):* The tank or group with no product X.

(a)(iv) *Dependent variable (1 pt):* Minnow health, for example number of fish dead (or survival rate) at the end of the test.

(b) *(1 pt)* If death rates were the same, or lower, in tanks with product X than in the control, the hypothesis would be rejected; the description must contradict the stated prediction.

(c)(i) *(3 pts)* IPM combines several tactics to keep pests below a damaging level while minimizing chemical pesticide use. Two approaches: introduce or attract natural predators or parasites, rotate crops, use pheromone traps, plant diverse crops, or use targeted low-persistence pesticides.

(c)(ii) *(1 pt)* Environmental gain, for example less pesticide runoff to waterways, fewer non-target kills, less biomagnification or slower pesticide resistance.

(d) *(2 pts)* Two practices, each tied to higher yield: apply fertilizer or organic matter, irrigate, plant high-yield varieties, rotate or intercrop, terrace or use contour plowing to protect soil.

Question 8 — 2014

(a) Two pollutants: nitrogen or phosphorus (fertilizer), sediment, pathogens such as E. coli, motor oil, road salt, pesticides, trash. *(2.)*

(b) Area = 200 m × 100 m = 20,000 m^2; depth 5 cm = 0.05 m. Volume = 20,000 m^2 × 0.05 m = **1,000 m^3**. *(1 setup, 1 answer.)*

(c) City area 10 km^2 = 10 × 10^6 m^2 = 1.0 × 10^7 m^2; impervious = 0.20 × 1.0 × 10^7 = 2.0 × 10^6 m^2. Runoff = 2.0 × 10^6 m^2 × 0.05 m = **1.0 × 10^5 m^3** (100,000 m^3). *(1 + 1.)*

(d) Flow arriving = 100,000 + 5,000 (domestic) = 105,000 m^3; plant capacity 10,000 m^3 per day. Bypass = 105,000 − 10,000 = **95,000 m^3 untreated**. *(1.)*

(e) Reduce runoff by: replacing pavement with permeable surfaces or planting vegetation; building retention basins, wetlands, rain barrels or green roofs. *(2.)*

(f) Extensive pavement causes habitat loss and fragmentation, an urban heat island effect, or reduced aquifer recharge. *(1.)*

Question 9 — 2015

(a) Two causes: heavy reliance on cars and new highways; cheaper land and larger lots outside cities; higher inner-city taxes or poor schools and crime; telecommuting. *(2.)*

(b) Two health effects: respiratory disease from traffic pollution; obesity and hypertension from inactive car-based living; crash injuries; longer emergency response times. *(2.)*

(c) The graph shows an inverse (negative) relationship: cities with lower population density use more petroleum per person, and use falls as density rises. *(1.)*

(d) Two smart-growth steps: invest in mass transit and bike paths; mixed-use zoning and urban growth boundaries; tax incentives for infill and brownfield redevelopment. *(2.)*

(e) Two wildlife measures: wildlife overpasses or underpasses and corridors linking habitat; fencing along migration routes, preserved green buffers, or routing highways away from prime habitat. *(2.)*

(f) Urban food production: rooftop or community gardens, vertical farms or hydroponics, converting vacant lots. *(1.)*

Question 10 — 2016

(a)(i) Rock waste. Waste = ore used − pig iron produced = 1.6 × 10^9 t − 1.2 × 10^9 t = 0.4 × 10^9 t = **4 × 10^8 tons**. *(1 point: answer with work)*

(a)(ii) Pig iron from all reserves. Yield = 1.2 ÷ 1.6 = 0.75 t pig iron per t ore. Pig iron = 0.75 × 800 × 10^9 t = **6.0 × 10^{11} tons (600 billion tons)**. *(2 points: 1 for correct setup, 1 for the answer)*

(a)(iii) Reserves going to steel. 95 % × 800 billion t = 0.95 × 800 × 10^9 = **7.6 × 10^{11} tons (760 billion tons)** of iron ore. *(1 point: answer with work)*

(b) Coal conserved. 0.7 t coal per t steel × 80 × 10^6 t steel = **5.6 × 10^7 tons (56 million tons)** per year. *(1 point)*

(c) Two problems at abandoned coal mines. Examples: acid mine drainage and heavy-metal pollution of streams; subsidence and sinkholes from collapsed shafts; habitat destruction that recovers slowly; erosion and sediment from bare tailings; underground fires and methane release; dust. *(2 points: 1 each, first two only)*

(d) Mitigation of one problem. Treat acid drainage with limestone or retain runoff in ponds; replant vegetation to stop erosion; fill or seal shafts; regrade and return tailings. *(1 point: method matched to a problem named in (c))*

(e) Surface mining cheaper. Reason plus discussion, e.g. underground mining needs more workers, higher wages and more safety measures (ventilation, supports), and has higher costs for accidents, insurance and illness such as black lung, whereas surface mining uses large machines with fewer workers. *(2 points: reason with linked economic explanation)*

Question 11 — 2016

(a)(i) Waste = ore in − pig iron out = 1.6 − 1.2 = 0.4 billion tons = **4×10^8 tons**. *(1.)* (ii) Pig iron per ton of ore = 1.2 ÷ 1.6 = 0.75; applied to reserves: 0.75 × 800 billion = **600 billion tons**. *(1 setup, 1 answer.)* (iii) Steel uses 95% of mined ore: 0.95 × 800 billion = **760 billion tons**. *(1.)*

(b) Coal saved = 0.7 tons coal/ton steel × 80 million tons recycled = **56 million tons per year**. *(1.)*

(c) Two problems at abandoned coal mines: acid mine drainage and heavy-metal runoff polluting streams; subsidence and sinkholes; habitat destruction and erosion; underground fires or methane release. *(2.)*

(d) One mitigation: neutralize acid drainage with limestone or retention ponds, replant and regrade slopes to stop erosion, or seal and fence shafts. *(1.)*

(e) Surface mining is cheaper because it needs fewer workers and less costly equipment and ventilation and has lower safety, insurance and legal liabilities than deep mining. *(2: reason plus discussion.)*

Question 12 — 2018

(a)(i) A carbon footprint is the total carbon dioxide (and other carbon compounds) released by an activity, product or person. *(1.)* (ii) Burning fossil fuel (oil, natural gas) in boilers emits CO_2 (CO if combustion is incomplete). *(1.)* (iii) Add insulation or sealing, lower the thermostat schedule, or switch to a renewable or more efficient heat source. *(1.)*

(b) Two green-roof benefits: insulation that cuts heating and cooling energy; stormwater absorption; habitat for pollinators; reduced urban heat island; CO_2 uptake. *(2.)*

(c) Two cafeteria practices: composting scraps, donating surplus food, using reusable trays, offering more plant-based or local food (lower transport and meat impact), bulk packaging. *(2.)*

(d) Two native-plant benefits: adapted to local conditions, so they need less water, fertilizer and pesticide (saving money); they support native food webs and biodiversity and crowd out invasives. *(2.)*

(e) Cork or bamboo flooring is a renewable, biodegradable product made with fewer fossil fuels and fewer VOC emissions than synthetic carpet, and the growing plants store carbon. *(1.)*

Question 13 — 2021

(a) Pesticides suppress disease vectors (mosquitoes, ticks, rodents) and raise food supply, improving health. *(1)*

(b) Entry routes: inhaling spray, drinking contaminated water, eating residues on food, or absorption through skin. *(1)*

(c)(i) 1975, when only 2 pests were captured. *(1)*

(c)(ii) Pests started at 700 (no pesticide), crashed to 2 after first use, then climbed steadily to 1,100 by 2015. *(1)*

(c)(iii) Even with the dose rising from 500 to 1,000 g/ha, catches rose from 2 to 1,100: survivors carried heritable resistance and reproduced, so each generation tolerates more pesticide (the pesticide treadmill). *(1)*

(d) Two non-lethal effects: contamination of soil and water; bioaccumulation and biomagnification; endocrine disruption and reproductive problems; thinner eggshells; impaired bee signaling; loss of food supply or predators upsetting the food web. *(2: 1 each)*

(e)(i) Rotation means growing different crops in sequence on the same field. *(1)*

(e)(ii) Integrated pest management, intercropping, timing application to the vulnerable stage, or pest-resistant crop varieties. *(1)*

(e)(iii) Justification: less pesticide residue in food, lower input cost and higher farm profit, or fewer exposed workers. *(1)*

Question 14 — 2022

(a) Dark pavement and roofs store heat, buildings block wind, there is little shade or evapotranspiration, and vehicles and air conditioners add waste heat. *(1)*

(b)(i) E.g. green roofs and more vegetation, reflective light-colored surfaces, or more efficient or less-used heat-producing equipment. *(1)*

(b)(ii) Extra benefit: stormwater capture, habitat, cleaner air, insulation or lower energy cost. *(1)*

(c)(i) Percent change = (44.5 − 15.9) ÷ 15.9 × 100 ≈ +180%. *(2)*

(c)(ii) Change = 12,900 kWh × ($0.13 − $0.11)/kWh = **$258** more per year. *(2)*

(c)(iii) 30 panels × 300 W × (1 kW/1,000 W) × 4 h/day × 365 days = 9 kW × 1,460 h = **13,140 kWh/year**. *(2)*

(d) June-August the Northern Hemisphere is tilted toward the Sun, so sunlight is more direct and days are longer. *(1)*

Question 15 — 2023

(a) Control group: Plot D (unmodified beans). *(1)*

(b) Does genetic modification change green-bean yield compared with unmodified beans? *(1)*

(c) Less water would lower yields overall, and unmodified beans would likely suffer most if modified types tolerate drought. *(1)*

(d) Lowest soil temperature: Plot D (17.6 °C). *(1)*

(e) Sediment, phosphorus and nitrogen runoff were all lower from modified-bean plots than from unmodified plot D (e.g. sediment 3.5 vs 0.9-2.4 mg/L). *(1)*

(f) Supported: plot B (Type 2) shows far lower P and N runoff than D and still yielded more, so more fertilizer was taken up by plants rather than washed off. *(1)*

(g) Secondary succession: pioneer, fast-growing species recolonized soil left intact after the fire. *(1)*

(h) More species means a flood removes a smaller share of species and interactions, and diversity includes flood-tolerant types that survive. *(1)*

(i) The newcomer could be a generalist or r-selected invader that reproduces quickly and exploits niches opened by the flood in a low-diversity plot. *(1)*

(j) Trap or remove beetles, apply pesticide, or introduce a biological control. *(1)*

Question 16 — 2023

(a) About 8-9 plant species in technically restored piles after 16-25 years (graph reading). *(1)*

(b) Plant species richness on spontaneous piles rises with pile age (direct relationship). *(1)*

(c) Claim: supports the hypothesis, because more plant species provide more food, shelter and habitat for animals. *(1)*

(d) Early colonizers are pioneer or generalist plants that tolerate poor, low-nutrient soil and full sun and spread fast. *(1)*

(e) Recovery in tundra would be slower (cold, short season, little rain) and involve tundra species with lower diversity. *(1)*

(f) Does ground cover (plants or wood chips) affect the quantity and clarity of runoff? *(1)*

(g) Dependent variable: volume of water collected (or its appearance). *(1)*

(h) More sand means more infiltration, so less runoff and probably clearer runoff because sand settles quickly. *(1)*

(i) Acidic, metal-laden or sediment-rich drainage can exceed tolerance, cause disease or death, or biomagnify metals. *(1)*

(j) Plant vegetation, cap waste with soil or lime, divert water, treat drainage, or store waste in basins or under water. *(1)*

Question 17 — 2024

(a) Chicken protein needs about 50 m² per kilogram. *(1)*

(b) Type III survivorship (many offspring, high early mortality). *(1)*

(c) r-selected beetles reproduce quickly and in large numbers, so large protein output in a short time. *(1)*

(d) Chicken causes less damage: lower GWP and less land than pork (less greenhouse gas, deforestation). *(1)*

(e) Beef's footprint is higher because cattle also emit methane, a strong greenhouse gas, and need more land. *(1)*

(f) Grazing near streams causes erosion and turbidity, and manure adds nutrients and bacteria. *(1)*

(g) Rotational grazing, fenced riparian zones, alternative water supply, or less beef in the diet. *(1)*

(h) Terraces, contour farming, no-till, cover crops or perennial crops. *(1)*

(i) Added advantage: more soil carbon, lower fuel and cost, farmable steep land, better infiltration. *(1)*

(j) Drought, flooding, heat, shifted rain seasons or expanded pests reduce yields. *(1)*

Unit 6 — Energy Resources and Consumption

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: E. The world's largest hydroelectric facility, the Three Gorges Dam, spans the Yangtze River, one of the longest in the world, in eastern China. That location is the keyed point E on the map. Dams convert the potential energy of stored water into electricity but also flood land and displace people.

Question 2 — 2008

Official key: E. Solar energy is renewable because the Sun's output is continually replenished on human time scales. Coal, natural gas and oil take millions of years to form, and uranium is a finite mined resource.

Question 3 — 2008

Official key: D. At the consumption rates of the time, proven oil reserves were projected to last only a few decades, whereas coal reserves were estimated to last far longer and natural gas somewhat longer than oil. The keyed answer is oil.

Question 4 — 2008

Official key: B. Making steam needs temperatures far above what flat plates can reach, so mirrors or lenses concentrate sunlight onto a small receiver (concentrating solar power). Flat-panel collectors, photovoltaic panels and passive designs cannot reach those temperatures.

Question 5 — 2008

Official key: E. A storage element holds collected heat for use when the Sun is not shining. The insulated tank of hot water in a solar water-heating system does that, while panels, overhangs, windows and tracking mirrors collect, block or redirect sunlight but do not store energy.

Question 6 — 2008

Official key: A. Ethiopia: 0.20×10^{10} kWh ÷ 70×10^{6} people ≈ 29 kWh per person. U.S.: $400 \times 10^{10} \div 300 \times 10^{6} \approx$ 13,300 kWh per person. The ratio is about 0.2%.

Question 7 — 2008

Official key: D. Natural gas burns more cleanly, giving lower CO_2, NO_x, particulates and CO per unit energy than gasoline. It is not renewable, still emits CO_2 (not just water), and fueling infrastructure is limited.

Question 8 — 2008

Official key: E. Reading the graph, oil use rose from roughly 1,500 to about 3,000 million tons of oil equivalent, a much larger absolute rise than for coal or natural gas. So the increase in oil outpaced both.

Question 9 — 2008

Official key: C. $1/16 = (½)^4$, so four half-lives have passed. 200 million years ÷ 4 = 50 million years.

Question 10 — 2008

Official key: B. Refining uses fractional distillation: heated crude is separated in a column according to the boiling points of its components, with light gases rising highest and heavy residues staying lowest.

Question 11 — 2008

Official key: B. Current demand: East 50,000×10 kW = 500 MW; West 75,000×4 kW = 300 MW; Center 200,000×5 kW = 1,000 MW; total 1,800 MW. In 5 years West ≈ 327 MW and Center ≈ 1,188 MW, so total ≈ 2,015 MW > 2,000. The plant lasts about 5 years.

Question 12 — 2008
Official key: C. West Fremont uses 75,000 × 4 kW = 300 MW. Capacity is 2,000 MW, so 300 ÷ 2,000 = 15%.

Question 13 — 2008
Official key: A. The recommended approach is deep geologic burial at a remote site (Yucca Mountain was designated), isolating waste for the thousands of years it stays hazardous.

Question 14 — 2008
Official key: B. The United States holds the largest proven coal reserves of the listed choices.

Question 15 — 2008
Official key: D. More cars on the road would raise total oil use. The other choices would reduce consumption.

Question 16 — 2010
Official key: B. Geothermal plants bring hot fluid up from underground, and that fluid can carry dissolved salts and gases that corrode equipment and must be handled. Steam release also makes noise. The other sources lack this combination.

Question 17 — 2010
Official key: C. Biomass, mainly wood, charcoal and dung burned for cooking and heat, is the main fuel for many poor households. Wealthier countries rely far more on fossil fuels, nuclear and large hydropower.

Question 18 — 2010
Official key: A. Photovoltaic cells use the photoelectric effect in semiconductors to turn light into an electric current with no heat engine in between. Biomass and hydropower store solar energy indirectly.

Question 19 — 2010
Official key: E. Energy = power × time = 2.0 W × 8,600 h = 17,200 Wh. Dividing by 1,000 gives 17.2 kWh, closest to 17.

Question 20 — 2010
Official key: E. The dam traps nutrient-rich sediment that once fertilized the floodplain and fed the delta's fisheries. Loss of that silt is an ecological harm; power supply is a benefit, and the other options are not ecological evidence.

Question 21 — 2010
Official key: B. In the graph the coal curve peaks near 1910, higher than at the other listed years. The 1880 and 1940 values are lower, and by 1970 and 2000 oil, gas and other sources had taken over.

Question 22 — 2010
Official key: C. The oil-and-gas curve crosses above coal around 1940. Before that, coal supplied the larger share, so earlier years do not fit.

Question 23 — 2010
Official key: C. Nuclear plants produce electricity, but oil is also needed for transportation fuels and petrochemicals that electricity does not directly replace. The other options exaggerate or misstate facts.

Question 24 — 2010
Official key: C. Widespread deforestation reduced wood supply, and coal became the replacement. Oil fields and warming did not cause the decline. Coal's energy density is actually higher than wood's, so option E is reversed.

Question 25 — 2010
Official key: C. Fish ladders allow fish to pass the dam. Repellents, harvesting before the dam or dredging do not let migrating fish move past the barrier.

Question 26 — 2010
Official key: E. Cogeneration captures waste heat from electricity production for another use such as heating. Using one plant to supply both electricity and heat raises overall energy efficiency.

Question 27 — 2016
Official key: E. The Middle East holds the largest concentration of proven oil reserves and has led global petroleum exports for decades. The keyed region is the one over the Arabian Peninsula and Persian Gulf area.

Question 28 — 2016
Official key: C. A hydrogen fuel cell combines hydrogen with oxygen electrochemically, producing electricity and water; the electricity runs the motor. The other descriptions fit hybrids, biodiesel, solar cells or steam turbines instead.

Question 29 — 2016
Official key: C. Regenerative braking turns the car's kinetic energy into electricity stored in the battery, which later helps drive the motor. This recaptures energy that conventional brakes waste as heat. Hybrids still burn gasoline and are not solar powered.

Question 30 — 2016
Official key: B. The data show coal generation falling around 2008 as natural gas and renewables grew; this competition is the supported explanation. The other options attribute trends to causes (shortages, costs, laws) that the graph cannot show.

Question 31 — 2016
Official key: E. Compared with coal, nuclear fission produces no CO_2 at the plant. Waste storage, accident risk, high capital cost and warm-water discharge are its drawbacks, which is why the other options are wrong.

Question 32 — 2016
Official key: A. Fracturing makes cracks in tight rock formations and the sand props them open so trapped hydrocarbons flow to the well. It does not pulverize rock, reopen mines, cause intended earthquakes or scrub rock by abrasion.

Question 33 — 2016
Official key: B. On the decay graph, find where the amount reaches half its starting value, here 0.20 to 0.10 mol. Reading across at 0.10 mol gives about 7 seconds. Check: at 14 seconds the amount should be about 0.05 mol, which matches the plotted curve.

Question 34 — 2016
Official key: B. Photovoltaic cells produce electricity only when light strikes them, so output drops under clouds and stops at night. They have no moving parts and emit no CO_2 in operation, which are benefits, and they can feed the grid.

Question 35 — 2016
Official key: C. Fuel use scales with distance divided by efficiency, and CO_2 scales with gallons burned. The more efficient car saves 1,000 gallons, which is 20,000 lb of CO_2 at 20 lb/gal.

Question 36 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Hot-spot and rift volcanism heat groundwater near the surface, supporting geothermal energy. Solar, wind and nuclear do not depend on local geology in this way.

Question 37 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Wood is widely available and cheap to gather, so it is the practical fuel where other sources are costly. It is not low-impact, does not convert to anthracite, and combustion does release greenhouse gases.

Question 38 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Fracking fluids can contaminate groundwater and the process releases VOCs. Mining waste called slag and overburden belong to other extraction methods, and waste heat does not help soil organisms.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Biomass is renewable but combustion releases CO_2 (and some other gases). Passive solar and hydropower have no combustion emissions, and nuclear is not classed as renewable.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Annual cost = monthly kWh × 12 months × price per kWh, with units kWh/month × month/year × $/kWh = $/year. Dividing by price or by days gives the wrong units.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Hot water from deep wells heated by magma and returned through an injection well is geothermal energy.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Groundwater is heated by near-surface magma or hot rock below it. Pressure release does not heat water and the injection water is cooler.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Hot water is brought up, converted to steam, and the steam turns a turbine connected to a generator. There is no waterwheel or fuel cell.

Question 44 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Photovoltaic cells produce electricity with minimal greenhouse gas release during operation. Other options are false or overstated, since solar farms can disrupt desert habitat and costs are not clearly lower.

Question 45 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The roof panels (A) contain photovoltaic cells that convert light directly into direct-current electricity. The battery, inverter and outlet handle storage, conversion to AC and use.

Question 46 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** An inverter changes direct current into alternating current, which household appliances use. Panels generate DC and the battery only stores it.

Question 47 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The storage battery holds energy produced when sunlight is available so it can be drawn on later. The panels produce nothing at night and the inverter and outlet do not store energy.

Question 48 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Wind power is intermittent, so no energy is generated when the wind does not blow. The other options describe minor or incorrect effects.

Question 49 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Fuel heats water to steam, steam turns a turbine, and the turbine drives a generator. The other options reverse the roles or omit the turbine.

Question 50 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Solar farms can degrade large tracts of desert habitat and carry high costs. Sun is renewable.

Question 51 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Hydroelectric, solar and geothermal are renewable. Oil, coal, natural gas and uranium are not.

Question 52 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Radioactive iodine concentrates in the thyroid, so thyroid cancer cases rise. Mesothelioma, malaria and ozone-related lung effects are unrelated.

Question 53 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** (90 − 25) ÷ 25 × 100 = 260%.

Section B — FRQ Solutions

Question 1 — 2002

(a) Two benefits, each with its reason. (i) Electric cars emit no tailpipe CO_2, so less greenhouse gas enters the atmosphere per mile driven at the vehicle, slowing warming if the grid is cleaner. (ii) Less crude oil is needed, so there is less drilling, spill risk and habitat disturbance. Also acceptable: no tailpipe NO_x/CO/VOC in cities, lower noise. *(commentary indicates 4 points for part (a): each benefit identified and explained; vague "less pollution" earns nothing)*

(b) Cars converted = 0.10 × 150,000,000 = 1.5×10^7 cars. Fuel per car per year = 10,000 mi ÷ 25 mi/gal = 400 gal/yr. Savings = $1.5 \times 10^7 \times 400$ gal = 6×10^9 gal of gasoline per year. *(setup with units, then answer; the official answer is gasoline, 6 billion gallons per year)*

(c) The statement is largely true: gasoline cars are many small *dispersed (mobile, non-point-like) sources*, whereas generating the electricity is concentrated at a few *point sources*, the power plants. A defender notes that a coal- or gas-fired grid still emits SO_2, NO_x, CO_2 and particulates; a refuter notes that a solar-, wind- or hydro-supplied grid would emit almost none, and that point sources are easier to regulate and fit with scrubbers. Either position earns credit if the point/dispersed distinction is made and defended. *(1–2 points)*

(d) Two policies, each explained: tax credits or rebates lowering the purchase price; funding for battery and charging-network research and build-out; preferential parking or lane access; stricter fleet fuel-economy or zero-emission mandates; higher fuel taxes. *(1 point each)*

Question 2 — 2004

(a) 12 MW = 12,000 kW; 12,000 kW × 8,000 h/yr = 9.6×10^7 kWh/yr. *(2 points)*

(b) 3,000 homes × 8,000 kWh/yr = 2.4×10^7 kWh/yr. *(2 points)*

(c) The plant could produce about four times what the homes use. Differences are expected: the plant is sized for peak demand and growth, supplies businesses, loses energy in transmission, and rarely runs at full capacity all year. *(1 point comparison/reason, plus an elaboration point)*

(d) Cost = 10 turbines × $\$3 \times 10^6 = \$3 \times 10^7$. Energy used over 25 yr = 2.4×10^7 kWh/yr × 25 = 6×10^8 kWh. $\$3 \times 10^7 \div 6 \times 10^8$ kWh = **$0.05 per kWh** (the answer based on community use). *(setup 1, answer 1)*

(e) Benefits: no SO_2/NO_x (less acid deposition and smog); no CO_2 from operation (slower warming); no mercury or ash; no coal mining damage. Costs: bird and bat collisions; noise; visual impact on ridges; land clearing, habitat fragmentation and erosion; need for backup power when wind is weak. *(1 point each, 4 total)*

Question 3 — 2004

(a) Low-level waste has low radioactivity and short half-lives (decays to safe levels within a few hundred years), so it can be kept in shielded drums on site or sent to a licensed landfill. High-level waste (spent fuel) is intensely radioactive for tens of thousands of years, so it needs heavily shielded casks or cooling pools and finally deep isolated storage. Isotope example: iodine-131 or technetium-99 (medical use, low level) or plutonium-239/cesium-137 from reactors. *(1 + 1 + 1)*

(b) Ideal site: stable geology away from faults and volcanoes; dry climate with little groundwater contact; deep (hundreds of meters) in non-reactive rock; remote from population; large enough. *(3 points, one each)*

(c) Options and limits: burial in polar ice (ice instability, treaties); launch into space (cost and launch-accident risk); seabed or subduction-zone disposal (leaks, treaties); reprocessing (cost, proliferation risk, law). *(1 identify + 2 feasibility points)*

(d) Cancer: ionizing radiation breaks DNA or damages chromosomes, causing mutations that lead to tumors. Other accepted: cataracts, sterility, birth defects, weakened immune system. *(1 point)*

Question 4 — 2005

(a) 10×10^9 barrels ÷ 20×10^6 barrels/day = 500 days. *(setup 1, answer 1)*

(b) Permafrost keeps meltwater and spills near the surface and is easily damaged by heat or vehicles; a short, cold growing season and thin nutrient-poor soil mean slow decomposition and recovery, so scars persist for decades. *(2 points)*

(c) Roads and drilling pads: destroy and fragment habitat, disrupt caribou migration and nesting. Pipelines or seismic surveys: spill risk, disturbance to dens, displacement of wildlife. *(2 activities + 2 impacts)*

(d) Transportation fuel: use mass transit, higher-efficiency or electric vehicles. Heating or plastics/petrochemicals: insulation, recycling, substitution with other energy. *(1 each, use linked to measure)*

Question 5 — 2006

(a) Benefit: no combustion emissions (CO_2, SO_2) or thermal pollution during operation. Cost: manufacturing uses energy and toxic materials, batteries or panels create waste, and large arrays need land. *(1 each plus elaboration)*

(b) Choose grid-connected: no batteries needed, grid supplies backup at night and in peak demand, a smaller system suffices, and net metering lets surplus be sold; cons: hookup cost and outage dependence. The stand-alone system's strength (independence) is outweighed by battery cost and capacity limits. *(3 points: clear choice, supported statements outweighing the other side)*

(c) Tax credits or subsidies; low-interest loans; research funding; mandatory net metering; lower panel prices by manufacturers. *(1 each)*

(d) Deciduous trees on the sun side shade in summer and admit winter sun; roof overhangs; thermal mass floors storing daytime heat; south-facing windows. Each reduces heating or cooling energy bills. *(1 + 1 each)*

Question 6 — 2008

(a) Two conditions, e.g., high birth rates because children provide farm labor and old-age support; high infant mortality encourages larger families; limited education and family-planning access for women; a larger share of the population is of reproductive age. *(1 point each; 1 extra for country-specific detail.)*

(b)(i) LDC energy: $(5.2 - 1.3) \times 10^{17} \div 1.3 \times 10^{17} \times 100 = 3.9 \div 1.3 \times 100 =$ **300% increase**. *(1 point setup; 1 point answer.)* (b)(ii) MDC energy: $(3.0 - 2.5) \times 10^{17} \div 2.5 \times 10^{17} \times 100 = 0.5 \div 2.5 \times 100 =$ **20% increase**. *(1 point setup; 1 point answer.)*

(c) LDC energy use grows faster because living standards rise (more appliances, vehicles, industry) and the LDC population grows far more (5.1 → 7.8 billion versus 1.2 → 1.3 billion). MDCs already use much energy and can reduce consumption with efficiency, conservation and higher prices. *(1 point each.)*

(d) Fuel is burned to heat water into high-pressure steam; the steam spins a turbine; the turbine turns a generator whose coils rotate in a magnetic field, inducing an electric current. *(up to 3 points.)*

(e) Two sustainable sources, e.g., wind turbines, photovoltaic or solar thermal, geothermal, tidal/wave, sustainable biomass, hydroelectric; each with a short description. *(1 point each; 1 extra for added detail.)*

Question 7 — 2011

(a) *(2 pts)* Hot water or steam from underground reaches the surface: either flashing high-pressure hot water to steam, piping dry steam directly, or using the hot water to boil a second fluid. The steam spins a turbine, and the turbine drives a generator that makes electric current.

(b) *(1 pt)* Total emissions are per-capita use multiplied by population; China's population is so large that a small per-person figure still gives the world's biggest total.

(c) *(issue 1 pt, linked health impact 1 pt)* A specific pollutant, for example SO_2 from burning coal (irritates lungs and worsens asthma and bronchitis), fine particulates (reduce lung function) or mercury (damages the nervous system). "Smog" alone is not specific enough.

(d)(i) *(2 pts, extraction + processing)* Sugarcane: harvest and crush the cane to extract sucrose juice, then ferment it to ethanol. Tar sands: surface-mine the sand, then heat it with steam or hot water to separate the thick bitumen, which is upgraded to fuel.

(d)(ii) *(2 pts)* Tar sands: habitat destruction from mining, high water and energy needs with toxic wastewater, greenhouse gases from a fossil fuel. Sugarcane: land clearing and monoculture, fertilizer runoff, water use, competition with food crops.

(d)(iii) *(2 pts)* Sugarcane, because it is renewable and replants each season, and the carbon it releases was recently taken up from the air, unlike the fossil carbon in tar sands, which also causes far more land and waste damage.

Question 8 — 2012

(a) *(4 possible points: each problem identified and described; the question's total is capped at 10)* Two water problems, each stated and explained: groundwater contamination (well leaks or lagoons let chemicals and methane reach aquifers); surface-water contamination (spilled or sprayed wastewater with salts and metals runs into streams); and heavy water use that can deplete aquifers and compete with farms and towns.

(b) *(2 pts)* Burning natural gas gives less SO_2 (less acid rain), less NO_x, fewer particulates and less mercury than coal, and about half the CO_2 per unit of energy; gas also avoids strip mining and mountaintop removal.

(c) *(2 pts, not about water)* Clearing land and roads fragments habitat; methane leaks add a strong greenhouse gas; drilling can trigger small earthquakes or land subsidence; diesel trucks and rigs emit CO_2, NO_x and particulates and make noise.

(d) *(1 pt)* A domestic energy supply, new jobs, or lease income for landowners.

(e) *(2 pts)* Long-lived radioactive waste with no permanent site; accident risk releasing radiation to air, soil and water; thermal pollution from cooling water harming aquatic life; uranium mining damage and tailings.

Question 9 — 2012

(a) *(source 1 pt, benefit 1 pt, drawback 1 pt)* One realistic source with both sides, for example: wind turbines (no air pollution, little land loss on farmland; but bird and bat deaths and visual or noise impact); solar panels (no emissions in use; but mining and toxic materials for cells); geothermal heat pump (little habitat disturbance; but pumping energy needed); biomass or biogas (reuses waste; but air pollutants and land demand).

(b) *(2 pts)* Habitat for organisms, biodiversity, water purification and flow regulation, reduced erosion and flooding, soil formation, nutrient cycling, climate moderation.

(c)(i) Carbon stored per hectare = 2.7×10^5 kg biomass/ha × 0.50 = **1.35×10^5 kg C/ha/yr**. *(1 pt.)*

(c)(ii) Carbon emitted = 3.0×10^5 gal × 10 kg CO_2/gal × 0.27 kg C/kg CO_2 = **8.1×10^5 kg C/yr**. *(1 pt setup, 1 pt answer.)*

(c)(iii) Area = 8.1×10^5 kg C ÷ 1.35×10^5 kg C/ha = **6 ha**. *(1 pt setup, 1 pt answer.)*

(c)(iv) Cost = 6 ha × $12,000/ha = **$72,000**. *(1 pt.)*

Question 10 — 2013

(a) Three federal strategies: tax credits or rebates for buying BEVs; funding or mandating public charging stations; higher fuel taxes on gasoline (also: subsidies for manufacturers, fleet purchase requirements, public education). *(3 points.)*

(b) Gasoline cost per mile = price ÷ efficiency = $3.75/gal ÷ 25 mi/gal = **$0.15 per mile**. *(1 point setup with units, 1 point answer.)*

(c)(i) Energy = power × time = 4.0 kW × 7.0 h = 28 kWh; cost = 28 kWh × $0.11/kWh = **$3.08**. *(setup, answer.)* (c)(ii) $3.08 ÷ 100 mi = **$0.031 per mile** (about 3 cents), roughly one fifth of the gasoline cost. *(1 point.)*

(d) ICE total = 72.8 + 17.7 = 90.5 lb CO2; BEV total = 63.6 lb (all upstream, from coal-fired electricity). Difference = 90.5 − 63.6 = **26.9 lb CO2 less for the BEV** per 100 miles. *(1 point.)*

(e) Two economic impacts: higher demand for electricity may raise power prices or require new generating capacity and grid upgrades; charging infrastructure creates jobs and investment, while gasoline stations, refiners and repair shops lose business and revenue. Highway fuel-tax revenue also falls. *(2 points; exclude climate costs and pump price.)*

Question 11 — 2014

(a)(i) Disagree: spent fuel rods and contaminated tools and clothing are radioactive wastes that must be isolated for very long periods. *(1.)* (ii) Either position can be defended if justified: agree because no fossil fuel is burned in operation; disagree because mining, enrichment, transport and construction use fossil energy, and cooling releases water vapor. *(1.)*

(b) Two construction problems: clearing and fragmenting riparian habitat, soil compaction, erosion and sediment runoff, noise and exhaust from machinery, construction waste. *(2.)*

(c)(i) The most likely river pollutant in normal operation is thermal pollution from heated cooling water (not radioactive release). *(1.)* (ii) Warm water holds less dissolved oxygen and raises metabolic rates, causing thermal shock or fish kills and shifting community composition. *(1.)* (iii) Cooling towers, cooling ponds or canals let the water shed heat before discharge. *(1.)*

(d) Two steps: replace old appliances and bulbs with efficient ones (LEDs); raise insulation or adjust thermostats; turn off or unplug idle devices. *(2.)*

(e) Accidents: Three Mile Island, Chernobyl or Fukushima Daiichi. One consequence: radioactive contamination spreading through food webs, killing plants and animals or causing mutations and loss of biodiversity. *(1 + 1.)*

Question 12 — 2017

(a) Water stored behind the dam has gravitational potential energy; released through intakes it falls and converts to kinetic energy, rotating turbine blades (mechanical energy); the turbine shaft spins a generator that converts the mechanical energy to electricity. *(3: one per step.)*

(b) Two benefits: recreation and tourism; reservoir water for irrigation, drinking and industry; jobs; fisheries or shipping. *(2.)*

(c) Seasonal floods deposit nutrient-rich sediment, recharge groundwater and maintain wetland and riparian habitat for fish and birds. *(1.)*

(d)(i) Removal reopens migration routes to spawning habitat and restores natural temperature and flow. *(1.)* (ii) Negatives: stored sediment and pollutants are released downstream, bank erosion, or loss of the lake habitat that has developed. *(1.)*

(e)(i) A keystone species has a disproportionately large effect on its ecosystem relative to its abundance. *(1.)* (ii) Beaver dams create ponds and wetlands, trap sediment, slow flow and reduce erosion, giving habitat to many other species. *(1.)*

Question 13 — 2018

(a) Benefit: displaces fossil fuel generation, reducing CO_2 and air pollutants and fuel extraction damage; turbine bases can act as artificial reefs. *(1.)*

(b) Identify and describe one effect: job creation in construction and maintenance (positive), or reduced tourism and property values or high construction costs (negative). *(1 identify, 1 describe.)*

(c) Another ocean energy: tidal turbines, wave energy converters, or ocean thermal energy conversion. *(1.)*

(d) $0.80 \times 2.0 \times 10^6$ MWh = **1.6×10^6 MWh**. *(1 + 1.)*

(e) Revenue = 1.6×10^6 MWh × 1,000 kWh/MWh × \$0.20/kWh = 1.6×10^9 kWh × \$0.20 = **\$3.2 × 10^8** (\$320 million). *(1 + 1.)*

(f) Total capacity = 200 turbines × 4 MW = 800 MW; hours = 1.6×10^6 MWh ÷ 800 MW = **2,000 hours**. *(1 + 1.)*

Question 14 — 2019

(a) One ecological service: carbon dioxide uptake, climate and water regulation, erosion control, nutrient cycling, soil formation or maintaining biodiversity. *(1 point)*

(b) One economic benefit: timber and other forest products, tourism and recreation jobs, or lower health costs from cleaner air. *(1 point)*

(c) Two non-habitat consequences: fossil-fuel combustion for machinery and processing releases CO_2, NO_x, SO_2 and particulates; toxic tailings need storage; spills or leaks from pipelines contaminate water; strip mining adds sediment to streams; large volumes of water are withdrawn. *(2 points: 1 each)*

(d) Days = 73×10^9 barrels ÷ (1×10^6 barrels/day) = 7.3×10^4 days = 73,000 days. *(2 points: setup; answer)*

(e) Years = 7.3×10^4 days ÷ 365 days/year ≈ 200 years (2×10^2). *(2 points; an error carried from (d) is not penalized again)*

(f) 30×10^6 barrels synthetic crude/month × 2 barrels water/barrel × 12 months/year = 7.2×10^8 barrels of freshwater per year. *(2 points: setup; answer)*

Question 15 — 2021

(a)(i) Independent variable: the compass direction the panels face. *(1)*

(a)(ii) Dependent variable: electricity produced (kWh). *(1)*

(a)(iii) E.g. "If panels face south (north hemisphere), then they will generate more kWh than panels facing other directions." A null statement of no effect is also acceptable. *(1: testable prediction relating the two variables)*

(a)(iv) An uncontrolled variable such as panel tilt, shading, dust or snow, cloud cover, season, or panel temperature. *(1)*

(b)(i) The turbine, where moving water's kinetic energy becomes rotational mechanical energy. *(1)*

(b)(ii) Water held high behind the dam has gravitational potential energy; releasing it downhill (greater head, more pressure) converts that energy to motion. *(1)*

(b)(iii) Solar and wind output varies with weather and time of day; hydro can be dispatched to fill gaps, and surplus solar or wind power can pump water back uphill into the reservoir. *(1)*

(b)(iv) Drought, evaporation or reduced snowpack lowers reservoir storage and generation; heavy storms increase sediment that shrinks storage; glacier loss first raises then lowers flow. *(1)*

(c)(i) Benefit besides power: flood control, water supply or irrigation, recreation, or lake habitat. *(1)*

(c)(ii) Silt reduces storage volume and the head available, lowering output, and abrasion damages turbine blades, raising repair cost. *(1)*

Question 16 — 2021

(a) Coal forms far more slowly than it is burned, so the supply is finite. *(1)*

(b) Gas emits less CO_2, no mercury or other trace metals, less NO_x, SO_x and particulates, and avoids strip-mining. *(1)*

(c) Economic edge: cheaper pipeline transport, fewer pollution controls, cheaper waste disposal, higher plant efficiency or lower health-care costs. *(1)*

(d) Install CO detectors, maintain and vent furnaces, mandate detectors, run public education, or change heating system. *(1)*

(e) Percent change = (188 − 85) ÷ 85 × 100 = +121% (the table gives 85 for 1990 and 188 for 2016). *(2)*

(f) Lost removal = (50,000 − 43,000) ha × 2.3 kg/ha = 7,000 × 2.3 = 16,100 kg PM2.5 per year. *(2)*

(g) A 23% rise ÷ 10% = 2.3 steps × 1% = 2.3% increase; 7,390 × 0.023 ≈ 170 additional admissions. *(2)*

Question 17 — 2022

(a)(i) Highest methane is 64 mg/L. *(1)*

(a)(ii) Methane concentration falls as distance from the fracking well increases (inverse, nonlinear). *(1)*

(a)(iii) Any distance of about 1,800-2,200 m, where the curve levels near zero. *(1)*

(a)(iv) High-pressure fluid cracks the shale, and sand in the fluid props the fractures open so gas and oil can flow to the well. *(1)*

(a)(v) Induced earthquakes or ground subsidence and sinkholes. *(1)*

(b)(i) Tragedy of the commons. *(1)*

(b)(ii) Falling water tables dry springs and groundwater-fed streams, harm habitats, increase erosion or cause desertification. *(1)*

(b)(iii) Pumping lowers the freshwater head in coastal aquifers so seawater flows in (saltwater intrusion), making water too salty to drink. *(1)*

(c)(i) Claim: raise vehicle fuel-economy standards, subsidize electric vehicles or transit, invest in renewables, or tax fuel. *(1)*

(c)(ii) Justify with a non-climate gain: less smog, particulates and acid rain, or fewer spills and less drilling disturbance. *(1)*

Question 18 — 2023

(a) Spills coat or poison marine life, block light, damage wetlands, or noise disrupts animals; fuel CO_2 adds to acidification. *(1)*

(b) CO_2, NO_x, SO_2, particulates, CO or VOCs. *(1)*

(c) Walk, bike, use transit, carpool, drive an efficient, hybrid or electric vehicle, or telework. *(1)*

(d) Fewer combustion pollutants lowers asthma, COPD and cardiovascular disease; active travel adds fitness. *(1)*

(e) 260×10^6 t × 5.0 g/t = 1.3×10^9 g of gold. *(2)*

(f) 1,000 t × 5.0 g/t = 5,000 g; × \$62.56/g = **\$312,800**. *(2)*

(g) Gold needed = 0.034 g × 100,000 = 3,400 g; ÷ 5.0 g/t = **680 t** of ore. *(2)*

Question 19 — 2024

(a) Pollutants: SO_2, NOx, particulates, mercury or CO. *(1 pt)* (b) Install scrubbers or use low-sulfur coal to reduce SO_2. *(1 pt)* (c) At X, steam turns a turbine (water is boiled to steam, or steam condenses). *(1 pt)* (d) Trains and trucks burn fossil fuels, emitting CO_2. *(1 pt)* (e) Photovoltaic cells convert sunlight directly to electricity, or mirrors concentrate heat to make steam. *(1 pt)* (f) At 60° latitude sunlight strikes at a lower angle and is spread over more area with longer atmospheric path, so output is lower. *(1 pt)* (g) Pesticides can harm nontarget species or cause resistance. *(1 pt)* (h) IPM practice: crop rotation, natural predators, resistant varieties or traps. *(1 pt)* (i) Solution: buffer strips, precise application timing or cover crops. *(1 pt)* (j) Justify: buffer strips also provide habitat and trap sediment. *(1 pt)*

Question 20 — 2024

(a) Uranium (U-235) or plutonium. *(1)*

(b) Cooling discharge causes thermal pollution, lowering dissolved oxygen and exceeding tolerance; intakes kill organisms; evaporation reduces flow. *(1)*

(c) 4.1×10^{12} kWh × 0.189 ≈ 7.7×10^{11} kWh (7.749×10^{11}). *(2)*

(d) Gas gives off far less NO_x, SO_x, ash, particulates, smog precursors and toxic metals. *(1)*

(e) Extra benefit: habitat and biodiversity, jobs, less erosion, shade cooling, or slower runoff. *(1)*

(f) 7.4×10^7 m³ × 4.76 kWh/m³ = 3.52×10^8 kWh; ÷ 10,715 kWh/home ≈ 32,874 homes. *(2)*

(g) Savings per kWh = 1.0 − 0.42 = 0.58 kg CO_2; × 8.99×10^{11} kWh = 5.2×10^{11} kg CO_2 (about 521 billion kg). *(2)*

Question 21 — 2025

(A) Strip mining, open-pit, or mountaintop removal. *(1)* **(B)** Habitat loss, erosion and sedimentation, acid mine drainage or dust. *(1)* **(C)** Downwind exposure to SO_2 and NO_x (acid rain), particulates (respiratory disease), mercury or smog. *(1)* **(D)** Require scrubbers and filters, shift to renewables or gas, fund efficiency, or tax emissions. *(1)* **(E)** Energy = 11,000 homes × 1.064×10^4 kWh = 1.1704×10^8 kWh; ÷ 0.88 kWh/lb = **1.33×10^8 lb of coal**. *(2)* **(F)** Percent change = (6.0 − 7.5) ÷ 7.5 × 100 = **−20%**. *(2)* **(G)** Doubling time = 70 ÷ 5.38 ≈ 13 years; 2022 + 13 = **2035** (26,250 → 52,500 is exactly a doubling). *(2)*

Unit 7 — Atmospheric Pollution

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: D. Ground-level ozone (O_3) is a secondary pollutant: sunlight drives reactions among nitrogen oxides and volatile organic compounds to form it, as in photochemical smog. The other gases are emitted directly or produced by other processes.

Question 2 — 2008
Official key: B. Coal contains sulfur and burning it releases SO_2 (and NO_x), which form sulfuric and nitric acids in the atmosphere. Midwestern coal-fired power plants and prevailing winds deliver this acid deposition to the Northeast. Natural gas and solar produce little or no SO_2.

Question 3 — 2008
Official key: A. Radon gas is produced by the radioactive decay chain of uranium found in rock and soil. Uranium-238 decays through radium to radon-222, which can seep into basements. Coal, gas, oil and sunlight are not the source of radon.

Question 4 — 2008
Official key: E. Unpolluted rain is slightly acidic, about pH 5.6, because atmospheric CO_2 dissolves to form weak carbonic acid. Lemon juice is far more acidic (~2), distilled water is ~7, ammonia is basic, and acid rain is below 5.

Question 5 — 2008
Official key: D. The pH scale is logarithmic: each unit is a tenfold change in H^+ concentration. A difference of 3 units is 10^3 = 1,000 times, so pH 4 has 1,000 times more H^+ than pH 7.

Question 6 — 2008
Official key: D. Wet scrubbers spray water mixed with limestone or lime through the exhaust gas, where SO_2 dissolves and reacts to form a solid. Electrostatic precipitators and filters remove particulates; catalytic converters and afterburners treat vehicle or combustion gases like CO and hydrocarbons.

Question 7 — 2008
Official key: D. The 1990 amendments set up a cap-and-trade program for SO_2 emissions from power plants to reduce acid rain. The act was passed in 1963/1970, has been amended several times, did not originally regulate CO_2, and does not expire.

Question 8 — 2008
Official key: E. Radon enters from soil through cracks and openings, so sealing entry points and ventilating (e.g., sub-slab depressurization) reduces indoor levels. Filtering water, avoiding the basement, monitoring only, or replacing soil does not address entry.

Question 9 — 2008
Official key: C. Photochemical smog needs strong sunlight and large vehicle emissions of NO_x and VOCs, which LA has. Sea-level elevation is not a factor; the basin's mountains and thermal inversions trap pollutants.

Question 10 — 2008
Official key: B. Electrostatic precipitators charge particles and collect them on oppositely charged plates. UV irradiation, catalytic converters, chromatography and aeration don't remove particulates.

Question 11 — 2008
Official key: E. Engines emit CO_2 and water from complete combustion, plus NO_x from high-temperature combustion of air and unburned VOCs. Ozone forms later in the atmosphere, and O, N, Ar and Pb are not direct hydrocarbon products.

Question 12 — 2010
Official key: C. Indoor pollutants such as radon, smoke, mold and volatile chemicals harm the lungs and cause headaches and fatigue. Skin cancer comes from ultraviolet exposure, not from indoor air, so it is the exception.

Question 13 — 2010
Official key: B. Nitrogen dioxide is the reddish-brown gas formed from vehicle and combustion emissions, and sunlight drives its reactions to form ozone. Ozone, CO and methane are colorless.

Question 14 — 2010
Official key: B. Acidification mobilizes aluminum and other heavy metals from soils, so metal levels rise. Fish richness drops and dissolved oxygen does not increase, so only statement II is expected.

Question 15 — 2010
Official key: D. Adding calcium carbonate (liming) neutralizes the acid and raises pH quickly. Taller smokestacks move pollution elsewhere, and fertilizer controls address eutrophication, not acidity.

Question 16 — 2010
Official key: A. Acid deposition leaches nutrients and damages needles, so trees are weakened and tolerate cold less well. It increases hydrogen ions and lowers soil buffering, the reverse of the other options.

Question 17 — 2010
Official key: E. Radon seeps from soil and rock into buildings and accumulates where ventilation is limited. Tightly insulated homes trap it indoors.

Question 18 — 2010
Official key: C. Sulfur oxides react with water in the atmosphere to form sulfuric acid, causing acid deposition. The other pairings are mismatched.

Question 19 — 2010
Official key: A. Primary pollutants are emitted directly, and incomplete combustion releases carbon monoxide. Ozone and sulfuric acid are secondary, formed later in the atmosphere.

Question 20 — 2010
Official key: A. Acid deposition mobilizes toxic metals such as aluminum in soil that harm roots, along with nutrient leaching. The other options are not the main factor.

Question 21 — 2010
Official key: C. pH 4.4 versus 5.4 differ by one unit, and each pH unit is a factor of ten in hydrogen ion concentration. Hydrogen ion concentration ratio = 10^(5.4 − 4.4) = 10.

Question 22 — 2010
Official key: B. Burning sulfur-containing fossil fuels, especially coal for electricity, releases SO2 that forms sulfuric acid.

Question 23 — 2016
Official key: A. Radon that collects in basements and poorly ventilated rooms is inhaled, and its radioactive decay products damage lung tissue. It is the largest source of lung cancer among non-smokers. The listed metals harm other organs, such as the nervous system, rather than being the main cause of lung cancer.

Question 24 — 2016
Official key: B. A catalytic converter uses metal catalysts to turn carbon monoxide and unburned hydrocarbons into CO_2 and water (and also to convert NO_x to N_2). It does not remove CO_2, lead, mercury or particulates, so the pair CO and hydrocarbons is correct.

Question 25 — 2016
Official key: C. A thermal inversion occurs when a warm layer sits above cooler air, stopping the cooler air from rising so pollutants stay near the ground in a shallow mixing depth. Albedo, greenhouse effect and heat islands do not describe this layering.

Question 26 — 2016
Official key: A. Acid rain is neutralized by bases, and carbonate rock is the common natural buffer. Soils over limestone resist acidification, whereas soils over granite are poorly buffered and lakes in those areas acidify.

Question 27 — 2016
Official key: D. Nitric oxide is emitted directly (primary pollutant) by engines, so it rises with morning traffic. It is then oxidized to NO_2 and consumed in smog chemistry, so its concentration falls through the day.

Question 28 — 2016
Official key: D. Hydrocarbons (VOCs) and NO_x react in sunlight to form ozone, so removing a reactant cuts ozone production. NO would not rise because it is not produced from hydrocarbons, and CO_2 is not tied to this change.

Question 29 — 2016
Official key: D. Ground-level ozone is a secondary pollutant made when NO_x and VOCs react in sunlight. Reducing NO_x lowers production, while sunny days, inversions and more VOCs raise it. CFCs destroy stratospheric ozone and do not affect ground-level ozone in this way.

Question 30 — 2016
Official key: D. Fine particles lodge deep in the lungs and trigger inflammation and breathing problems such as asthma and bronchitis. Cataracts and melanoma are linked to UV radiation, not particulates.

Question 31 — 2016
Official key: A. Fine particulate matter (PM2.5) from burning coal and other fuels penetrates deep into lungs causing respiratory disease. The other pairs link unrelated causes and effects, such as sewage with skin cancer.

Question 32 — 2016
Official key: A. During inversions, pollutants including fine particulates are trapped near the surface, and breathing them irritates the respiratory system. Ozone, mold, CO_2 and pressure are not the main inversion problems.

Question 33 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** A thermal inversion is a layer of warm air trapping cooler air below it, reversing the usual cooling with height. It traps pollutants near the ground.

Question 34 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Radon, a radioactive gas seeping from soil and rock into buildings, is the second leading cause of lung cancer after smoking and acts through chronic exposure. CFCs, mercury and carbon monoxide are not lung-cancer causes in this way.

Question 35 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Limiting vehicle traffic in parks and preserves cuts noise where the birds live. Electric urban buses and efficiency do not target habitats, and swapping solar for wind does not reduce noise.

Question 36 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Geothermal fluids commonly carry hydrogen sulfide gas. Uranium-235 is nuclear fuel, NO comes from combustion and mercury is not typical here.

Question 37 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** 200,000 tons × 900,000 g/ton = 1.8×10^{11} g; ÷ 2 g/gal = 9.0×10^{10} gallons.

Question 38 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Multiply 0.021 g/mile × 250 miles/month × 12 months/year ≈ 63 g/year. Other options have inverted fractions.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The graph shows all six pollutants lower than their starting index, supporting regulation effectiveness. PM2.5 did exist before 2000 and the other options make claims the graph does not show.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** CO comes from incomplete combustion at the stove, so replacing it with an electric stove removes the source. Catalytic converters, glass cookware or smoke detectors do not stop CO.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The Clean Air Act set national air-quality standards and required regulation of emissions from power plants, factories and vehicles. The Montreal Protocol and Kyoto Protocol are international agreements.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Noise causes animals to avoid noisy areas, altering migration and habitat use. The other options attribute unrelated effects such as litter size or vegetation loss.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Acid rain forms from SO_2 and NOx, and coal-fired plants are a major source. Nuclear, wind and natural gas generation emit little or none.

Question 44 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Highways show the tallest NO bar.

Question 45 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** O_3 is higher in areas with less traffic.

Question 46 — 2024

(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Catalytic converters reduce NOx in exhaust.

Section B — FRQ Solutions

Question 1 — 2018

(a)(i) Two pollutants: carbon monoxide, particulate matter (also NO_x, SO_x, VOCs). *(2.)* (ii) Illness: particulates are linked to asthma, bronchitis, COPD or lung cancer; CO causes poisoning. *(1.)*

(b) Approach: install a chimney or ventilation, or switch to a cleaner-burning stove or fuel. Why it works: pollutants are vented out or less is produced, so indoor exposure and respiratory illness fall. *(1 + 1.)*

(c) Young children breathe more air per kilogram of body mass, have developing lungs and immune systems and narrower airways, and spend more time indoors. *(1 for one reason.)*

(d) Pick two. Asbestos: source is old insulation, tiles or building materials; reduce exposure by sealing or leaving undisturbed, or professional removal. Radon: source is uranium-bearing rock and soil seeping through the foundation; reduce by sealing cracks and installing a sub-slab vent. Mold: source is damp or flooded areas; reduce by repairing leaks, ventilating or dehumidifying and cleaning. *(1 per source, 1 per method; 4.)*

Question 2 — 2018

(a)(i) Any two indoor combustion pollutants: carbon monoxide, particulate matter (soot), nitrogen oxides, sulfur oxides, volatile organic compounds, methane, or trace metals. *(2 points: 1 each)*

(a)(ii) Link one pollutant to a named respiratory illness: fine particulates to COPD, asthma, bronchitis or lung cancer; CO to carbon monoxide poisoning. *(1 point: specific illness matched to a pollutant you listed)*

(b) Pair an approach with its health effect, e.g. a chimney or vented stove sends smoke outdoors, so less is inhaled; a cleaner, more efficient stove burns less fuel and releases fewer pollutants; switching to biogas, solar or electric cooking removes the smoke source; cooking outdoors moves the smoke away from the home. *(2 points: realistic approach; how it lowers respiratory illness)*

(c) Young children breathe in more air per kilogram of body mass, have immature immune systems and narrow airways, and spend more time indoors near the stove, so the dose and effect are larger. *(1 point: one valid reason)*

(d) Choose two: asbestos source = old insulation, floor tiles, cement or ceiling products; control = leave undisturbed, seal or enclose, or have it professionally removed. Radon source = uranium-bearing soil and rock whose gas seeps through foundations; control = seal cracks and install subfloor venting. Mold source = leaks, flooding or damp, poorly ventilated rooms; control = fix the leak, dry or ventilate the space, clean or replace damaged material. *(4 points: a source and a control for each of two pollutants)*

Question 3 — 2019

(a) Sequoia National Park shows the largest loss of visibility versus the natural baseline. *(1 point)*

(b)(i) Primary pollutant example: SO_2, NO_x, CO, particulate matter or VOCs. *(1 point)*

(b)(ii) Primary pollutants enter the air directly from a source such as a smokestack, tailpipe or burning fuel. *(1 point)*

(b)(iii) Secondary pollutant example: ozone, sulfuric acid, nitric acid or PANs. *(1 point)*

(b)(iv) Secondary pollutants form when primary pollutants react in the air, e.g. NO_x and VOCs react in sunlight to make ozone, or SO_2 and water form sulfuric acid. *(1 point)*

(c)(i) Percent increase = (45 mi − 25 mi) ÷ 25 mi × 100 = 80%. *(1 point)*

(c)(ii) Two actions: tighten emission standards or cap-and-trade for utilities; incentivize renewable energy; limit private vehicles with shuttles or parking fees; support electric vehicles; restrict campfires or burning. *(2 points: 1 each)*

(d) Two non-air impacts: litter harming wildlife or water; trampling and off-trail travel causing erosion and compaction; roads and facilities fragmenting habitat; noise or light disrupting animal behavior; visitors bringing invasive species; wastewater pollution. *(2 points: 1 each)*

Question 4 — 2025

(A) E.g. coal-fired power plants, industrial combustion, construction, agriculture tilling or open burning. *(1)* **(B)** The nozzle captures gasoline vapors (VOCs) during fueling so they do not escape to form smog. *(1)* **(C)** Fewer car trips burn less fuel, cutting NO_x and SO_2 emissions, which form nitric and sulfuric acids in rain. *(1)* **(D)** Percent change = (36 − 22) ÷ 22 × 100 ≈ +63.6%. *(2)* **(E)** 14 gal × (36 − 22) mi/gal = **196 more miles**. *(2)* **(F)** E.g. insulation, efficient HVAC or smart thermostats, window upgrades, reflective roofing, or passive solar design. *(1)* **(G)** 2.8×10^4 bulbs × 0.0085 kW × 2,340 h/yr = **5.57×10^5 kWh per year** (about 556,920). *(2)*

Unit 8 — Aquatic and Terrestrial Pollution

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: C. Decomposers consume oxygen as they break down sewage, so dissolved oxygen is lowest where its curve bottoms out, at point C, which is where fish suffocate. Point B is peak BOD, and farther downstream the stream recovers as DO climbs back.

Question 2 — 2008

Official key: E. Eutrophication is caused by excess nutrients, so adding nitrates would worsen it. Herbicides, dredging, aeration with oxygen and biological control of nuisance plants are all ways of managing the symptoms.

Question 3 — 2008

Official key: C. Waste supplies nutrients that let algae multiply, but the large algal population then dies and decomposes, and aerobic decomposers consume the dissolved oxygen. More waste means more decomposition and lower DO.

Question 4 — 2008

Official key: C. Repeating the experiment several times and comparing results (replication) reduces the effect of chance and shows whether findings are consistent. Dropping the control, changing the water type, varying algae or waste type would introduce extra variables.

Question 5 — 2008

Official key: C. Phosphorus is typically the scarcest nutrient in freshwater, so adding it (from detergents, fertilizer, sewage) triggers algal blooms. Oxygen is not a nutrient for production, and calcium, carbon and iron are rarely limiting there.

Question 6 — 2008

Official key: B. Fertilizer nitrogen runs into the Mississippi River and flows to the Gulf of Mexico, where it fuels algal blooms. Their decomposition depletes oxygen, producing a hypoxic dead zone.

Question 7 — 2008

Official key: E. CERCLA (Superfund) authorizes the EPA to clean up abandoned or uncontrolled hazardous waste sites and to bill responsible parties. The other acts regulate air, water, drinking water, or current hazardous waste management.

Question 8 — 2008

Official key: C. Burning coal releases mercury that occurs naturally in coal, so coal-fired power plants are the largest human source of atmospheric mercury. The other sources are smaller.

Question 9 — 2008

Official key: E. The sequence is screening out debris (P), adding a flocculant (S), letting the clumps settle (R), then disinfecting (Q) last to kill remaining pathogens.

Question 10 — 2008
Official key: C. Biomagnification is the increase in concentration of persistent fat-soluble toxins at each higher trophic level, because predators eat many contaminated prey.

Question 11 — 2008
Official key: C. Standard treatment is designed for solids, organic matter and pathogens, not for removing pharmaceutical compounds, which pass through into receiving waters. Plants remove solids first, are regulated by the Clean Water Act, and have not eliminated eutrophication in Chesapeake Bay.

Question 12 — 2008
Official key: E. Wetlands are identified by all three: water saturating or covering the land for part of the year, hydric soils, and vegetation adapted to saturated conditions.

Question 13 — 2008
Official key: D. The idea implies harm grows with dose. Here the response decreases as concentration increases, which is the opposite, so the data do not support the concept.

Question 14 — 2008
Official key: B. As concentration rises, percent induction falls (an inverse relationship), so the statement saying induction decreases with higher concentration is correct.

Question 15 — 2008
Official key: D. Smaller particles have a larger surface area per unit volume, exposing more material to water, so more contaminant dissolves.

Question 16 — 2008
Official key: A. Paper and paperboard are the largest category of MSW by weight, larger than yard waste, food, plastics or glass.

Question 17 — 2008
Official key: B. Landfilling handles more municipal solid waste than incineration, composting or recycling, and ocean dumping is banned.

Question 18 — 2008
Official key: E. Agricultural runoff comes from diffuse areas, not a single identifiable pipe or site. The other options are single, identifiable sources.

Question 19 — 2008
Official key: A. RCRA regulates hazardous waste from generation through disposal. CERCLA deals with cleanup of abandoned sites.

Question 20 — 2008
Official key: A. More people moving quickly between continents spreads pathogens, so infectious disease spread is the most direct effect. Sprawl, biodiversity loss, hypoxia and fertility are indirect or unrelated.

Question 21 — 2008
Official key: C. DDT is persistent and fat-soluble, and imported produce from countries still using it carries residues. DDT is not water soluble, and illegal use in the U.S. is not the main explanation.

Question 22 — 2008
Official key: C. Survival is 0 at pH 3.2 and 0.6 at pH 5.3, so interpolating, 0.5 falls at roughly 4.5–5. The closest choice is 4.5.

Question 23 — 2008
Official key: A. A control provides baseline conditions to compare with, showing the effect of random environmental factors apart from the independent variable.

Question 24 — 2008
Official key: D. Survival is greatest near pH 7.4, so the species is best adapted to near-neutral water.

Question 25 — 2010
Official key: A. Organic waste feeds decomposers, which consume dissolved oxygen, so biochemical oxygen demand rises. Dissolved oxygen then falls and oxygen-sensitive fish decline, and coliform counts rise.

Question 26 — 2010

Official key: B. Gray water is wastewater from sinks, showers and laundry, which has no toilet waste and can be reused for irrigation. Toilet and garbage-disposal water is black water.

Question 27 — 2010

Official key: D. A steady rise in effect with dose is a linear dose-response relationship. A threshold would show no effect up to a certain dose, which the steady increase does not suggest.

Question 28 — 2010

Official key: D. Persistent chemicals biomagnify, so concentrations are highest in higher trophic levels. Secondary consumers carry larger doses than producers or decomposers, and hormone mimics disrupt their reproduction.

Question 29 — 2010

Official key: C. Plastics break down very slowly, if at all, so they persist in landfills and the environment. They do not readily release methane, ozone or radioactive isotopes.

Question 30 — 2010

Official key: D. Erosion and runoff from farmland come from a broad area with no single outlet. Landfill seepage, effluent pipes and sewage overflow are point sources. Point sources, by contrast, discharge through an identifiable pipe or leak.

Question 31 — 2010

Official key: D. Neurotoxin exposure affects nervous or endocrine function and does not spread infection. Flooding, warming, insect vectors and urban crowding all do.

Question 32 — 2010

Official key: D. It is spread by bites from infected mosquitoes. It is not acquired from food, water or direct contact with sick animals.

Question 33 — 2010

Official key: D. Animal models may not reflect human responses. Differences in metabolism and body size between rats and people make extrapolation uncertain.

Question 34 — 2016

Official key: C. Tetraethyl lead was added to gasoline for decades to prevent engine knock, and exhaust spread lead widely until leaded fuel was phased out. Lead is the metal that matches. Radon is a gas, mercury is tied to coal burning and fish, and copper and iron were never fuel additives.

Question 35 — 2016

Official key: A. Radon-222 forms when uranium decays in rock and soil and is a colorless gas that can move up into buildings. None of the metals listed are gases at ordinary temperatures, so radon is the only fit.

Question 36 — 2016

Official key: B. Mercury, released mostly from coal burning, deposits in water, is converted to methylmercury and biomagnifies up the food chain. Top predators such as swordfish and tuna carry the highest concentrations, so heavy consumers risk neurological damage.

Question 37 — 2016

Official key: E. Household hazardous wastes contain toxic, flammable or corrosive substances and need special disposal. Food scraps are organic and can be composted or landfilled, so they are the exception.

Question 38 — 2016

Official key: C. Some herbicides mimic or interfere with hormones; as endocrine disruptors they can alter amphibian development and reproduction. The thin permeable skin of amphibians makes them especially sensitive.

Question 39 — 2016

Official key: C. Compare each site with the unaffected reference sites (B and D), which have clear water and low solids. Rain on bare construction soil also carries sediment but not acid, so pH stays near neutral, unlike the acidified Site E (pH 4.0).

Question 40 — 2016
Official key: C. Runoff from roofs, roads and construction carries sediment and suspended solids that cloud the water and bury spawning beds. Mercury biomagnification, malaria, and ozone as a drinking-water contaminant are unrelated to storm runoff.

Question 41 — 2016
Official key: C. Phosphorus from fertilizer and manure running off fields into Lake Erie drives eutrophication and cyanobacterial blooms that produce toxins. Warmer, not cooler, summers favor blooms, and chlorine would kill rather than feed algae.

Question 42 — 2016
Official key: D. The independent variable is what the experimenter changes (concentration of ammonium sulfate), and the dependent variable is what is measured (seeds that fail to germinate). A proper control receives seeds without the chemical, not no seeds.

Question 43 — 2016
Official key: B. Human nitrogen sources include fertilizer, combustion (vehicle and industrial NO_x) and sewage or animal waste entering water. Refrigeration units, trash dumping and oil spills are not major nitrogen sources.

Question 44 — 2016
Official key: C. Animal manure raises nitrates, phosphorus and fecal coliform bacteria; bacterial decomposition of the extra organic matter lowers dissolved oxygen and macroinvertebrate diversity. Site C has the highest nutrients and bacteria and the lowest oxygen and diversity.

Question 45 — 2016
Official key: B. Mercury released in coal smoke falls to water, is converted to methylmercury, taken up by plankton, and magnified in fish. The other pathways describe routes that do not involve atmospheric transport into water food chains.

Question 46 — 2016
Official key: B. Nutrient enrichment causes algal blooms; when the algae die, bacteria decomposing them consume dissolved oxygen. Fish and other aerobic organisms suffocate, rather than being poisoned by pH, mercury or salt.

Question 47 — 2016
Official key: E. Compostable fraction is 0.60 of waste. Weekly waste per person is 2 kg/day × 7 days = 14 kg, and 0.60 × 14 kg = 8.4 kg.

Question 48 — 2016
Official key: B. Bioaccumulation is the buildup of a chemical within one organism over time because intake exceeds elimination. It differs from biomagnification, which is increase up trophic levels.

Question 49 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The author assumes plastic is persistent and does not readily break down, saying it could last decades or centuries. Cost, manufacturing difficulty and recycling rates are not discussed.

Question 50 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The sustained three-year mapping effort by an international team of scientists is direct evidence of thoroughness. Area covered, journal and publication date describe results or timing but not research depth.

Question 51 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Preventing more plastic from entering the ocean requires reducing waste at the source, and incentives for recycling do that. Replacing paper with plastic, closing landfills or cheaper plastic would increase leakage.

Question 52 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Weight per piece = 88,000 tons ÷ 1.8×10^{12} pieces ≈ 4.9×10^{-8} ton. The other values are off by orders of magnitude.

Question 53 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Summing ratings gives permethrin 13, DDT 11, lindane 11 and malathion 9, so permethrin is highest overall.

Question 54 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Eroded sediment increases turbidity and sedimentation, reducing light and so primary productivity. Other statements invert the effect.

Question 55 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Nutrients trigger algal blooms; decomposition of dead algae by bacteria consumes oxygen, producing hypoxia. Nutrients do not bind oxygen or consume CO_2.

Question 56 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Drip irrigation applies water at roots in small amounts, cutting nutrient-laden runoff compared with flood irrigation. Integrated pest management targets pests, annual crops leave soil bare and cattle ranching adds manure.

Question 57 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Toxins biomagnify up the food web, so lower trophic levels carry lower concentrations and less exposure for humans.

Section B — FRQ Solutions

Question 1 — 2002
Data (100 shrimp per dish): ≤0.0001% → 10 dead; 0.001% → 10; 0.01% → 20; 0.1% → 55; 1% → 90; 10% → 100.

(a) Put concentration on the logarithmic x-axis and number dead on the linear y-axis; plot the six points and draw a smooth S-shaped (sigmoid) curve. *(1 point axes/points, 1 point smooth curve)*

(b) LD50 is the dose that kills half of the test population. Reading the curve at 50 dead gives a concentration just below 0.1%, about 0.07–0.08% (interpolating on the log scale between 0.01% and 0.1%). Mark it on the graph. *(1 point definition, 1 point value read from the graph; "about 0.1%" with no marking was judged too vague)*

(c) The threshold is the highest dose that produces no response above the background effect. The 10% mortality at the lowest two concentrations is background, so response first rises above it between 0.001% and 0.01%; the threshold is about 0.001%. *(1 point definition, 1 point value)*

(d) For: both are animals with similar basic physiology, the dose–response pattern probably holds, testing on people is unethical so animal tests give estimates. Against: humans differ in size, metabolism and exposure route (swallowed or touched, not immersed in water), a short 48-hour test on an invertebrate cannot capture chronic effects, and species vary in sensitivity. *(2 points for, 2 for against; maximum 4)*

Question 2 — 2004
(a) Source: coal burning in power plants. Mercury leaves with the flue gas, travels on wind, and returns by rain or dry fallout onto land and water; microbes convert it to methylmercury that enters food webs. *(source 1, transport 1, entry to water 1; up to 3)*

(b) Reduce emissions by fuel switching or efficiency (lower-mercury coal, natural gas, wind/solar, conservation), by washing coal before burning, or by post-combustion capture (activated-carbon sorbent with filters, scrubbers). Describe how each cuts mercury. *(1 point each)*

(c) Mercury biomagnifies: it is stored in tissue and not excreted, so concentrations increase at each trophic level; big predators eat many contaminated prey and live long, bioaccumulating more over a lifetime. *(up to 3 points)*

(d) Lead: from mining and smelting, old paint, leaded gasoline, plumbing, batteries. Acute sublethal effects: abdominal pain, headaches, anemia, and nervous-system effects such as learning problems or irritability. Cadmium, arsenic also accepted. *(1 point each: metal, release route, effect)*

Question 3 — 2005

(a) Pick, e.g., SARS (new) and malaria (old). SARS: virus spread by respiratory droplets and close contact; control by quarantine, masks, handwashing. Malaria: parasite spread by infected mosquito bites; control with nets, repellent, drained standing water, spraying or biological control. *(1 transmission + 1 control per disease = 4)*

(b) For malaria: warmer or wetter climate expands mosquito breeding range, or habitat change creates standing water, raising infections. *(factor 1, causal explanation 1)*

(c) Diseases spread quickly because travel and trade move infected people, animals and vectors worldwide; deforestation, crowding and climate change bring humans into contact with new pathogens. *(mechanism 1, matching explanation 1)*

(d) HIV/AIDS: spread through unprotected sex, shared needles, blood and mother-to-child; no cure or vaccine and long latency. (Influenza/pneumonia in an aging population also accepted.) *(1 disease, 1 cause)*

Question 4 — 2006

(a) Excavation is costly; transporting soil risks spills and dust; contaminated soil is only moved to a landfill; the site is disrupted and erosion/groundwater issues may remain. *(2)*

(b) Phytoremediation: plants take up or break down contaminants through roots. Advantage: cheap, less disruptive, controls erosion. Disadvantage: slow; contaminated plant matter needs disposal; only works to root depth. *(1 + 1 + 1)*

(c) Environmental: green space/habitat and less runoff and groundwater contamination. Societal: higher property values, jobs, reduced health risk, reduced sprawl. *(1 + 1)*

(d)(i) Reduction: substitute less toxic materials, recycle or reuse solvents, or improve process efficiency, each with description. **(ii)** Legal disposal: lined secure landfill, incineration, deep-well injection or stabilization, with a brief description. *(2 + 2)*

Question 5 — 2008

(a) *Environmental benefit (1 point):* a local landfill removes a 50-mile haul, so fewer truck emissions and less risk of spills in transit. *Economic benefit (1 point):* the city saves fuel and the private dump's tipping fees (one-third of the sanitation budget) and can create local jobs or sell captured landfill methane.

(b) *Any two (1 point each):* a compacted clay or synthetic liner with a leachate collection system, so leachate does not reach groundwater; daily soil cover to control odor, pests and litter; methane vents or gas capture; monitoring wells around the site; fencing and a tree buffer to hide the site; siting away from streams and aquifers.

(c) Tract A is the strongest answer because it is next to the highway and urban area. *Choice (2 points). Economic justification (1 point):* shorter access roads and shorter truck trips. *Environmental justification (1 point):* the two tracts that remain (B and C) stay contiguous, so the turtle habitat is not fragmented and the turtles are not forced to cross the highway. Tract C is also accepted for 1 point with justifications; tract B earns no choice point but can still earn justification points.

(d) *Law or treaty (1 point):* for example the Endangered Species Act, because the wood turtle is a rare species; *application (1 point):* the project would need to avoid destroying habitat of a listed species or get a permit/mitigation. Other acceptable answers: Clean Water Act (protects streams/wetlands from leachate), RCRA (landfill standards), Clean Air Act (landfill gas emissions).

Question 6 — 2010

(a) Choose lead as the pollutant. (i) A source other than the chemical plants: old paint or lead water pipes. *(1 point: one specific source.)* (ii) Entry: children swallow paint chips or dust, or drink water that has passed through lead plumbing. Effect: lead damages the developing nervous system and lowers learning ability. *(2 points: 1 for the route, 1 for the health effect.)* (iii) Two steps short of a ban: set and enforce limits on lead in drinking water and replace old pipes; test and remove lead-based paint from older housing. Public education on avoiding exposure would also earn credit. *(2 points: 1 per specific step.)* (b) Children take in more food, water and air per kilogram of body weight, and their growing organ systems are more sensitive than adult ones. *(1 point: one valid reason.)* (c) Cost = 40 kg CO_2 × \$0.75 per kg = \$30 per MP3 player. *(2 points: 1 for the setup, 1 for the answer.)* (d) Choose sulfur oxides: a flue-gas scrubber removes SO_2 from the exhaust. Alternatives are

electrostatic precipitators or baghouse filters for particulates, and selective catalytic reduction for NO_x. *(1 point: a technology matched to the pollutant.)* (e) Two reasons: labor is cheaper, and environmental or workplace regulations are less strict or less strictly enforced, which lowers compliance costs. Other accepted reasons are cheaper land, lower taxes and access to large markets. *(2 points: 1 per reason, with brief discussion.)*

Question 7 — 2010

(a) Choose one pollutant and stay with it for all three sub-parts.

- (i) *Source (1 pt).* Any specific non-plant source: mercury from coal burning, dental fillings or broken fluorescent bulbs; lead from old paint, solder, plumbing or leaded ceramics; PCBs from old transformers and capacitors.
- (ii) *Entry route (1 pt) and one health effect (1 pt).* Example for mercury: eating contaminated fish (it biomagnifies as methylmercury) or breathing vapor; effect: nervous-system damage and developmental delay. Lead: swallowing paint chips or dust; effect: lowered IQ and learning problems. PCBs: eating contaminated fatty fish; effect: immune and developmental problems, possible cancer.
- (iii) *Two control steps, neither a ban (2 pts).* For example: public education on exposure, safe collection and disposal programs, scrubbers or filters on emission sources, treatment of water supplies, cleanup or capping of contaminated soil and sediment, substituting safer materials. Each must be a concrete action that lowers exposure.

(b) *(1 pt)* Children take in more air, water and food per kilogram of body mass than adults, and their organs and nervous systems are still developing, so the same exposure does more damage. (Hand-to-mouth behavior or a weaker detoxifying liver are also accepted.)

(c) Cost = 40 kg CO_2 × \$0.75/kg CO_2 = **\$30**. *(1 pt setup with units cancelling, 1 pt answer.)*

(d) *(1 pt)* Pick one pollutant and match a real control: SO_x → flue-gas scrubber or removing sulfur before burning; NO_x → selective catalytic reduction or lower-temperature combustion; particulates → electrostatic precipitator or baghouse filter. Fluidized-bed combustion or coal gasification are also accepted.

(e) *(2 pts, one per reason, each discussed)* Typical reasons: weaker or less strictly enforced pollution and workplace rules lower compliance cost; lower wages and benefits; cheaper land, raw materials and taxes or subsidies; access to a large local market. Name the reason and say why it saves the company money.

Question 8 — 2012

(a) *(2 pts)* Two of: soil flooded or saturated for part of the year; vegetation adapted to waterlogged conditions; hydric (water-formed) soils.

(b)(i) *(2 pts)* Draw arrows pointing from the organism eaten to the eater (producer to consumer). At least two valid food chains that share a species, for example algae → snail → fish and algae → fish → eagle, show a web. Every box must be linked and no arrow may point the wrong way.

(b)(ii) *(2 pts)* Only about 10% of energy passes up each trophic level; the rest is lost as heat or used for metabolism (second law of thermodynamics), and some material is indigestible, so a top predator needs a very large base of producers and prey, hence many hectares.

(c) *(2 pts, benefit + linked description)* Examples: tourism and hunting or fishing permits bring revenue; nursery areas support commercial fisheries; flood and storm buffering reduces property damage and insurance costs; groundwater recharge reduces water-treatment costs.

(d) *(1 pt)* Draining or filling for farms or building, polluted runoff, damming or diking, peat or mineral mining, or invasive species.

(e) *(1 pt each)* (i) Like primary treatment, wetland plants and slow water physically trap and settle solids and debris. (ii) Like secondary treatment, microbes in the sediment and roots break down dissolved organic waste.

Question 9 — 2013

(a) Two activities, each with its mechanism. Levees and channelization confine the river so sediment is shot out to deep water instead of spreading over the delta; dams and reservoirs trap sediment upstream, so less travels downstream; dredged canals and navigation channels divert flow and let seawater in. *(1 point per activity named, 1 per correct explanation of the effect on sediment flow; 4 points.)*

(b) Two harms. Without fresh sediment the marsh surface cannot build up as the land sinks, so wetlands drown and convert to open water; the lost marsh and barrier land no longer buffers storm surge, and nursery habitat for fish, shrimp and birds shrinks, lowering biodiversity and productivity. *(2 points.)*

(c)(i) Two impacts of fertilizer reaching the Gulf: nitrogen and phosphorus drive algal blooms; when the algae die, decomposers use up dissolved oxygen, producing a hypoxic "dead zone" where fish and shellfish die or flee; reduced light penetration also damages seagrass. *(2 points.)* (ii) Two economic consequences: lower commercial and recreational fishing catches (lost income, jobs); reduced tourism and higher costs for water treatment or monitoring. *(2 points.)* (iii) Another strategy: restore or build wetlands and vegetated buffer strips that trap nutrients before they enter the river, or upgrade wastewater treatment to remove nutrients. *(1 point.)*

Question 10 — 2015

(a) Two activities that reduce inflow: diverting water for irrigation or flood-control canals; withdrawals for homes, industry and lawns; roads and filled wetlands that block sheet flow. *(2; must be described, not just named.)*

(b)(i) Phosphorus source: fertilizer runoff from farms or lawns, septic seepage, or discharge from plants lacking advanced treatment. *(1.)* (ii) Excess phosphorus triggers algal blooms and eutrophication, or favors invasive cattails over native sawgrass adapted to low-nutrient water. *(1.)* (iii) Link the fix to the source: buffer strips and constructed treatment wetlands for farm runoff, precision fertilizer use, or tertiary treatment. *(1.)*

(c) One example for each: water quantity: higher evaporation or sea-level rise lowers freshwater levels (or heavy storms increase flooding); water quality: warmer water holds less dissolved oxygen, or saltwater intrusion raises salinity; habitat: saltwater pushes mangroves inland, displacing freshwater communities. *(3.)*

(d)(i) Restoring flow and water quality re-establishes native vegetation and habitat, stabilizing food webs and reducing algal blooms. *(1.)* (ii) Restoration increases tourism and recreation revenue and jobs, and avoids water-treatment costs. *(1.)*

Question 11 — 2015

(a) Percent increase = (new − old) ÷ old × 100 = (180 − 30) ÷ 30 × 100 = **500%**. *(1 setup, 1 answer.)*

(b) 1.8×10^8 devices × 0.03 g/device = **5.4×10^6 g** (5,400,000 g) of gold. *(1 + 1.)*

(c) Total mass = $1.8 \times 10^8 \times 0.1$ kg = 1.8×10^7 kg; 90% not recycled = $0.90 \times 1.8 \times 10^7$ = **1.62×10^7 kg**. *(1 + 1.)*

(d) Mercury exposure can damage the nervous system, brain or kidneys, impair development (birth defects, learning problems) or vision and hearing. *(1.)*

(e)(i) Two reasons the U.S. ships e-waste abroad: labor and disposal are cheaper there; environmental and worker-safety rules are weaker or less enforced; avoiding liability and public opposition at home. *(2; reasons must explain the sender's motive, and "cheaper" counts once.)* (ii) Take-back or trade-in programs, modular repairable designs, longer-lasting products, or reduced planned obsolescence. *(1.)*

Question 12 — 2015

(a) On the bar chart the highest estimate for the Deepwater Horizon (Gulf of Mexico, 2010) is just below 200 million gallons, accepted as any value from 185 to 199 million gallons. *(1 point with units.)*

(b) Two problems: oil coats birds and mammals, destroying insulation and buoyancy; smothered or poisoned marsh, beach and reef habitat disrupts food webs and loses nursery grounds; oil blocks light for photosynthesis. *(2.)*

(c) One economic impact: lost fishing and tourism income, or cleanup costs (a positive: temporary cleanup jobs). *(1.)*

(d)(i) Advantage: breaks slicks into droplets that dilute and disperse, protecting birds and shorelines. Disadvantage: dispersed oil and the chemicals can be toxic to marine life and the oil is not removed, harming the water column and seafloor. *(1 + 1.)* (ii) Physical: floating booms and skimmers contain and collect surface oil; or biological: adding oil-degrading microbes. *(1.)*

(e) Other source: leaking boat engines and bilge discharge, stormwater runoff carrying oil from roads and cars, natural seeps, or leaks in drilling and pipelines; give how it reaches the water. *(1 + 1.)*

(f) Substitute: plant-derived plastic (corn-based bottles), paper bags, or cotton and hemp fibers in place of synthetic fabrics. *(1.)*

Question 13 — 2016

(a)(i) Cause of the change in per-capita generation, 2000–2012. Per-capita waste fell slightly; probable causes include more recycling and reuse, lighter packaging materials (plastic in place of glass), less paper as news and magazines go online, or reduced consumption during the 2008–2009 recession. *(1 point: one cause explained)*

(a)(ii) Percent increase in total MSW, 1980–2012. Read from the graph: about 150 million tons in 1980 and 250 million tons in 2012. Percent increase = (250 − 150) ÷ 150 × 100 = **≈ 67 %** (66–67 % accepted). *(1 point: answer with work)*

(b)(i) Disadvantage of incineration. Air pollutants (CO_2, dioxins, particulates, SO_x, NO_x), toxic ash needing disposal, high building and operating cost, or neighbors' complaints about trucks. *(1 point)*

(b)(ii) Disadvantage of landfills. Leachate contaminating groundwater or soil, methane and CO_2 release, odors, vermin, habitat loss and the land they take up for decades. *(1 point)*

(c) Electricity from landfill gas. Step 1: methane produced by decomposing buried waste is collected through pipes. Step 2: the gas is burned, converting chemical energy to heat that makes steam (or hot gas). Step 3: the steam or hot gas turns a turbine connected to a generator, producing electricity. *(3 points: 1 per step)*

(d) Tire piles. Standing water in tires breeds mosquitoes that carry disease, or tire fires release smoke and toxic fumes causing respiratory problems. *(1 point)*

(e)(i) Advantage of composting. Produces soil amendment that can be used or sold, lowers disposal costs, less odor in trash, creates jobs. *(1 point)*

(e)(ii) Disadvantage of composting. Odors, attracts pests, can release methane, nutrient runoff, or requires labor and time. *(1 point)*

Question 14 — 2016

(a)(i) One cause: more reduction, reuse and recycling; lighter packaging; economic recession lowering consumption; digital media replacing paper. *(1.)* (ii) From the graph, total MSW rose from about 150 to about 250 million tons: (250 − 150) ÷ 150 × 100 = **≈ 67%** (accepted 66% to 67%). *(1.)*

(b)(i) Incineration: releases air pollutants (particulates, dioxins, CO_2, NO_x), leaves toxic ash, expensive to build. *(1.)* (ii) Landfills: leachate can contaminate groundwater, methane and odors escape, land is lost for other uses. *(1.)*

(c) Landfill gas electricity: (1) collect the methane produced by decomposing waste through wells and pipes; (2) burn the methane, converting chemical energy to heat that produces steam or drives an engine; (3) the steam or exhaust spins a turbine connected to a generator, making electricity. *(3.)*

(d) Tire piles collect rainwater that breeds mosquitoes that carry disease, and tire fires release toxic smoke harming respiration. *(1.)*

(e)(i) Compost provides a free soil amendment/fertilizer and can create jobs. *(1.)* (ii) Compost can attract pests, smell bad, or release methane if poorly managed; it requires labor and space. *(1.)*

Question 15 — 2017

(a) Two effects: beads clog or fill the digestive tract so animals feel full and starve; beads carry toxic chemicals that accumulate in tissues and may magnify up the food chain. *(2.)*

(b) Nitrates fertilize algae, causing blooms (cultural eutrophication); when the algae die, decomposition by microbes consumes dissolved oxygen, creating hypoxic water that kills fish; blooms also cloud water and block light. *(1 for response, 1 for effect linked to it.)*

(c)(i) Primary treatment removes large plastic by screens or bar racks, settling tanks and skimmers. *(1.)* (ii) Disinfect with chlorine, ultraviolet light or ozone. *(1.)* (iii) Advantage: biosolids are a nutrient-rich, low-cost fertilizer and avoid landfilling; disadvantage: they may contain heavy metals, pathogens or drug residues that contaminate soil and water, and may smell. *(1 + 1.)*

(d)(i) Mangroves are cut for shrimp farms, coastal development, agriculture, or wood and charcoal. *(1.)* (ii) Intact mangroves protect shorelines from storms and erosion and provide nursery habitat for fisheries. *(1.)*

Question 16 — 2022

(a)(i) Area A (upstream, no mercury). *(1)*

(a)(ii) Nesting success. *(1)*

(a)(iii) Does mercury level in nesting habitat affect snapping-turtle nest success or nest location? *(1)*

(a)(iv) Upstream sites are the control: baseline comparison with mercury-exposed downstream nests. *(1)*

(a)(v) Without bank vegetation, less mercury is taken up or filtered, soil near the river erodes, and nests may be placed nearer contaminated ground or in hotter, drier soil, so fewer succeed. *(1)*

(b)(i) Persistent mercury bioaccumulates and is a neurotoxin that damages organs, reproduction and survival. *(1)*

(b)(ii) It biomagnifies up the food chain, harming top predators and potentially causing trophic cascades and lower diversity. *(1)*

(b)(iii) The factory releases inorganic mercury; bacteria in sediment convert it to methylmercury. *(1)*

(b)(iv) The map shows nests near the river in Area B failed while those farther away succeeded, supporting the claim that mercury-rich soil near the river reduces success. *(1)*

(c) E.g. tilling loosens and erodes soil; monocropping depletes nutrients; fertilizer alters chemistry; irrigation can salinize; heavy machinery compacts soil; legumes add nitrogen. *(1)*

Question 17 — 2022

(a)(i) Urban land covers the least area. *(1)*

(a)(ii) Fertilizer or manure runoff (or sewage) from farms enters rivers, carries nutrients to the Gulf, and fuels algae. *(1)*

(a)(iii) Cities release treated or overflow wastewater, and impervious surfaces speed lawn-fertilizer and detergent runoff into the river. *(1)*

(b)(i) Algae die and decompose, depleting dissolved oxygen (and block light), so organisms flee or die. *(1)*

(b)(ii) Smaller fish catch and fishing income, lower tourism, or higher fuel and consumer costs. *(1)*

(b)(iii) Summer brings heavy fertilizer runoff and spring flows, and warmer water holds less dissolved oxygen. *(1)*

(c)(i) Hypothesis: stream sites downstream of riparian buffers will have lower nitrate and phosphate than sites without buffers. *(1)*

(c)(ii) Control: sites 100 m downstream from areas lacking a vegetation buffer. *(1)*

(c)(iii) Another test: turbidity, dissolved oxygen, temperature, fecal coliform, pH, conductivity, BOD or total suspended solids. *(1)*

(c)(iv) A feeding operation adds manure nutrients, bacteria, solids or antibiotics that would worsen water quality independent of the buffer, confounding results. *(1)*

Question 18 — 2023

(a) Methane is a potent greenhouse gas that traps heat, driving warming and sea-level rise. *(1)*

(b) Factors: share of organic matter, oxygen and moisture, microbe population, temperature or climate. *(1)*

(c) Capture and burn or use landfill gas, compost organics, divert and recycle, or incinerate waste. *(1)*

(d) Extra benefit: electricity or heat, compost fertilizer, longer landfill life, or less land and raw material use. *(1)*

(e) 150 L/cow/day × 31.2×10^6 cows × 365 days = 1.7×10^{12} L per year (the guideline scores it as an annual amount). *(2)*

(f) 31.2×10^6 cows × 11.8 kg/cow/day ÷ 26.2 kg/ha ≈ 1.4×10^7 ha (14,051,908 ha). *(2)*

(g) $(30.4 \times 10^6 - 2.7 \times 10^6)$ kcal/ha × 150 ha ÷ 2,250 kcal/person = $27.7 \times 10^6 \times 150 \div 2{,}250 \approx 1.8 \times 10^6$ more people (1,846,667). *(2)*

Question 19 — 2024

(a) Zone C has the lowest dissolved oxygen. *(1)*

(b) Inverse: as BOD rises, dissolved oxygen falls. *(1)*

(c) Zone B, where BOD jumps and oxygen starts to drop, likely marks the discharge point. *(1)*

(d) Dependent variable: macroinvertebrate species richness. *(1)*

(e) E.g. "Lower dissolved oxygen will be associated with fewer macroinvertebrate species." *(1)*

(f) Zone A is upstream of the pollution source, so it shows unaffected baseline conditions. *(1)*

(g) Winter water is colder, holds more oxygen (more species) but may be below some species' tolerance or have less photosynthesis (fewer species). *(1)*

(h) Sewage supplies nutrients and bacteria, so bacterial populations rise. *(1)*

(i) Abiotic factor: temperature, pH, light, turbidity, salinity or nutrient level. *(1)*

(j) Persistent organic pollutants accumulate in fatty tissue and biomagnify, so top consumers carry the highest doses and suffer toxicity. *(1)*

Unit 9 — Global Change

Section A — MCQ Answers and Explanations

Question 1 — 2008

Official key: A. Water vapor is by far the most abundant greenhouse gas and is entirely natural in origin in terms of its baseline concentration. CO_2 and CH_4 are less abundant, and ozone is a trace gas; CFCs are synthetic. Water vapor mostly acts as a feedback because warmer air holds more of it.

Question 2 — 2008

Official key: E. CCl_2F_2 is a chlorofluorocarbon (CFC-12), a synthetic compound manufactured for refrigerants and propellants and with no natural source. H_2O, CO_2, CH_4 and O_3 all have natural sources as well as human ones.

Question 3 — 2008

Official key: C. Fossil fuels hold carbon that was removed from the active carbon cycle for millions of years. Burning moves that stored carbon into the atmosphere as CO_2, adding to the carbon cycling between atmosphere and biosphere that natural sinks cannot fully absorb.

Question 4 — 2008

Official key: A. Greenhouse gases in the troposphere absorb outgoing infrared radiation emitted by Earth's surface and re-radiate some downward, warming the lower atmosphere. They do not primarily absorb incoming UV.

Question 5 — 2008

Official key: C. CFC molecules absorb infrared radiation far more strongly per molecule than CO_2 does, and they persist for decades to centuries, so small amounts have a large warming effect. The other options either describe weak greenhouse properties or irrelevant features.

Question 6 — 2008

Official key: A. Percent increase = (383 − 278) ÷ 278 × 100 = 105 ÷ 278 × 100 ≈ 38%.

Question 7 — 2008

Official key: B. Ozone depletion in the stratosphere lets more UV-B radiation reach the surface, and UV-B damages skin cells and raises skin cancer rates. Australia is close to the Antarctic ozone hole, so it is strongly affected. Water pollution, immigration, solar cycles and tropospheric warming do not act on skin cancer directly.

Question 8 — 2008

Official key: C. Invasive species often outcompete native species for resources and lack natural predators, so they reduce native populations.

Question 9 — 2008

Official key: C. Melting Arctic sea ice exposes darker ocean, which absorbs more sunlight, causing more warming and more melting.

Question 10 — 2010
Official key: A. Habitat conservation plans are a provision of the Endangered Species Act: a landowner agrees to offset harm to a listed species and, in return, may develop part of the property. The other listed measures contain no such provision.

Question 11 — 2010
Official key: C. The Montreal Protocol committed nations to phase out chlorofluorocarbons and related ozone-depleting chemicals. Kyoto addressed greenhouse gases rather than ozone.

Question 12 — 2010
Official key: D. Reading the bars, mammal extinctions were about 135 in the last century and about 50 the century before. The difference is roughly 85 to 90, so the choice of 90 fits.

Question 13 — 2010
Official key: B. Human land clearing, hunting, pollution and introduced species altered or destroyed habitats at an increasing rate. The other options describe changes that did not drive the twentieth-century rise.

Question 14 — 2010
Official key: A. Mitigation reduces the cause by cutting greenhouse-gas emissions. Replacing fossil-fuel electricity with wind power does so. More air conditioners adapt to heat, and cheaper fuel or coal power would raise emissions.

Question 15 — 2010
Official key: E. Invasive species typically succeed because they reproduce quickly, which means high, not low, reproductive rates. The other statements about competition, predation, human introduction and variety of taxa are true.

Question 16 — 2010
Official key: D. Warming shrinks ice sheets, shifts plant ranges, boosts insect numbers and raises sea level. Ground-level ultraviolet radiation depends on stratospheric ozone, not on greenhouse warming.

Question 17 — 2010
Official key: C. In the diagram the gases absorb surface infrared and re-emit about 60 units back down toward the surface; this return flow is their greenhouse contribution. The other values in the diagram belong to different flows, such as the 70 units escaping to space or the 120 units radiated from the surface.

Question 18 — 2010
Official key: E. The surface absorbs 50 units of sunlight plus 70 units of infrared returned from the atmosphere, for 120 units. That matches the 120 units it radiates upward.

Question 19 — 2010
Official key: D. Outgoing energy balances incoming: 30 units of reflected sunlight plus 70 units of emitted infrared make 100 units, equal to the 100 arriving.

Question 20 — 2010
Official key: D. Rail moves people and freight using much less fuel per unit than road vehicles, so fossil-fuel CO_2 falls. Burning coal for heat raises emissions, and ozone protection concerns UV, not CO_2.

Question 21 — 2010
Official key: A. Chlorine- and bromine-containing compounds release atoms that destroy ozone. Carbon dioxide has neither and affects climate, not ozone.

Question 22 — 2010
Official key: D. Chlorofluorocarbons are synthetic chemicals that never occurred naturally. Nitrous oxide, CO_2 and methane have natural sources as well.

Question 23 — 2010
Official key: D. Photosynthesis was far higher when UV was filtered and fell when UV was added, so more UV through a thinner ozone layer would depress productivity further. The data do not address tropics, CFC causes or future UV levels.

Question 24 — 2010
Official key: B. The results show UV strongly limits phytoplankton productivity, so ozone thinning could affect the base of the polar food web. Other options claim things the experiment did not test.

Question 25 — 2010
Official key: A. Tank 1 has natural UV and tank 2 has none. The large increase when UV is removed isolates the effect of natural UV.

Question 26 — 2010
Official key: A. Reliability means repeatability, so repeating under identical conditions with similar results is the strongest support. Different conditions or more variables test generality or add new questions instead.

Question 27 — 2010
Official key: D. Ground-level ozone harms lungs and stratospheric ozone absorbs some ultraviolet radiation. Ozone is not a precursor of acid rain, so only II and III are true.

Question 28 — 2010
Official key: C. Burning biomass releases CO2 as the main carbon-containing product. Water vapor and ozone are not the main products of burning plant material.

Question 29 — 2010
Official key: D. Small populations are vulnerable to unpredictable events such as storms or disease and have low genetic diversity. Larger or generalist populations are buffered, which is the opposite of what the other options claim.

Question 30 — 2010
Official key: E. Sulfate aerosols reflect sunlight back to space, cooling Earth.

Question 31 — 2016
Official key: D. The Arctic warms roughly two to three times faster than the global average because loss of reflective ice and snow exposes darker surfaces that absorb more sunlight. The keyed region is the Arctic one in the northwest of the map.

Question 32 — 2016
Official key: D. As climate warms, suitable habitat for cold-adapted species shifts poleward and uphill. Red pine's range should therefore move northward into Canada, while southern areas become too warm.

Question 33 — 2016
Official key: A. A positive feedback amplifies the original change: melting ice exposes darker water that absorbs more sunlight, warming the surface, which melts more ice. Only the choice in which melting leads to warmer water and more melting shows this reinforcement.

Question 34 — 2016
Official key: E. Sea ice already floats, so melting it does not change sea level, and flooding would not create farmland. The other four statements describe real or plausible Arctic consequences, including methane release from permafrost.

Question 35 — 2016
Official key: A. CO_2 absorbs outgoing infrared radiation and warms the planet, while sulfate particulates in the stratosphere scatter sunlight and cool it for a few years. They have opposing effects.

Question 36 — 2016
Official key: C. Water vapor cycles quickly through evaporation and precipitation, staying only days in the atmosphere, so human additions don't accumulate. It matters mostly as a feedback to warming by other gases.

Question 37 — 2016
Official key: B. Warming lets mosquitoes and ticks survive in latitudes and elevations that were too cold, widening disease range. Population growth is actually slowing, and rainforests are shrinking.

Question 38 — 2016
Official key: B. Warming from higher CO_2 is predicted to cause loss of species in tropical ecosystems, along with more severe storms, more heat waves and earlier spring events. The other options say the opposite.

Question 39 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The article stresses urgency: the debris must be removed before it breaks into pieces too small to collect and dangerous to marine life. The other options contradict or are not stated in the passage.

Question 40 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The independent variable is what the investigator manipulates, here the water temperature of each tank. Volume and nutrients are controlled, and dissolved oxygen is the measured response.

Question 41 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Biological control uses a natural enemy or pathogen specific to the pest, such as a pathogen that targets rabbits without harming other mammals. Pesticides, poison fumigation and planting native plants are not biological control.

Question 42 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Extra CO_2 dissolves to lower pH and reduces carbonate ions that corals need to build calcium carbonate skeletons. It does not stop reproduction through algae or involve ozone.

Question 43 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** UV splits oxygen to atoms that form ozone, and UV also breaks ozone apart, the continual production and destruction of stratospheric ozone. It is not photosynthesis or tropospheric ozone formation.

Question 44 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Chlorofluorocarbons release chlorine atoms in the stratosphere that catalytically destroy ozone. Methane, VOCs and lead do not do this.

Question 45 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** A population bottleneck, a sharp drop in numbers from habitat fragmentation, removes alleles and reduces genetic diversity. Demographic transition and biomagnification are unrelated.

Question 46 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The article says trout fishing has so far been resilient but continued warming could cost millions in revenue. It does not say the industry is destroyed.

Question 47 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** The author cites current closures and a projected 35% loss of suitable cold-water habitat by 2080 as grounds for future impact.

Question 48 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Moving toward higher latitudes lets species track suitable temperatures. Later mating or becoming a specialist would not improve survival.

Question 49 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** A generalist with a varied diet and rapid, repeated reproduction can exploit new areas quickly. Specialists or slow breeders rarely spread aggressively.

Question 50 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Requiring boats to be drained and cleaned prevents transfer of larvae and eggs. Collecting larvae or posting lists does not stop spread.

Question 51 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Warming thaws permafrost, releasing stored carbon as CO_2 and methane, which warms the climate further (positive feedback).

Question 52 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The Montreal Protocol (1987) is the international treaty that phased out production of CFCs and related chemicals. The Kyoto Protocol addresses greenhouse gases, and the Clean Air Act and Endangered Species Act are US laws.

Question 53 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** The Endangered Species Act protects listed species and their critical habitat and restricts trade in them in the United States. The other options regulate air pollution or global emissions.

Question 54 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** Ozone (O_3) in the stratosphere absorbs most incoming UV-B. CFCs destroy it, and VOCs and NOx are pollutants that form ozone only in the troposphere.

Question 55 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Larvae that feed inside the cactus are shielded from predators and from sprays, so control is very hard. Long flight ability or dormancy would make the pest spread or hide but not shield it the same way.

Question 56 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Habitat loss affects 1,800 species, far more than any other threat, so paying landholders not to develop habitat protects the most species.

Question 57 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** CFC propellants release chlorine in the stratosphere, destroying ozone. Fossil fuel burning, methane and sewage lagoons do not target ozone.

Question 58 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The largest saltwater loss (101,000 acres) is from sea-level rise submerging habitat.

Question 59 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: D.** The average distance moved north increases over time, which fits birds wintering farther north as higher-latitude temperatures rise.

Question 60 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: B.** Kudzu can be grazed by goats and harvested as feed for livestock. The other options claim benefits that are false or harmful.

Question 61 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The largest segment is fish.

Question 62 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: C.** Percent change = (final − initial) ÷ initial × 100 = (10,119 − 5,742)/5,742 × 100 ≈ 76%.

Question 63 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** Warming allows mosquito habitat to expand upward in elevation.

Question 64 — 2024
(The source copy had no published key; this answer was derived by Success4Sure Academy.) **Answer: A.** The robot cannot reproduce or become invasive.

Section B — FRQ Solutions

Question 1 — 2006

(a)(i) CO_2 about 200 ppm at 140,000 yr ago and about 280 ppm at 125,000 yr ago: increase ≈ **80 ppm**. *(1)*

(ii) Temperature went from about −8 °C to +2 °C: change = +10 °C. Ratio = 10 °C ÷ 80 ppm = **0.125 °C per ppm** (1 °C per 8 ppm). *(1 + 1)*

(iii) 200 ppm × 0.125 °C/ppm = **25 °C**. *(1)*

(iv) Assumption: CO_2 alone controls temperature with a constant linear relationship. Not valid: other greenhouse gases and aerosols, feedbacks, and the rapid rate of modern change mean the relationship is not constant (and the answer is unrealistically large). *(1 + 1)*

(b) Fossil fuel burning for electricity, transport and industry as population and per capita use grow; deforestation and land clearing. *(2 points)*

(c) Methane from rice paddies, livestock, landfills; N_2O from fertilizer use and combustion; CFCs/HFCs from refrigeration. *(1 each, gas + activity)*

Question 2 — 2010

(a)(i) Highest consumption in the table is 0.27 mg per day per termite, at 30 °C and 90% relative humidity. *(1 point: both values.)* (ii) CH_4 = 2,000 ha × 4.5 × 10^7 termites/ha × 70 kg CH_4 per year per 1,000 termites = 6.3 × 10^9 kg CH_4 per year. *(2 points: 1 for setup with units, 1 for the answer.)* (iii) At 35 °C and 50% humidity consumption is 0.09, so emissions scale by 0.09/0.27 = one third: 6.3 × 10^9 × 0.09/0.27 = 2.1 × 10^9 kg CH_4 per year. *(2 points: 1 for setup, 1 for the answer.)* (iv) Temperature and humidity are limiting factors with a range of tolerance. At 40 °C consumption stops, so termites and their gut microbes likely die; outside the tolerance range, reproduction and colony founding fall, so population size falls. *(2 points: 1 for the reason, 1 for the linked explanation.)* (b)(i) Clearing leaves a large amount of dead wood and plant material, which raises food supply and supports more termites. *(1 point.)* (ii) The extra food is used up, the population exceeds carrying capacity and crashes; loss of shade also raises surface temperature beyond termite tolerance. *(1 point.)* (c) Removing trees reduces photosynthesis, so less CO_2 is taken up and the carbon sink is lost; burning cleared trees also releases CO_2. *(2 points: 1 for the change, 1 for the climate link.)*

Question 3 — 2010

(a) The mussels were introduced in the East and are still spreading west; the Rocky Mountains and scarcer connected waters in the West slow their spread, and boating traffic is denser in the East. *(1 point: one valid explanation.)* (b) Mussels hitch rides on boat hulls and trailers or in bilge and bait water, and they move on waterfowl. Prevention: require inspection, cleaning and draining of boats before they enter a new lake. *(2 points: 1 for a mechanism, 1 for a prevention method.)* (c) Zebra mussels filter plankton from the water, which increases water clarity; more light reaches the bottom and aquatic plants grow more, while competition for plankton reduces native mollusks and zooplankton feeders. *(2 points: 1 for the impact, 1 for the linked explanation.)* (d) Kudzu vine smothers and shades out native plants. Other accepted: emerald ash borer, which kills ash trees. *(2 points: 1 for the species, 1 for the impact.)* (e) Example: the cane toad was introduced to control beetle pests; it became invasive itself and poisoned native predators that ate it. *(2 points: 1 for a real example, 1 for a negative impact.)* (f) Two traits: a high reproductive rate (many offspring, early maturity) and generalist habits or strong dispersal, so they can use many foods and habitats and spread quickly. *(2 points: 1 per trait.)*

Question 4 — 2010

(a) Rise = 3.0 mm/yr × 50 yr = 150 mm = 0.15 m. *(2 points: 1 for the setup, 1 for the answer.)* (b) First: melting of land-based ice such as the Greenland ice sheet or mountain glaciers adds water to the ocean. Second: thermal expansion, because warmer water has greater volume. *(4 points: 1 for each process, 1 for each explanation.)* Floating sea ice does not raise sea level when it melts. (c) Two impacts: flooding converts marshes and mangroves to open water, removing nursery habitat for fish and shellfish; saltwater moves farther inland, changing salinity and eliminating species with narrow salt tolerance. Loss of the buffer also raises storm damage inland. *(2 points: 1 per impact.)* (d)(i) Rising seas damage or destroy property, lowering property values and raising insurance and repair costs. *(1 point.)* (ii) Two strategies: raise insurance premiums or deny coverage in repeatedly flooded areas; zone to restrict new building near the shore. Ending subsidies for beach protection is another option. *(2 points: 1 per strategy.)*

Question 5 — 2010

(a) *(1 pt)* The mussels arrived in the East and are still spreading; the East has far more connected surface waters and more boat traffic, while western mountain ranges act as barriers.

(b) *(1 pt mechanism, 1 pt prevention)* Spread: larvae or adults riding on boat hulls, trailers and bilge or live-well water, in stocked fish water, or through newly built canals. Prevention: inspect and clean or dry boats before moving between lakes, drain water, or run public-education and boat-wash programs.

(c) *(impact + linked explanation, 2 pts)* Filter-feeding clears the water, so more light reaches the bottom and aquatic plants grow more; or the mussels remove plankton, outcompeting native mollusks and zooplankton and weakening food chains.

(d) *(species + impact, 2 pts)* Name a specific organism, for example kudzu (smothers plants), cane toad (poisons predators) or emerald ash borer (kills ash trees), and state one harm.

(e) *(example 1 pt, negative effect 1 pt)* A real case, such as the mongoose introduced to eat rats or cane toads introduced against beetles; the introduced control often eats non-target native species, competes with natives or disrupts food webs.

(f) *(2 pts)* Two specific traits, such as generalist diet and habitat, rapid reproduction with many offspring (r-strategy), strong dispersal, or lack of natural predators and parasites in the new range.

Question 6 — 2010

(a) Rise = 3.0 mm/yr × 50 yr = 150 mm; 150 mm × (1 m ÷ 1,000 mm) = **0.15 m**. *(1 pt setup, 1 pt answer.)*

(b) *(2 identifications + 2 linked explanations)* (1) Thermal expansion: warmer seawater takes up more volume. (2) Melting of land-based ice, such as the Greenland and Antarctic ice sheets or mountain glaciers, which adds water that was not in the ocean before. (Floating sea ice does not raise sea level when it melts, and "ice caps" alone is too vague.)

(c) *(2 pts)* Loss of marsh, mangrove and intertidal habitat as water floods inland, including nursery areas for fish and shellfish and stopover habitat for birds; saltwater pushing farther up the estuary (higher salinity) that eliminates salt-sensitive species; weaker buffering of storm surges and more shoreline erosion.

(d)(i) *(1 pt)* Property loss or costly repairs, higher insurance, falling land values, lost tourism or fishing income, or saltwater contaminating water supplies.

(d)(ii) *(2 pts)* Two practicable policies: stop subsidizing flood insurance in high-risk zones, zone against new development on flood-prone land, tax or fee coastal building, fund buyouts and relocation away from the shore, or publish flood-risk maps so buyers see the danger.

Question 7 — 2011

(a) *(1 pt)* CO_2 reacts with water to form carbonic acid, which releases hydrogen ions, so pH falls.

(b) *(2 pts)* Atmospheric CO_2 concentration has risen, and that rise comes mostly from burning fossil fuels; more CO_2 in the air means more dissolves into the surface ocean.

(c)(i) Volume = area × growth: 2.5×10^{11} m² × 3 mm/yr × (1 m ÷ 1,000 mm) = $2.5 \times 10^{11} \times 0.003$ = **7.5×10^{8} m³/yr**. *(1 pt setup, 1 pt answer.)*

(c)(ii) Mass = volume × density: 7.5×10^{8} m³/yr × 2×10^{3} kg/m³ = **1.5×10^{12} kg/yr**. *(2 pts.)*

(c)(iii) Shortfall = $0.20 \times 1.5 \times 10^{12}$ kg = **3×10^{11} kg** less per year. *(2 pts.)*

(d) *(impact 1 pt, linked description 1 pt)* Loss of reef habitat removes nursery, feeding and shelter areas, so fish diversity and populations fall; or reefs no longer buffer shorelines from waves and storm surge.

(e) *(1 pt)* Any other global marine problem, for example overfishing, warming seawater, plastic debris, nutrient-driven dead zones or invasive species.

Question 8 — 2013

(a) Stratospheric ozone absorbs ultraviolet (mainly UV-B and UV-C) radiation; the health benefit is lower rates of skin cancer, sunburn or cataracts. *(1 + 1.)*

(b) UV splits ozone: $O_3 + UV \rightarrow O_2 + O$. The free oxygen atom meets an O_2 molecule: $O + O_2 \rightarrow O_3$. The two steps continuously recycle ozone and absorb UV. *(1 point for each process.)*

(c)(i) UV breaks chlorine atoms off CFC molecules (e.g., $CCl_3F + UV \rightarrow CCl_2F + Cl$); each free Cl then attacks ozone: $Cl + O_3 \rightarrow ClO + O_2$, and the chlorine is regenerated so one atom destroys many ozone molecules. *(1 + 1.)* (ii) Ozone recovery lags because CFCs are very long-lived and take years to reach the stratosphere, chlorine already present stays aloft for decades, and other ozone-depleting substances are still emitted. *(1 point for one.)*

(d) Burning fossil fuels (e.g., vehicle exhaust) emits NO_x and VOCs, which react to form ozone. Ozone peaks in the daytime because the reactions are photochemical and need sunlight. *(1 + 1.)*

(e) Ecological: ozone damages plant tissue and reduces photosynthesis/crop growth. Health: ozone irritates the respiratory system and aggravates asthma. *(1 + 1.)*

Question 9 — 2016

(a) One reason: genetic variation means some individuals resist the pathogen and reproduce; or deaths reduce host density so transmission slows; or pathogens that kill every host lose their own habitat. *(1.)*

(b)(i) Two threats to a very small population: trouble finding mates; loss of genetic diversity (inbreeding, one disease or change affecting all); competition from similar species; reduced group protection, such as shared roost warmth. *(2.)* (ii) Bats reproduce slowly (often one pup a year), mature late and have long generation times, so numbers rebound slowly. *(1.)*

(c) Two effects: insect prey increase (more crop pests and disease vectors); plants relying on bat pollination or seed dispersal decline; predators of bats lose food; guano-dependent cave communities shrink. *(2.)*

(d) Two forest services with human benefits: timber and fuel; carbon storage that slows climate change; soil protection and flood control; water purification; recreation and tourism. *(1 + 1 each pair.)*

(e) A likely reason: growing contact with wildlife through habitat clearing or wildlife trade spreads zoonotic disease to people; or increased global travel and dense populations spread pathogens quickly; or climate change widens vector ranges. *(1 for reason, 1 for explanation.)*

Question 10 — 2017

(a) Plot year (independent variable) on the x-axis and elephant population on the y-axis, with labeled axes and even scales, then place each table point; connect or not as instructed. *(1 point axes labeled and scaled; 1 point data plotted with at most one error.)*

(b) Percent loss = (initial − final) ÷ initial × 100 = (2,000,000 − 400,000) ÷ 2,000,000 × 100 = **80%**. *(1 + 1.)*

(c) 20% decline from the 2015 population of 600,000: 600,000 × 0.80 = **480,000 elephants** in 2022. *(1 + 1.)*

(d) K-strategists have few offspring and long gestation, mature late and give much parental care, so each lost adult is hard to replace and populations rebound slowly after losses, making them vulnerable to poaching or habitat change. *(1 trait, 1 explanation.)*

(e) Two strategies, each with consequence: anti-poaching enforcement and trade bans lower killing and demand; protected reserves or corridors secure habitat; ecotourism gives communities a stake in conservation; captive breeding and reintroduction restore numbers. *(1 + 1.)*

Question 11 — 2018

(a) Producer: diatom (phytoplankton); primary consumer: krill (zooplankton); secondary consumer: cod or seal (as drawn). *(3.)*

(b) The arrows also show the direction of energy (and matter) flow between trophic levels. *(1.)*

(c) With less ice there are fewer platforms and seal breathing holes, so bears must swim farther, expend more energy and wait longer between kills. *(1.)*

(d) Melting sea ice lowers surface albedo; dark open water absorbs more solar energy and warms, which melts more ice: a positive feedback that amplifies warming. *(1 for reduced reflectivity and more absorption, 1 for completing the loop.)*

(e)(i) Reasons to migrate: follow food, seek a milder climate, or reach breeding grounds. *(1.)* (ii) Commercial activity: shipping lanes cause ship strikes; fishing gear entangles whales; seismic surveys and engines create disruptive noise. *(1.)* (iii) Linked strategy: slow-speed zones and rerouting ships; modified or weak-link fishing gear; limits on seismic surveys or quieter engines. *(1.)*

Question 12 — 2019

(a)(i) Open, unvegetated sandy beach and storm-washover areas. *(1 point)*

(a)(ii) Nests fell from 44 in 1999 to 25 in 2009, a drop of about 43%. *(1 point: direction plus a number or percent)*

(a)(iii) Plant cover spread over formerly bare sand, so the open sandy nesting habitat shrank. *(1 point: habitat loss from revegetation)*

(b)(i) Fence or sign off nesting zones, cover nests with protective enclosures, restrict vehicles, kites or fireworks, or require leashed pets. *(1 point: one realistic restriction)*

(b)(ii) Barrier islands absorb wave energy and wind, which reduces mainland erosion and storm-surge flooding; their lagoons and marshes also trap sediment and shelter nursery habitat. *(1 point: one function with its protective effect)*

(b)(iii) E.g. beachfront development removes habitat; or trash and plastic entangle or are swallowed by wildlife; or oil spills coat marine life. Name the action and its effect on a species other than the plover. *(2 points: action; effect on species)*

(c) Lost tourism revenue, lower property values, repair costs for damaged infrastructure, higher insurance, or loss of farmland or fisheries income. *(1 point: one economic impact)*

(d) Two of: raise buildings; set new construction back from the shore; install pumps; build seawalls or groins; plant shoreline vegetation; nourish beaches with sand. *(2 points: 1 each)*

Question 13 — 2019

(a)(i) About 380 ppm CO_2 in 2005 (±5). *(1 point)*

(a)(ii) About pH 8.08 in 2005 (±0.02). *(1 point)*

(b)(i) More atmospheric CO_2 means more CO_2 dissolves in the ocean. *(1 point)*

(b)(ii) Inverse relationship: as atmospheric CO_2 climbs, ocean pH falls (more acidic). *(1 point)*

(b)(iii) $CO_2 + H_2O \rightleftharpoons H_2CO_3$ (carbonic acid), which ionizes to $H^+ + HCO_3^-$. *(1 point: balanced equation)*

(b)(iv) Ocean acidification. *(1 point)*

(c)(i) Added H^+ lowers the carbonate available and can dissolve calcium carbonate, so shells and coral skeletons form more slowly or erode; weaker organisms have lower reproduction and survival and are easier prey. *(2 points: chemical effect on shell or skeleton; consequence for survival)*

(c)(ii) E.g. warming causes bleaching (loss of symbiotic algae); nutrient runoff drives algal blooms that block light and deplete oxygen; destructive fishing or tourism physically breaks reefs; invasive species outcompete natives. *(2 points: threat; how it damages reefs)*

Question 14 — 2021

(a) Specialists have narrow niches, low adaptability and often small, genetically limited populations, so losing or splitting habitat harms them more than flexible generalists. *(1)*

(b) Brood parasitism (cowbird benefits, thrush harmed). *(1)*

(c) Retained forest keeps some species, reduces erosion, provides shade and transpiration cooling, and can buffer edge effects. *(1)*

(d) E.g. wildlife corridors, conservation easements, buffer zones, offsetting preserved land, clustered or vertical housing, shared green space. *(1)*

(e) Area = 275 territories × 800 m²/territory = 220,000 m². *(2: setup; answer)*

(f) Set-aside = 1,000 lots × 1,100 m² × 0.10 = 110,000 m²; ÷ 800 m² = 137.5 (about 137) territories. *(2)*

(g) Needed fraction = 220,000 m² ÷ (1,000 × 1,100 m² = 1,100,000 m²) × 100 = 20% of each lot (twice the developer's 10%). *(2)*

Question 15 — 2021

(a)(i) Greatest richness lies between the Tropics of Cancer and Capricorn (roughly 30° N-30° S). *(1)*

(a)(ii) Tropical rainforest. *(1)*

(a)(iii) Warm, wet conditions all year and high productivity supply many niches, food and habitat. *(1)*

(b) Clearing land for farming, housing, fuel or timber. *(1)*

(c)(i) A species-rich ecosystem is more resilient: some species survive a stressor and help restore function. *(1)*

(c)(ii) Other drivers: pesticide or drug pollution and endocrine disruptors, climate-driven loss of moist habitat, illegal collection, invasive species, habitat fragmentation, chytrid fungus. *(1)*

(d)(i) Endangered Species Act, CITES or the Lacey Act. *(1)*

(d)(ii) ESA: lists species, bans harming them, protects critical habitat and requires recovery plans; CITES: regulates or bans international trade; Lacey Act: bans trade in illegally taken wildlife. *(1)*

(e)(i) E.g. relocate frogs, reserve part of the wetland, build corridors, require a setback, or cluster buildings. *(1)*

(e)(ii) Justify with a non-frog benefit, e.g. buffer strip filters runoff, preserve raises rental appeal, clustering frees land and limits fragmentation. *(1)*

Question 16 — 2022

(a)(i) Clearings and gardens had the most lizards. *(1)*

(a)(ii) Inverse: lizard counts are lower where mongooses live than on mongoose-free islands. *(1)*

(b) Largest island located close to the mainland has highest species diversity. *(1)*

(c)(i) Invasive animals outcompete or prey on natives, cutting biodiversity and disrupting food webs. *(1)*

(c)(ii) Claim: biological control, trapping or hunting, manual removal, or chemicals. *(1)*

(c)(iii) Justify with a benefit matched to the method, e.g. trapping avoids synthetic chemicals; pesticides act quickly. *(1)*

(d)(i) Provisioning service: timber, food or medicine. *(1)*

(d)(ii) The graph shows more lizards in cleared land than forest on mongoose-free islands, so converting forest to cane likely increased counts (more food, suitable habitat, fewer predators). *(1)*

(d)(iii) Monocultures lower species and genetic diversity, raise pest vulnerability, deplete nutrients and need more inputs. *(1)*

(e) Legumes host nitrogen-fixing bacteria, enriching the soil and reducing nitrogen-fertilizer needs. *(1)*

Question 17 — 2023

(a) Any labeled invaded water body, e.g. Mississippi, Missouri or Ohio River, Lake Erie. *(1)*

(b) Carp now occupy a much larger area, having moved up the Mississippi, Missouri and Ohio rivers. *(1)*

(c) Possible impacts: reduced fishing or recreation, or changed water quality by displacing native filter feeders. *(1)*

(d) Reward fishing for carp, promote eating carp, barriers or electric dispersal, carp-specific deterrents or sound, ballast and bait rules. *(1)*

(e) Justify with an added benefit, e.g. new jobs or income, little harm to natives, or protecting food webs. *(1)*

(f) Zooplankton (eats algae, the producer). *(1)*

(g) Arrows show flow of energy (and matter) from prey to predator. *(1)*

(h) Carp compete for plankton, lowering food for fish higher in the chain. *(1)*

(i) Blue pike were an open-access shared resource; each fisher maximized catch, and the stock collapsed. *(1)*

(j) Aquaculture drawbacks: waste or antibiotic pollution, escapes and disease spread to wild fish, forage-fish feed demand, habitat conversion. *(1)*

Question 18 — 2024

(a) Cultural service: recreation, aesthetic or spiritual value, ecotourism. *(1 pt)* (b) Forest roots and soil slow runoff and filter water, trapping sediment and pollutants. *(1 pt)* (c) Urbanization adds impervious surface, causing runoff, flooding and pollutant loading, which reduces water quality; or it lowers infiltration and groundwater recharge. *(1 pt)* (d) Dependent variable: crude-oil concentration remaining. *(1 pt)* (e) Hypothesis: warmer water increases bacterial breakdown, so oil concentration will be lower at higher temperatures. *(1 pt)*

(f) Four replicate tanks per treatment reduce the effect of chance or tank-specific differences and allow averaging. *(1 pt)* (g) Aerated tanks supply more oxygen to the aerobic bacteria, so oil would break down faster and concentrations would be lower. *(1 pt)* (h) As CO_2 rises, pH falls: an inverse relationship from 1850 to 2100. *(1 pt)* (i) CO_2 rise from fossil-fuel burning and deforestation, absorbed by oceans. *(1 pt)* (j) Supports hypothesis: falling pH lowers carbonate ions, making calcification harder for corals, so reefs would shrink. *(1 pt)*

Question 19 — 2024

(a) 1700 had the highest forest share. *(1)*

(b) Wild grassland decreased as grazing land increased (inverse). *(1)*

(c) Overgrazing causes loss of vegetation, erosion, soil compaction, desertification or sediment in water. *(1)*

(d) Refuted: forest share fell, so less photosynthetic carbon uptake and storage, which would raise CO_2, not lower it. *(1)*

(e) Impervious surfaces raise runoff and flooding, groundwater overuse, polluted runoff, or lost aquatic habitat. *(1)*

(f) Matched response: permeable pavement, green roofs, retention basins, preserved green space, groundwater management, less fertilizer and salt, cleanup. *(1)*

(g) Forest clearing removed food, shelter and mates; hunting or trapping also reduces bobcats. *(1)*

(h) Small populations have low genetic diversity, inbreeding risk and vulnerability to disease or disturbance. *(1)*

(i) Link patches with habitat corridors, or restore and protect more forest. *(1)*

(j) Extra benefit: other species use corridors or restored forest, improving biodiversity, water, air and carbon storage. *(1)*

Question 20 — 2025

(A) About 1,400 breeding pairs in 1995 (tallest bar). *(1)* **(B)** Pairs generally declined from 1995 to 2008. *(1)* **(C)** Sea level rose over the period while tern pairs fell, contradicting an increase. *(1)* **(D)** Warming melts land ice and causes thermal expansion of ocean water. *(1)* **(E)(i)** E.g. "As sea level rises, the number of wading-bird species increases." *(1)* **(E)(ii)** Dependent variable: number of wading-bird species per location. *(1)* **(F)** PCBs bioaccumulate and biomagnify in fish-eating birds, likely lowering survival or reproduction at A and B so species counts there would drop or rise less. *(1)* **(G)** Regulating service examples: flood and storm buffering, water filtration, carbon storage, erosion control. *(1)* **(H)** Turbid water hinders visual hunters, so prey capture falls, reducing birds' food, survival and population size. *(1)* **(I)** Habitat loss, pollution, runoff, hydrological change or invasive species. *(1)*

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